q-Pochhammer Symbols and q-Binomial Coefficients

Algebra, Combinatorics, Analysis, Arithmetic, and Summation Theory

A self-contained, proof-oriented monograph

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Abstract

The finite and infinite q-Pochhammer symbols and the Gaussian, or q-binomial, coefficients form a small set of notational primitives from which a remarkable amount of mathematics can be built. They simultaneously encode weighted subsets, partitions inside rectangles, lattice paths, subspaces over finite fields, Schubert cells in Grassmannians, noncommutative binomial expansions, q-difference operators, basic hypergeometric summations, theta products, partition identities, and congruences at roots of unity.

This monograph develops those themes from first principles. It proves the algebraic laws of shifted factorials; the polynomial, reciprocal, and cyclotomic structure of Gaussian coefficients; finite and infinite q-binomial theorems; weighted generalizations through elementary and complete symmetric functions; combinatorial and finite-geometric models; unimodality by $\mathfrak{sl}_2$ representation theory; the q-Gauss, q-Chu–Vandermonde, and q-Pfaff–Saalschütz summations; Jackson calculus and the q-gamma and q-beta functions; Jacobi's triple product, Euler's pentagonal theorem, and the Rogers–Ramanujan identities through Bailey pairs; cyclotomic factorization, q-Lucas and q-Babbage congruences; negative upper indices and q-Newton interpolation; a recent family of central q-binomial/Lambert-series evaluations; geometric Lagrange filters with exact Gaussian residual moments; half-base Mersenne normalizations and Prouhet–Thue–Morse cancellation; and geometric uniform sums, q-cumulants, finite spline approximants, and the Fabius–Rvachev specializations. Every displayed theorem used in the main development is accompanied by a proof, and the final appendices collect formulas, limiting transitions, computational recipes, and a per-result record of what has and has not yet been formalized in Lean.

Preface and source map

The aim is not merely to list identities, but to expose the mechanisms that repeatedly generate them. Three such mechanisms dominate:

1. finite products split into residue classes, blocks, or initial and tail factors;
2. coefficient extraction turns product factorizations into convolution identities;
3. functional equations determine analytic q-series from one initial value.

Once these mechanisms are visible, many formulas that initially appear unrelated become instances of the same principle.

The supplied archive was used throughout. In particular, the weighted subset framework of Konvalina motivates Chapter 4; Kupershmidt's treatment of the q-Newton binomial informs Chapters 11 and 17; Straub's seminar on q-binomial congruences is reflected in Chapter 16; the formalized q-series project of Lau, Lee, and Ono motivates the algebraic-versus-analytic distinctions in Chapter 20 and supplies a modern Bailey-pair route to the Rogers–Ramanujan identities; and the central q-binomial paper of Chern, Dilcher, and Jiu is developed, with a complete proof, in Chapter 18. The archive's reference chapters and encyclopedia snapshots were checked against the standard sources listed in the bibliography. The current *ProveIt* Fabius-function documentation [38], including its main exposition and non-formalized research-frontier drafts, supplies the geometric-node interpolation rows, exact residual moments, half-base identities, formal filters, and dyadic Thue–Morse connections expanded independently in Chapters 21 to 23. Online research also led to the recent automated identity-translation framework of Zhong and Zhao [40] and the Isabelle formalization of combinatorial q-analogues by Eberl [36]. Additional material was drawn from the classical books and articles of Andrews, Askey, Bailey, Gasper, Rahman, Rota, Goldman, Reiner, Stanton, White, and others, as well as the NIST Digital Library of Mathematical Functions.

Two regimes for q

Throughout the finite algebraic chapters, q may be an indeterminate and identities are understood in a polynomial or rational-function ring. In analytic chapters, unless stated otherwise,

$$q \in \mathbb{C}, \qquad 0 < |q| < 1.$$

For the Jackson integral and positivity statements we usually assume $0 < q < 1$. A capital Q is used for the cardinality of a finite field, so Q is a prime power and normally $Q > 1$.

Conventions

Empty products equal 1, empty sums equal 0, and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = 0 \qquad \text{when } k < 0 \text{ or } k > n$$

for nonnegative integral n. Multiple parameters are compressed as

$$(a_1, \ldots, a_r; q)_n = \prod_{j=1}^{r} (a_j; q)_n.$$

The notation $(a; q)_\infty$ is analytic when $|q| < 1$; when formal power series are intended, that will be said explicitly.

Contents

Part I

The finite algebra

Chapter 1

The algebra of q-shifted factorials

1.1 Definitions and first examples

Definition 1.1. For $n \in \mathbb{N}_0$ the *q-shifted factorial*, or *q-Pochhammer symbol*, is

$$(a;q)_n = \prod_{j=0}^{n-1}(1-aq^j), \qquad (a;q)_0 = 1.$$

For parameters $a_1, \dots, a_r$ we write

$$(a_1, \dots, a_r; q)_n = \prod_{i=1}^{r}(a_i;q)_n.$$

When $0 < |q| < 1$, the infinite symbol is

$$(a;q)_\infty = \prod_{j=0}^{\infty}(1-aq^j).$$

The word "shifted" refers to the geometric progression $a, aq, aq^2, \dots$. The first few values are

$$(a;q)_1 = 1-a, \qquad (a;q)_2 = (1-a)(1-aq), \qquad (a;q)_3 = (1-a)(1-aq)(1-aq^2).$$

Important specializations include

$$(q;q)_n = \prod_{j=1}^{n}(1-q^j), \qquad (-q;q)_n = \prod_{j=1}^{n}(1+q^j), \qquad (a;q^r)_n = \prod_{j=0}^{n-1}(1-aq^{rj}).$$

1.2 Concatenation, dissection, and reversal

Proposition 1.2 (Concatenation law)**.** *For $m, n \in \mathbb{N}_0$,*

$$(a;q)_{m+n} = (a;q)_m(aq^m;q)_n.$$

Consequently, whenever the denominator is nonzero,

$$\frac{(a;q)_{m+n}}{(a;q)_m} = (aq^m;q)_n.$$

Proof. Split the defining product at $j = m$:

$$\prod_{j=0}^{m+n-1} (1 - aq^j) = \left(\prod_{j=0}^{m-1} (1 - aq^j)\right)\left(\prod_{j=m}^{m+n-1} (1 - aq^j)\right).$$

In the second product put $j = m + r$, where $0 \le r < n$. □

Proposition 1.3 (Dissection into residue classes)**.** *For integers $r, n \ge 1$,*

$$(a; q)_{rn} = \prod_{s=0}^{r-1} (aq^s; q^r)_n.$$

If $|q| < 1$, then also

$$(a; q)_\infty = \prod_{s=0}^{r-1} (aq^s; q^r)_\infty.$$

Proof. Every integer j with $0 \le j < rn$ has a unique representation $j = s + rt$ with $0 \le s < r$ and $0 \le t < n$. Reordering the finite product gives the first identity. For the infinite identity, the factors are absolutely summably close to 1 because $\sum_{j\ge0} |aq^j| < \infty$, so the product may be regrouped into the r residue classes. □

Corollary 1.4 (Dissection with a remainder)**.** *Let $r \ge 1$ and $n \ge 0$ be integers, let $0 \le u < r$, and set $N = rn + u$. Then*

$$(a; q)_N = \prod_{s=0}^{u-1} (aq^s; q^r)_{n+1} \prod_{s=u}^{r-1} (aq^s; q^r)_n.$$

For $u = 0$ this is Proposition 1.3.

Proof. Division with remainder by r is a bijection between the nonnegative integers j and the pairs (s, ℓ) with $0 \le s < r$ and $\ell \ge 0$, given by $j = s + r\ell$. Fix a residue s and count the ℓ for which $j = s + r\ell$ lies in the exponent set $\{0, \ldots, N - 1\}$ of the defining product. The condition $s + r\ell < rn + u$ reads

$$r\ell < rn + (u - s).$$

If $s < u$, then $0 < u - s \le u < r$, so

$$rn < rn + (u - s) \le rn + r = r(n + 1).$$

As $r\ell$ runs through the multiples of r, the inequality $r\ell < rn + (u - s)$ therefore holds exactly for $\ell = 0, 1, \ldots, n$, which is $n + 1$ values.

If $s \ge u$, then $u - s \le 0$, so the inequality implies $r\ell < rn$, that is $\ell \le n - 1$; conversely every such ℓ satisfies $s + r\ell \le s + r(n - 1) = rn + (s - r) < rn \le N$. Hence ℓ ranges over $0, \ldots, n - 1$, which is n values.

The defining product is finite and its factors commute, so it may be reordered according to the residue of j modulo r:

$$(a; q)_N = \prod_{s=0}^{u-1} \prod_{\ell=0}^{n} \bigl(1 - aq^s (q^r)^\ell\bigr) \prod_{s=u}^{r-1} \prod_{\ell=0}^{n-1} \bigl(1 - aq^s (q^r)^\ell\bigr).$$

Each inner product is by definition a q^r-shifted factorial with argument aq^s, namely $(aq^s; q^r)_{n+1}$ in the first family and $(aq^s; q^r)_n$ in the second. This is the asserted identity. When $u = 0$ the first family is empty and the second has all r residues, recovering Proposition 1.3. □

Remark 1.5. Two consistency checks are worth recording. The total number of linear factors on the right is

$$u(n+1)+(r-u)n = rn+u = N,$$

as it must be, since $(a;q)_N$ is a product of N linear factors in a. At the other extreme $n=0$ the corollary reads $(a;q)_u = \prod_{s=0}^{u-1}(aq^s;q^r)_1 = \prod_{s=0}^{u-1}(1-aq^s)$, again the definition. Combining the corollary with Proposition 1.2 in the form $(a;q)_{rn+u} = (a;q)_{rn}(aq^{rn};q)_u$ and dividing by Proposition 1.3 recovers the identity $\prod_{s=0}^{u-1}(1-aq^{s+rn}) = (aq^{rn};q)_u$, so the two decompositions agree.

Proposition 1.6 (Base reversal)**.** *For $n \geq 0$ and $a \neq 0$,*

$$(a;q)_n = (-a)^n q^{\binom{n}{2}}(a^{-1};q^{-1})_n.$$

Equivalently,

$$(a;q^{-1})_n = (-a)^n q^{-\binom{n}{2}}(a^{-1};q)_n.$$

Proof. Factor $-aq^j$ from each term:

$$1-aq^j = -aq^j\left(1-a^{-1}q^{-j}\right).$$

Multiplying for $0 \leq j < n$ gives the factor $(-a)^n q^{0+1+\cdots+(n-1)} = (-a)^n q^{\binom{n}{2}}$. The remaining product is $(a^{-1};q^{-1})_n$. □

1.3 Negative indices

The concatenation law suggests the only extension to negative integers that preserves $(a;q)_{m+n} = (a;q)_m(aq^m;q)_n$.

Definition 1.7. For $n \geq 1$, define

$$(a;q)_{-n} = \frac{1}{(aq^{-n};q)_n},$$

provided no factor in the denominator vanishes.

Proposition 1.8 (Closed form for a negative index)**.** *For $n \geq 1$ and $a \neq 0$,*

$$(a;q)_{-n} = \frac{(-1)^n a^{-n} q^{\binom{n+1}{2}}}{(q/a;q)_n}.$$

Moreover, for all integers m,n for which both sides are defined,

$$(a;q)_{m+n} = (a;q)_m(aq^m;q)_n.$$

Proof. By definition,

$$(aq^{-n};q)_n = \prod_{j=0}^{n-1}(1-aq^{-n+j}).$$

Factor $-aq^{-n+j}$ from each factor. Their product is

$$(-a)^n q^{-n^2+\binom{n}{2}} = (-a)^n q^{-\binom{n+1}{2}},$$

and the remaining product is

$$\prod_{j=0}^{n-1}(1-a^{-1}q^{n-j}) = (q/a;q)_n.$$

Taking the reciprocal proves the formula. The extended concatenation identity follows by cancelling the overlapping finite factors; the cases in which one of $m,n,m+n$ is negative reduce directly to the definition. □

Corollary 1.9 (A terminating factor)**.** *If* $0 \le k \le N$*, then*

$$(q^{-N};q)_k = (-1)^k q^{-Nk+\binom{k}{2}} \frac{(q;q)_N}{(q;q)_{N-k}}.$$

In particular, $(q^{-N};q)_k = 0$ *for* $k > N$*.*

Proof. Apply base reversal to $(q^{-N};q)_k$:

$$(q^{-N};q)_k = (-1)^k q^{-Nk+\binom{k}{2}} (q^{N-k+1};q)_k.$$

The last factor is $(q;q)_N/(q;q)_{N-k}$. For $k > N$, the defining product contains the factor $1 - q^0$. □

1.4 Partial fractions and logarithmic derivatives

Theorem 1.10 (Finite partial-fraction decomposition)**.** *Assume* $q^i \neq q^j$ *for* $0 \le i < j \le n$*. Then*

$$\frac{1}{(z;q)_{n+1}} = \sum_{j=0}^{n} \frac{(-1)^j q^{\binom{j+1}{2}}}{(q;q)_j (q;q)_{n-j}} \frac{1}{1 - zq^j}.$$

Proof. The left side has simple poles at $z = q^{-j}$, $0 \le j \le n$. The coefficient of $(1 - zq^j)^{-1}$ is

$$A_j = \left.\frac{1 - zq^j}{(z;q)_{n+1}}\right|_{z=q^{-j}} = \frac{1}{\prod_{r \neq j}(1 - q^{r-j})}.$$

For $r < j$, put $s = j - r$ and obtain

$$\prod_{r=0}^{j-1}(1 - q^{r-j}) = \prod_{s=1}^{j}(1 - q^{-s}) = (-1)^j q^{-\binom{j+1}{2}} (q;q)_j.$$

The factors with $r > j$ multiply to $(q;q)_{n-j}$. Thus A_j is the displayed coefficient. Subtracting the proposed right side from the left gives a rational function with no poles and tending to 0 at infinity, hence it is identically zero. □

Proposition 1.11 (Logarithmic derivatives)**.** *Away from its zeros,*

$$\frac{\partial}{\partial a} \log(a;q)_n = -\sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j}.$$

If $a = a(x)$ *is differentiable, then*

$$\frac{\mathrm{d}}{\mathrm{d}x} \log(a(x);q)_n = -a'(x) \sum_{j=0}^{n-1} \frac{q^j}{1 - a(x)q^j}.$$

Proof. Differentiate the logarithm of the finite product term by term. □

1.5 Infinite products and Lambert series

Theorem 1.12 (Convergence and zero set)**.** *Let $0 < |q| < 1$. The product $(a;q)_\infty$ converges locally uniformly as a function of $a \in \mathbb{C}$ and therefore defines an entire function. Its zeros are precisely*

$$a = q^{-j}, \qquad j = 0, 1, 2, \ldots,$$

and all are simple.

Proof. On a compact set $K \subset \mathbb{C}$ let $M = \max_{a \in K} |a|$. Then

$$\sum_{j=0}^{\infty} \sup_{a \in K} |aq^j| \le \frac{M}{1 - |q|} < \infty.$$

The standard convergence criterion for infinite products shows that $\prod_j (1 - aq^j)$ converges uniformly on K; equivalently, one may apply uniform convergence to the tails after choosing an analytic branch of the logarithm near 1. Uniform limits of holomorphic functions are holomorphic, so the limit is entire.

If $a = q^{-j}$, one factor vanishes. Conversely, if no factor vanishes, then the tail product is nonzero because the sum of the absolute deviations from 1 converges, while the finite initial product is nonzero. At $a = q^{-j}$ exactly one factor vanishes and its derivative is $-q^j \neq 0$, whereas all other factors are nonzero; the zero is simple. □

1.5.1 Complex orders

Finite and negative integral indices extend naturally to a complex order once an infinite product is available. Fix $0 < |q| < 1$ and a branch of the logarithm at q, denoted by $\log q$, and write $q^\alpha = \exp(\alpha \log q)$.

Definition 1.13. For $\alpha \in \mathbb{C}$, define

$$(a;q)_\alpha := \frac{(a;q)_\infty}{(aq^\alpha;q)_\infty}, \tag{1.1}$$

whenever the denominator is nonzero.

Proposition 1.14 (Complex concatenation)**.** *Whenever the displayed quantities are defined,*

$$(a;q)_{\alpha+\beta} = (a;q)_\alpha (aq^\alpha;q)_\beta. \tag{1.2}$$

In particular,

$$(a;q)_{\alpha+1} = (1 - aq^\alpha)(a;q)_\alpha. \tag{1.3}$$

For integral α, (1.1) agrees with the finite and negative-index definitions already given.

Proof. By definition,

$$(a;q)_\alpha (aq^\alpha;q)_\beta = \frac{(a;q)_\infty}{(aq^\alpha;q)_\infty} \frac{(aq^\alpha;q)_\infty}{(aq^{\alpha+\beta};q)_\infty} = \frac{(a;q)_\infty}{(aq^{\alpha+\beta};q)_\infty}.$$

This is (1.2). Taking $\beta = 1$ gives (1.3). If $\alpha = n \ge 0$, the tail factorization $(a;q)_\infty = (a;q)_n (aq^n;q)_\infty$ recovers the finite product. If $\alpha = -n < 0$, the same relation, applied after shifting by $-n$, gives $(a;q)_{-n} = 1/(aq^{-n};q)_n$. □

Proposition 1.15 (Derivative with respect to the order)**.** *On every domain avoiding the poles of* (1.1)*,*

$$\frac{\partial}{\partial\alpha}\log(a;q)_\alpha = (\log q)\sum_{j=0}^{\infty}\frac{aq^{\alpha+j}}{1-aq^{\alpha+j}}. \tag{1.4}$$

The series is locally uniformly convergent.

Proof. Only the denominator in (1.1) depends on α. Differentiate its locally uniformly convergent logarithm term by term:

$$-\frac{\partial}{\partial\alpha}\sum_{j\geq 0}\log(1-aq^{\alpha+j}) = (\log q)\sum_{j\geq 0}\frac{aq^{\alpha+j}}{1-aq^{\alpha+j}}.$$

On a compact set avoiding the poles, the denominators of the tail are bounded away from zero and the numerators form a geometric majorant, proving local uniform convergence. □

Theorem 1.16 (Lambert-series expansion)**.** *If* $|q| < 1$ *and* $|a| < 1$*, then*

$$\log(a;q)_\infty = -\sum_{m=1}^{\infty}\frac{a^m}{m(1-q^m)}.$$

Consequently,

$$\frac{\partial}{\partial a}\log(a;q)_\infty = -\sum_{m=1}^{\infty}\frac{a^{m-1}}{1-q^m} = -\sum_{j=0}^{\infty}\frac{q^j}{1-aq^j}.$$

All series converge locally uniformly in $|a| < 1$*.*

Proof. For $|a| < 1$, use $\log(1-u) = -\sum_{m\geq 1}u^m/m$ and absolute convergence:

$$\sum_{j\geq 0}\sum_{m\geq 1}\frac{|a|^m|q|^{jm}}{m} = \sum_{m\geq 1}\frac{|a|^m}{m(1-|q|^m)} < \infty.$$

Thus the order of summation may be exchanged:

$$\sum_{j\geq 0}\log(1-aq^j) = -\sum_{m\geq 1}\frac{a^m}{m}\sum_{j\geq 0}q^{jm}.$$

The geometric sum gives the first formula. Local uniform convergence permits termwise differentiation. Expanding $(1-aq^j)^{-1}$ geometrically gives the second form of the derivative. □

Corollary 1.17 (Divisor sums)**.** *As a formal power series,*

$$-q\frac{\mathrm{d}}{\mathrm{d}q}\log(q;q)_\infty = \sum_{n=1}^{\infty}\sigma_1(n)q^n,$$

where $\sigma_1(n) = \sum_{d|n} d$*.*

Proof. Logarithmic differentiation gives

$$-q\frac{\mathrm{d}}{\mathrm{d}q}\log(q;q)_\infty = \sum_{j\geq 1}\frac{jq^j}{1-q^j} = \sum_{j\geq 1}\sum_{r\geq 1}jq^{jr}.$$

The coefficient of q^n is the sum of all j dividing n. □

1.6 The dilogarithmic limit

The singular regime $q \to 1^-$ turns an infinite product into an exponential of order $(1-q)^{-1}$. The dilogarithm is the natural leading term.

Theorem 1.18 (Asymptotic expansion near $q = 1$)**.** *Fix $0 < \rho < 1$ and an integer $N \geq 1$. Uniformly for $|a| \leq \rho$, as $t \to 0^+$,*

$$\begin{aligned}\log(a; \mathrm{e}^{-t})_\infty = &-\frac{\mathrm{Li}_2(a)}{t} + \frac{1}{2}\log(1-a)\\ &- \sum_{r=1}^{N-1} \frac{B_{2r}}{(2r)!} t^{2r-1}\,\mathrm{Li}_{2-2r}(a) + O_{\rho,N}(t^{2N-1}),\end{aligned}$$

where B_{2r} are Bernoulli numbers and $\mathrm{Li}_s(a) = \sum_{m\geq 1} a^m/m^s$ for $|a| < 1$.

Proof. By Theorem 1.16,

$$\log(a; \mathrm{e}^{-t})_\infty = -\sum_{m\geq 1} \frac{a^m}{m(1-\mathrm{e}^{-mt})}.$$

The Bernoulli expansion at the origin is

$$\frac{1}{1-\mathrm{e}^{-x}} = \frac{1}{x} + \frac{1}{2} + \sum_{r=1}^{N-1} \frac{B_{2r}}{(2r)!} x^{2r-1} + R_N(x),$$

with $R_N(x) = O_N(x^{2N-1})$ for $0 \leq x \leq 1$. Substitution of the displayed polynomial terms gives respectively

$$-\frac{1}{t}\sum_{m\geq 1}\frac{a^m}{m^2}, \qquad -\frac{1}{2}\sum_{m\geq 1}\frac{a^m}{m} = \frac{1}{2}\log(1-a),$$

and the polylogarithmic terms in the statement.

It remains to bound the total remainder. Split the m-sum at $m \leq t^{-1}$. In this range,

$$\sum_{m\leq t^{-1}} \frac{|a|^m}{m}|R_N(mt)| \leq C_N t^{2N-1} \sum_{m\geq 1} \rho^m m^{2N-2} = O_{\rho,N}(t^{2N-1}).$$

For $m > t^{-1}$, subtracting finitely many asymptotic terms leaves a quantity bounded by a polynomial in mt plus $(1-\mathrm{e}^{-mt})^{-1}$; the factor ρ^m makes the tail exponentially small in $1/t$, hence smaller than every power of t. This proves the uniform remainder estimate. □

1.7 Classical shifted factorials as a limit

Proposition 1.19 (Finite classical limit)**.** *For fixed $x \in \mathbb{C}$ and $n \in \mathbb{N}_0$,*

$$\lim_{q\to 1} \frac{(q^x; q)_n}{(1-q)^n} = x(x+1)\cdots(x+n-1) = (x)_n,$$

where the limit is taken through values for which a consistent branch of q^x is chosen.

Proof. Write

$$\frac{(q^x; q)_n}{(1-q)^n} = \prod_{j=0}^{n-1} \frac{1-q^{x+j}}{1-q}.$$

Each factor tends to $x + j$, since $q^u = 1 + u(q-1) + o(q-1)$ for fixed u. □

Chapter 2

q-integers and Gaussian coefficients

2.1 Definitions

Definition 2.1. For $n \in \mathbb{N}_0$ define the q-integer and q-factorial by

$$[n]_q = \frac{1-q^n}{1-q} = 1 + q + \cdots + q^{n-1}, \qquad [n]_q! = \prod_{j=1}^{n} [j]_q.$$

For $0 \le k \le n$, the *Gaussian coefficient* is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_k (q;q)_{n-k}} = \frac{[n]_q!}{[k]_q! [n-k]_q!}.$$

Outside $0 \le k \le n$ it is defined to be 0.

The notation is initially a rational expression, but the recurrences below show that it is a polynomial in q with nonnegative integral coefficients.

Proposition 2.2 (Product forms)**.** *For* $0 \le k \le n$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1};q)_k}{(q;q)_k} = \prod_{j=1}^{k} \frac{1-q^{n-k+j}}{1-q^j}.$$

Proof. Cancel $(q;q)_{n-k}$ from $(q;q)_n$ using Proposition 1.2. □

2.2 The two Pascal recurrences

Theorem 2.3 (q-Pascal recurrences)**.** *For* $1 \le k \le n-1$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q, \tag{2.1}$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k \end{bmatrix}_q. \tag{2.2}$$

Both identities remain valid at the boundary under the zero convention.

Proof. Put the two terms on the right of (2.1) over the common denominator $(q;q)_k(q;q)_{n-k}$. Their numerator is

$$(q;q)_{n-1}\big((1-q^k)+q^k(1-q^{n-k})\big)=(q;q)_{n-1}(1-q^n)=(q;q)_n.$$

The proof of (2.2) is the same, using $q^{n-k}(1-q^k)+(1-q^{n-k})=1-q^n$. □

Corollary 2.4 (Polynomiality and positivity)**.** *For all* $0\le k\le n$*, the Gaussian coefficient* $\begin{bmatrix} n\\ k\end{bmatrix}_q$ *belongs to* $\mathbb{N}_0[q]$.

Proof. Induct on n. The boundary values are 1. Either recurrence in Theorem 2.3 expresses an interior coefficient as a sum of two polynomials with nonnegative integral coefficients. □

2.3 Symmetry, reciprocity, and degree

Theorem 2.5 (Elementary structure)**.** *For* $0\le k\le n$*,*

(i) $\begin{bmatrix} n\\ k\end{bmatrix}_q=\begin{bmatrix} n\\ n-k\end{bmatrix}_q$;

(ii) $\begin{bmatrix} n\\ k\end{bmatrix}_{q^{-1}}=q^{-k(n-k)}\begin{bmatrix} n\\ k\end{bmatrix}_q$;

(iii) $\begin{bmatrix} n\\ k\end{bmatrix}_q$ *has degree* $k(n-k)$*, constant term* 1*, and leading coefficient* 1*;*

(iv) *its coefficient sequence is palindromic.*

Proof. Symmetry is immediate from the factorial definition. For reciprocity, use

$$(q^{-1};q^{-1})_m=(-1)^m q^{-\binom{m+1}{2}}(q;q)_m$$

for $m=n,k,n-k$. The signs cancel and the exponent is

$$-\binom{n+1}{2}+\binom{k+1}{2}+\binom{n-k+1}{2}=-k(n-k).$$

Since $\begin{bmatrix} n\\ k\end{bmatrix}_q$ is a polynomial with constant term 1, reciprocity says

$$\begin{bmatrix} n\\ k\end{bmatrix}_q=q^{k(n-k)}\begin{bmatrix} n\\ k\end{bmatrix}_{q^{-1}}.$$

Hence its degree is at most $k(n-k)$ and its leading coefficient equals its constant coefficient. The product form shows that the order at infinity is exactly $k(n-k)$, so the degree is exact. The same reciprocal identity equates coefficients at complementary degrees and proves palindromicity. □

Corollary 2.6 (Classical limit)**.** *For fixed* n,k*,*

$$\lim_{q\to 1}\begin{bmatrix} n\\ k\end{bmatrix}_q=\binom{n}{k}.$$

Proof. Use the q-factorial form and $[j]_q\to j$. □

2.4 Multinomial coefficients and flags

Definition 2.7. If $n_1, \ldots, n_r \geq 0$ and $n_1 + \cdots + n_r = n$, define

$$\begin{bmatrix} n \\ n_1, \ldots, n_r \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_{n_1} \cdots (q;q)_{n_r}}.$$

Proposition 2.8 (Factorization into binomials)**.** *Let $N_j = n_j + n_{j+1} + \cdots + n_r$. Then*

$$\begin{bmatrix} n \\ n_1, \ldots, n_r \end{bmatrix}_q = \prod_{j=1}^{r-1} \begin{bmatrix} N_j \\ n_j \end{bmatrix}_q.$$

In particular, every q-multinomial coefficient is a polynomial with nonnegative integral coefficients.

Proof. Write out the factorial ratios. Consecutive numerator and denominator factors telescope:

$$\prod_{j=1}^{r-1} \frac{(q;q)_{N_j}}{(q;q)_{n_j}(q;q)_{N_{j+1}}} = \frac{(q;q)_n}{\prod_{j=1}^{r}(q;q)_{n_j}}.$$

Positivity follows from Corollary 2.4. □

2.5 Adjacent ratios and efficient recurrences

Proposition 2.9. *For $0 \leq k < n$,*

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}_q}{\begin{bmatrix} n \\ k \end{bmatrix}_q} = \frac{1-q^{n-k}}{1-q^{k+1}}.$$

For $0 \leq k \leq n-1$,

$$\frac{\begin{bmatrix} n \\ k \end{bmatrix}_q}{\begin{bmatrix} n-1 \\ k \end{bmatrix}_q} = \frac{1-q^{n}}{1-q^{n-k}}.$$

Proof. Cancel common factors in the product or factorial definitions. □

These ratios are useful numerically, but near roots of unity they hide cancellations. The polynomial recurrences of Theorem 2.3 are then preferable.

2.6 Coefficient moments

Write

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=0}^{D} c_j q^j, \qquad D = k(n-k),$$

and choose an exponent X with probability $\Pr(X = j) = c_j / \binom{n}{k}$. The combinatorial meaning of X will be established in Chapter 5; its moments can already be found algebraically.

Theorem 2.10 (Mean and variance)**.** *For the preceding distribution,*

$$\mathbb{E}X = \frac{k(n-k)}{2}, \qquad \operatorname{Var}(X) = \frac{k(n-k)(n+1)}{12}.$$

Equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}q} \begin{bmatrix} n \\ k \end{bmatrix}_q \bigg|_{q=1} = \frac{k(n-k)}{2} \binom{n}{k}.$$

Proof. The palindromic identity $c_j = c_{D-j}$ immediately gives $\mathbb{E}X = D/2$ and the derivative formula.

For the variance, use the moment-generating function

$$M(t) = \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{\mathrm{e}^t}}{\binom{n}{k}} = \prod_{i=1}^{k} \frac{\mathrm{e}^{(n-k+i)t} - 1}{(n-k+i)t} \frac{it}{\mathrm{e}^{it} - 1}.$$

The local expansion

$$\log \frac{\mathrm{e}^u - 1}{u} = \frac{u}{2} + \frac{u^2}{24} + O(u^3)$$

gives

$$\log M(t) = \frac{t}{2} \sum_{i=1}^{k} (n-k) + \frac{t^2}{24} \sum_{i=1}^{k} \big((n-k+i)^2 - i^2\big) + O(t^3).$$

The first sum is $k(n-k)$. The second equals

$$k(n-k)^2 + 2(n-k)\frac{k(k+1)}{2} = k(n-k)(n+1).$$

Since the first and second derivatives of $\log M$ at 0 are the mean and variance, respectively, the result follows. $\square$

Chapter 3

Finite q-binomial theorems and inversion

3.1 Euler's finite product expansions

Theorem 3.1 (Finite q-binomial theorem)**.** *For $n \geq 0$,*

$$(z;q)_n = \sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k, \tag{3.1}$$

$$(-z;q)_n = \sum_{k=0}^{n} q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \tag{3.2}$$

Proof. It is enough to prove the first identity; replace z by $-z$ for the second. The case $n = 0$ is clear. Suppose the formula holds for n. Then

$$(z;q)_{n+1} = (1 - zq^n)(z;q)_n.$$

The coefficient of z^k on the right is

$$(-1)^k q^{\binom{k}{2}} \left(\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q \right).$$

By the second q-Pascal recurrence, the parenthesis is $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$. This completes the induction. □

Corollary 3.2 (Alternating sums)**.** *For $n > 0$,*

$$\sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q = 0.$$

More generally,

$$\sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}+mk} \begin{bmatrix} n \\ k \end{bmatrix}_q = (q^m;q)_n.$$

Proof. Set $z = 1$ and $z = q^m$, respectively, in (3.1). □

Corollary 3.3 (Weighted subset sum)**.** *For $0 \leq k \leq n$,*

$$\sum_{0 \leq i_1 < \cdots < i_k \leq n-1} q^{i_1 + \cdots + i_k} = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Equivalently,

$$\sum_{1 \leq j_1 < \cdots < j_k \leq n} q^{j_1 + \cdots + j_k} = q^{\binom{k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Proof. Expand $\prod_{i=0}^{n-1}(1+zq^i)$ by choosing either 1 or zq^i from each factor, and compare the coefficient of z^k with (3.2). □

3.2 Reciprocal finite products

Theorem 3.4 (Reciprocal finite q-binomial theorem)**.** *For $n \geq 0$ and $|z| < 1$,*

$$\frac{1}{(z;q)_{n+1}} = \sum_{k=0}^{\infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k.$$

The identity also holds formally in $\mathbb{Z}[q][[z]]$.

Proof. Let $H_n(z) = \sum_{k \geq 0} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$. The second q-Pascal recurrence gives

$$\begin{bmatrix} n+k \\ k \end{bmatrix}_q - q^n \begin{bmatrix} n+k-1 \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

Therefore

$$(1-zq^n)H_n(z) = H_{n-1}(z),$$

where $H_0(z) = \sum_{k \geq 0} z^k = (1-z)^{-1}$. Iterating yields

$$H_n(z) = \frac{1}{(1-z)(1-zq)\cdots(1-zq^n)}.$$

All operations are valid formally; analytically they are justified for $|z| < 1$. □

3.3 q-Vandermonde convolution

Theorem 3.5 (q-Vandermonde convolution)**.** *For $m, n, r \in \mathbb{N}_0$,*

$$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_{j \in \mathbb{Z}} q^{(m-j)(r-j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q, \tag{3.3}$$

$$= \sum_{j \in \mathbb{Z}} q^{j(n-r+j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q. \tag{3.4}$$

Only finitely many terms are nonzero.

Proof. Factor

$$(-z;q)_{m+n} = (-z;q)_m(-zq^m;q)_n$$

and use (3.2). The coefficient of z^r on the left is $q^{\binom{r}{2}} \begin{bmatrix} m+n \\ r \end{bmatrix}_q$. The term with j chosen from the first factor and $r-j$ from the second has exponent

$$\binom{j}{2} + m(r-j) + \binom{r-j}{2}.$$

Subtracting $\binom{r}{2}$ gives $(m-j)(r-j)$, proving (3.3). Splitting the product in the reverse order gives (3.4). □

Corollary 3.6 (Central form)**.** *For $m, n \geq 0$,*

$$\begin{bmatrix} m+n \\ m \end{bmatrix}_q = \sum_j q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q.$$

Proof. Set $r = m$ in (3.3) and put $\ell = m - j$. Then the exponent becomes ℓ^2, while symmetry gives $\begin{bmatrix} m \\ m-\ell \end{bmatrix}_q = \begin{bmatrix} m \\ \ell \end{bmatrix}_q$ and the second factor is $\begin{bmatrix} n \\ \ell \end{bmatrix}_q$. □

Corollary 3.7 (Shifted central form)**.** *For $N \in \mathbb{N}_0$ and $k \in \mathbb{Z}$,*

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \sum_{\ell \in \mathbb{Z}} q^{\ell^2 + \ell k} \begin{bmatrix} N \\ \ell \end{bmatrix}_q \begin{bmatrix} N \\ \ell + k \end{bmatrix}_q, \tag{3.5}$$

where a Gaussian coefficient whose lower index falls outside $\{0, 1, \ldots, \text{upper index}\}$ is read as 0, so that only finitely many terms are nonzero. The case $k = 0$ is Corollary 3.6 with $m = n = N$.

Proof. If $|k| > N$ both sides vanish: the left side because $N + k \notin \{0, \ldots, 2N\}$, and the right side because a nonzero term requires both $0 \le \ell \le N$ and $0 \le \ell + k \le N$, which forces $|k| \le N$.

Assume therefore $-N \le k \le N$ and set $r = N + k \in \{0, \ldots, 2N\}$. Theorem 3.5, in the form (3.3) with $m = n = N$, gives

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \sum_{j \in \mathbb{Z}} q^{(N-j)(N+k-j)} \begin{bmatrix} N \\ j \end{bmatrix}_q \begin{bmatrix} N \\ N+k-j \end{bmatrix}_q.$$

Substitute $\ell = N - j$. Then $N + k - j = \ell + k$, so the exponent becomes $\ell(\ell + k) = \ell^2 + \ell k$; the symmetry $\begin{bmatrix} N \\ N-\ell \end{bmatrix}_q = \begin{bmatrix} N \\ \ell \end{bmatrix}_q$ of Theorem 2.5 (i) turns the first Gaussian coefficient into $\begin{bmatrix} N \\ \ell \end{bmatrix}_q$, and the second is $\begin{bmatrix} N \\ \ell+k \end{bmatrix}_q$. This is (3.5). □

3.4 q-binomial inversion

Theorem 3.8 (q-binomial inversion)**.** *For sequences (a_n) and (b_n) in a commutative ring containing q, the relations*

$$b_n = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q a_k$$

are equivalent to

$$a_n = \sum_{k=0}^{n} (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k.$$

Proof. Substitute the first relation into the proposed inverse. The coefficient of a_j becomes

$$\sum_{k=j}^{n} (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q.$$

The factorial definition gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} n-j \\ k-j \end{bmatrix}_q.$$

Put $r = n - k$. The coefficient is

$$\begin{bmatrix} n \\ j \end{bmatrix}_q \sum_{r=0}^{n-j} (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n-j \\ r \end{bmatrix}_q.$$

By Corollary 3.2, this equals 0 when $n > j$ and 1 when $n = j$. Thus the composition is the identity. The same argument in the opposite order proves equivalence. □

Corollary 3.9 (Matrix orthogonality)**.** *For $0 \le j \le n$,*

$$\sum_{k=j}^{n} (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \delta_{n,j}.$$

Proof. This is the kernel computation in the preceding proof. □

3.5 A q-Cauchy product identity

Theorem 3.10 (q-Cauchy identity)**.** *For u, v in a commutative ring,*

$$(uv; q)_n = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q (u; q)_k v^k (v; q)_{n-k}.$$

Proof. Induct on n. The assertion is clear at $n = 0$. Let the right side be $C_n(u, v)$. Use

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n \\ k \end{bmatrix}_q$$

and split C_{n+1} accordingly. In the first part shift $k \mapsto k + 1$, use $(u; q)_{k+1} = (u; q)_k (1 - uq^k)$, and combine with the second part after using $(v; q)_{n+1-k} = (v; q)_{n-k}(1 - vq^{n-k})$. The bracket that remains is

$$(1 - vq^{n-k}) + vq^{n-k}(1 - uq^k) = 1 - uvq^n,$$

independent of k. Therefore

$$C_{n+1}(u, v) = (1 - uvq^n) C_n(u, v) = (uv; q)_{n+1}.$$

□

Remark 3.11. Taking $u = 0$ recovers the tautology $(0; q)_k = 1$ inside a weighted expansion; taking special limits of u or v produces several finite convolution formulas used later in the proof of the q-Pfaff–Saalschütz summation.

Chapter 4

Weighted binomial coefficients and symmetric functions

The ordinary and Gaussian binomial coefficients are instances of a broader construction in which each available object carries a weight. This point of view, developed systematically by Konvalina, places subset selection, multiset selection, Stirling numbers, and Gaussian coefficients in one framework.

4.1 Elementary and complete weighted coefficients

Definition 4.1. Let $\mathbf{w} = (w_1, \dots, w_n)$ be elements of a commutative ring. Define

$$C_k^n(\mathbf{w}) = \sum_{1 \le i_1 < \cdots < i_k \le n} w_{i_1} \cdots w_{i_k},$$
$$S_k^n(\mathbf{w}) = \sum_{1 \le i_1 \le \cdots \le i_k \le n} w_{i_1} \cdots w_{i_k}.$$

We set $C_0^n = S_0^n = 1$. Thus C_k^n is the elementary symmetric function $e_k(w_1, \dots, w_n)$ and S_k^n is the complete homogeneous symmetric function $h_k(w_1, \dots, w_n)$.

Theorem 4.2 (Weighted Pascal recurrences)**.** *For $n, k \ge 1$,*

$$C_k^n(\mathbf{w}) = C_k^{n-1}(\mathbf{w}) + w_n C_{k-1}^{n-1}(\mathbf{w}), \tag{4.1}$$
$$S_k^n(\mathbf{w}) = S_k^{n-1}(\mathbf{w}) + w_n S_{k-1}^{n}(\mathbf{w}). \tag{4.2}$$

Proof. A k-element subset of $\{1, \dots, n\}$ either omits n, contributing C_k^{n-1}, or contains n, in which case removal of n leaves a $(k-1)$-element subset and the weight acquires a factor w_n. This proves (4.1).

For multisets, separate those with no occurrence of n from those with at least one occurrence. Removing one occurrence of n from the latter leaves an arbitrary multiset of size $k-1$ drawn from all n types. This proves (4.2). □

Theorem 4.3 (Generating products)**.** *One has*

$$\prod_{i=1}^{n} (1 + w_i z) = \sum_{k=0}^{n} C_k^n(\mathbf{w}) z^k, \tag{4.3}$$
$$\prod_{i=1}^{n} \frac{1}{1 - w_i z} = \sum_{k=0}^{\infty} S_k^n(\mathbf{w}) z^k. \tag{4.4}$$

The second identity is formal in z, or analytic when $|w_i z| < 1$ for all i.

Proof. In the first product, choosing $w_i z$ from exactly the factors indexed by a subset I produces $z^{|I|} \prod_{i \in I} w_i$. Summing over subsets gives (4.3). For the second, expand each factor geometrically:

$$\frac{1}{1 - w_i z} = \sum_{r_i \geq 0} w_i^{r_i} z^{r_i}.$$

The coefficient of z^k sums over weak compositions $r_1 + \cdots + r_n = k$, which are equivalent to multisets of size k. □

Corollary 4.4 (Elementary–complete orthogonality). *For $m \geq 0$,*

$$\sum_{j=0}^{m} (-1)^j C_j^n(\mathbf{w}) S_{m-j}^n(\mathbf{w}) = \delta_{m,0}.$$

Proof. Multiply $\prod_i (1 - w_i z)$ by its reciprocal and compare the coefficient of z^m, using Theorem 4.3 with z replaced by $-z$ in the elementary generating function. □

4.2 Classical, Stirling, and Gaussian specializations

Theorem 4.5 (A specialization table). *The following identities hold.*

(i) *If $w_i = 1$, then*

$$C_k^n = \binom{n}{k}, \qquad S_k^n = \binom{n+k-1}{k}.$$

(ii) *If $w_i = q^{i-1}$, then*

$$C_k^n = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q, \qquad S_k^n = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

(iii) *For the unsigned Stirling numbers of the first kind $c(n,k)$ and Stirling numbers of the second kind $\left\{ {n \atop k} \right\}$,*

$$c(n+1, k+1) = C_{n-k}^n(1, 2, \ldots, n),$$

$$\left\{ {n+k \atop n} \right\} = S_k^n(1, 2, \ldots, n).$$

Proof. For $w_i = 1$, C_k^n counts k-subsets. The multiset count is the stars-and-bars number $\binom{n+k-1}{k}$.

For $w_i = q^{i-1}$, the elementary identity is Corollary 3.3. For the complete identity, Theorem 4.3 gives

$$\prod_{i=0}^{n-1} \frac{1}{1 - zq^i} = \frac{1}{(z;q)_n}.$$

The coefficient of z^k is $\begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q$ by Theorem 3.4.

Finally,

$$x(x+1)\cdots(x+n) = \sum_{j=0}^{n+1} c(n+1, j) x^j.$$

The coefficient of x^{k+1} is $e_{n-k}(1, 2, \ldots, n)$. The standard ordinary generating function

$$\prod_{i=1}^{n} \frac{1}{1 - iz} = \sum_{k \geq 0} \left\{ {n+k \atop n} \right\} z^k$$

follows from the recurrence for Stirling numbers of the second kind, or by induction on n; comparison with (4.4) proves the last formula. □

4.3 Principal specialization of Schur polynomials

The two Gaussian entries of Theorem 4.5(ii) are the one-column and the one-row case of a single product formula. Stating it requires no symmetric-function theory beyond determinants, because a Schur polynomial can be *defined* by its bialternant.

Definition 4.6. Let $m \geq 1$ and let $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m \geq 0)$ be a partition with at most m parts, padded by zeros. Put

$$\mu_j = \lambda_j + m - j \quad (1 \leq j \leq m), \qquad \delta = (m-1, m-2, \ldots, 0),$$

so that $\mu_1 > \mu_2 > \cdots > \mu_m \geq 0$, and δ is the value of μ at $\lambda = 0$. For indeterminates $x_1, \ldots, x_m$ set

$$a_\mu(x_1, \ldots, x_m) = \det\bigl(x_i^{\mu_j}\bigr)_{i,j=1}^m,$$

and define the *Schur polynomial*

$$s_\lambda(x_1, \ldots, x_m) = \frac{a_\mu(x_1, \ldots, x_m)}{a_\delta(x_1, \ldots, x_m)}.$$

Finally put $n(\lambda) = \sum_{j=1}^m (j-1)\lambda_j$.

Lemma 4.7 (Vandermonde alternant)**.** *For indeterminates $y_1, \ldots, y_m$,*

$$\det\bigl(y_i^{j-1}\bigr)_{i,j=1}^m = \prod_{1 \leq i < j \leq m} (y_j - y_i).$$

Consequently $a_\delta(x_1, \ldots, x_m) = \prod_{1 \leq i < j \leq m}(x_i - x_j)$, and a_δ divides a_μ in $\mathbb{Z}[x_1, \ldots, x_m]$, so s_λ is a polynomial with integer coefficients.

Proof. Induct on m; for $m = 1$ both sides equal 1. Regard $D = \det(y_i^{j-1})_{i,j=1}^m$ as a polynomial in y_m of degree at most $m-1$. If $y_m = y_i$ for some $i < m$, two rows coincide and $D = 0$; hence $\prod_{i<m}(y_m - y_i)$ divides D, and by the degree bound $D = c \prod_{i<m}(y_m - y_i)$ with c free of y_m. Expanding along the last row, the coefficient of y_m^{m-1} in D is the cofactor at position (m, m), namely $\det(y_i^{j-1})_{i,j=1}^{m-1}$, which by induction equals $\prod_{1 \leq i < j \leq m-1}(y_j - y_i)$. Multiplying proves the formula.

Reversing the order of the columns of $\bigl(x_i^{m-j}\bigr)_{i,j}$ turns the exponents $m - j$ into $j - 1$ and multiplies the determinant by the sign $(-1)^{\binom{m}{2}}$ of the reversal permutation, so

$$a_\delta = (-1)^{\binom{m}{2}} \prod_{1 \leq i < j \leq m} (x_j - x_i) = \prod_{1 \leq i < j \leq m} (x_i - x_j).$$

For the divisibility, interchanging x_i and x_j interchanges two rows of a_μ, so a_μ changes sign; setting $x_i = x_j$ therefore gives $a_\mu = 0$, and $x_i - x_j$ divides a_μ in $\mathbb{Z}[x_1, \ldots, x_m]$. The polynomials $x_i - x_j$ with $i < j$ are pairwise non-associate primes in that unique factorization domain, so their product, which is a_δ, divides a_μ. ◻

Theorem 4.8 (Principal specialization of a Schur polynomial)**.** *Let λ have at most m parts. Then, as an identity in $\mathbb{Q}(q)$,*

$$s_\lambda(1, q, q^2, \ldots, q^{m-1}) = q^{n(\lambda)} \prod_{1 \leq i < j \leq m} \frac{1 - q^{\lambda_i - \lambda_j + j - i}}{1 - q^{j-i}}.$$

The left side lies in $\mathbb{Z}[q]$, so by Corollary 20.11 the identity persists at every complex q once the right-hand quotient has been reduced; as displayed it is to be read at those q with $q^d \neq 1$ for $1 \leq d \leq m-1$.

Proof. Specialize $x_i = q^{i-1}$, so that $x_i^{\mu_j} = q^{(i-1)\mu_j}$. Writing $y_j = q^{\mu_j}$, the matrix $\left(x_i^{\mu_j}\right)_{i,j}$ has entries y_j^{i-1}, hence is the transpose of $\left(y_i^{j-1}\right)_{i,j}$. A determinant is invariant under transposition, so Lemma 4.7 gives

$$a_\mu(1, q, \ldots, q^{m-1}) = \prod_{1\le i<j\le m} \left(q^{\mu_j} - q^{\mu_i}\right).$$

Because λ is weakly decreasing we have $\mu_1 > \cdots > \mu_m \ge 0$, so $\mu_i - \mu_j > 0$ whenever $i < j$, and we extract from each difference the *smaller* power:

$$q^{\mu_j} - q^{\mu_i} = q^{\mu_j}\left(1 - q^{\mu_i - \mu_j}\right).$$

Hence

$$a_\mu(1, q, \ldots, q^{m-1}) = q^{A(\mu)} \prod_{1\le i<j\le m} \left(1 - q^{\mu_i-\mu_j}\right), \qquad A(\mu) = \sum_{1\le i<j\le m} \mu_j = \sum_{j=1}^{m} (j-1)\mu_j,$$

the last equality because μ_j is contributed once for each $i < j$. Applying this to $\mu = \delta$, where $\delta_i - \delta_j = j - i$, gives

$$a_\delta(1, q, \ldots, q^{m-1}) = q^{A(\delta)} \prod_{1\le i<j\le m} \left(1 - q^{j-i}\right),$$

a nonzero element of $\mathbb{Q}(q)$. Dividing,

$$s_\lambda(1, q, \ldots, q^{m-1}) = q^{A(\mu)-A(\delta)} \prod_{1\le i<j\le m} \frac{1 - q^{\mu_i-\mu_j}}{1 - q^{j-i}}.$$

Finally $\mu_i - \mu_j = \lambda_i - \lambda_j + j - i$ and

$$A(\mu) - A(\delta) = \sum_{j=1}^{m} (j-1)(\mu_j - \delta_j) = \sum_{j=1}^{m} (j-1)\lambda_j = n(\lambda).$$

The left side is a polynomial with integer coefficients by Lemma 4.7; the transfer to arbitrary complex q is Lemma 20.10 together with Corollary 20.11. □

Corollary 4.9 (The one-column and one-row shadows)**.** *Let* $m = n \ge 1$.

(i) *For* $0 \le k \le n$ *and* $\lambda = (1^k) = (1, \ldots, 1, 0, \ldots, 0)$,

$$s_{(1^k)}(1, q, \ldots, q^{n-1}) = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

(ii) *For* $k \ge 0$ *and* $\lambda = (k)$,

$$s_{(k)}(1, q, \ldots, q^{n-1}) = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

Since $s_{(1^k)} = e_k$ *and* $s_{(k)} = h_k$ *[21], these are exactly the two Gaussian entries of Theorem 4.5(ii), that is,* $C_k^n(\mathbf{w})$ *and* $S_k^n(\mathbf{w})$ *at* $w_i = q^{i-1}$.

Proof. (i) Here $\lambda_i = 1$ for $i \le k$ and $\lambda_i = 0$ for $i > k$, so $n(\lambda) = \sum_{i\le k}(i-1) = \binom{k}{2}$, and $\lambda_i - \lambda_j$ vanishes except when $i \le k < j$, where it equals 1. The product of Theorem 4.8 therefore collapses to

$$\prod_{i=1}^{k} \prod_{j=k+1}^{n} \frac{1 - q^{j-i+1}}{1 - q^{j-i}} = \prod_{i=1}^{k} \frac{1 - q^{n-i+1}}{1 - q^{k-i+1}},$$

the inner product telescoping. The numerators are $(1-q^n)(1-q^{n-1})\cdots(1-q^{n-k+1}) = (q;q)_n/(q;q)_{n-k}$ and the denominators are $(1-q^k)\cdots(1-q) = (q;q)_k$, so the product is $\left[{n \atop k}\right]_q$.

(ii) Here $\lambda_1 = k$ and $\lambda_j = 0$ for $j \geq 2$, so $n(\lambda) = 0$ and only the pairs $(1, j)$ contribute:

$$\prod_{j=2}^{n} \frac{1-q^{k+j-1}}{1-q^{j-1}} = \frac{(q;q)_{n+k-1}/(q;q)_k}{(q;q)_{n-1}} = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q. \qquad \square$$

Remark 4.10. Both hypotheses of Theorem 4.8 are needed as stated. If λ had more than m parts, μ would not be strictly decreasing and a_μ would vanish identically. And the displayed quotient is meaningful only where $q^{j-i} \neq 1$ for all $1 \leq i < j \leq m$; at the excluded roots of unity, and at $q = 1$, one must cancel first, exactly as prescribed by Corollary 20.11. After cancellation the limit $q \to 1$ is

$$s_\lambda(1, 1, \ldots, 1) = \prod_{1 \leq i < j \leq m} \frac{\lambda_i - \lambda_j + j - i}{j - i},$$

by $\lim_{q\to 1}(1-q^a)/(1-q^b) = a/b$; this is Weyl's dimension formula for GL_m, and it stands to Theorem 4.8 exactly as Corollary 2.6 stands to the Gaussian coefficient.

4.4 Weighted inversion

Theorem 4.11 (Symmetric-function inversion)**.** *Let (a_m) and (b_m) be sequences. The transformations*

$$b_m = \sum_{j=0}^{m} S^n_{m-j}(\mathbf{w}) a_j$$

and

$$a_m = \sum_{j=0}^{m} (-1)^{m-j} C^n_{m-j}(\mathbf{w}) b_j$$

are inverse to each other.

Proof. The generating functions $A(z) = \sum a_m z^m$ and $B(z) = \sum b_m z^m$ satisfy

$$B(z) = A(z) \prod_{i=1}^{n} (1 - w_i z)^{-1}.$$

Multiplication by $\prod_i (1 - w_i z)$ gives the proposed inverse coefficient formula. Equivalently, use the orthogonality in Corollary 4.4. $\square$

4.5 The subset–subspace analogy

The choice $w_i = q^{i-1}$ is more than a convenient specialization. It replaces the Boolean lattice of subsets by the lattice of subspaces over a finite field. Ordinary binomial coefficients count subsets by cardinality; Gaussian coefficients count subspaces by dimension. The two-term weighted recurrence mirrors the decomposition of a subspace according to its relation with a fixed hyperplane. This analogy is made exact in Chapter 6.

Chapter 5

Partitions, words, paths, and inversion statistics

5.1 Binary words

Let $\mathcal{W}_{n,k}$ be the set of binary words of length n containing k ones. Define

$$\mathrm{inv}(w) = \#\left\{(i,j) : i < j,\ w_i = 1,\ w_j = 0\right\}.$$

Theorem 5.1 (Inversion generating function)**.** *For* $0 \le k \le n$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{w \in \mathcal{W}_{n,k}} q^{\mathrm{inv}(w)}.$$

Proof. Let $F_{n,k}(q)$ denote the right side. If the last letter is 1, deletion of it gives a word in $\mathcal{W}_{n-1,k-1}$ and creates no new inversion. If the last letter is 0, deletion gives a word in $\mathcal{W}_{n-1,k}$ and the final zero forms an inversion with each of the k ones, contributing a factor q^k. Hence

$$F_{n,k} = F_{n-1,k-1} + q^k F_{n-1,k}.$$

The boundary values are 1, so $F_{n,k}$ and $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q$ obey the same recurrence and initial conditions. □

Corollary 5.2. *The random variable in Theorem 2.10 is the inversion number of a uniformly random word in* $\mathcal{W}_{n,k}$*.*

Proof. The coefficient of q^j in Theorem 5.1 counts words with j inversions, while $|\mathcal{W}_{n,k}| = \binom{n}{k}$. □

5.2 Partitions inside a rectangle

A partition $\lambda = (\lambda_1, \ldots, \lambda_k)$ is said to fit in a $k \times m$ rectangle if

$$m \ge \lambda_1 \ge \cdots \ge \lambda_k \ge 0.$$

Its size is $|\lambda| = \lambda_1 + \cdots + \lambda_k$.

Theorem 5.3 (Rectangular partition generating function)**.** *For* $m, k \ge 0$*,*

$$\begin{bmatrix} m+k \\ k \end{bmatrix}_q = \sum_{\lambda \subseteq k \times m} q^{|\lambda|}.$$

Proof. Take a word $w \in \mathcal{W}_{m+k,k}$ and let the positions of its ones be $1 \le i_1 < \cdots < i_k \le m+k$. Define

$$\lambda_j = m + j - i_j \qquad (1 \le j \le k).$$

Then $0 \le \lambda_j \le m$ and

$$\lambda_j - \lambda_{j+1} = i_{j+1} - i_j - 1 \ge 0.$$

Thus λ fits in the rectangle. Conversely, $i_j = m + j - \lambda_j$ recovers a strictly increasing sequence of positions, so this is a bijection. The number λ_j is exactly the number of zeros to the right of the jth one, hence $|\lambda| = \mathrm{inv}(w)$. Apply Theorem 5.1. □

Corollary 5.4 (Conjugation and complement)**.** *Conjugating Ferrers diagrams proves* $\begin{bmatrix} m+k \\ k \end{bmatrix}_q = \begin{bmatrix} m+k \\ m \end{bmatrix}_q$*. Taking the complement in the rectangle proves*

$$\begin{bmatrix} m+k \\ k \end{bmatrix}_{q^{-1}} = q^{-mk} \begin{bmatrix} m+k \\ k \end{bmatrix}_q .$$

Proof. Conjugation exchanges a $k \times m$ rectangle with an $m \times k$ rectangle without changing size. Complementation sends size r to $mk - r$, so the generating polynomial is reciprocal of degree mk. □

5.3 Lattice paths and area

Encode a binary word by a path from $(0,0)$ to (m,k), reading 0 as an east step and 1 as a north step. With a suitable choice of side, the number of unit squares between the path and the boundary is the inversion number.

Corollary 5.5 (Area generating function)**.** *The polynomial* $\begin{bmatrix} m+k \\ k \end{bmatrix}_q$ *is the generating function of monotone lattice paths from* $(0,0)$ *to* (m,k) *by the area under the path:*

$$\begin{bmatrix} m+k \\ k \end{bmatrix}_q = \sum_P q^{\mathrm{area}(P)}.$$

Proof. Under the word–path encoding, every north step contributes the number of later east steps, which is precisely the number of squares in its row between the path and the chosen boundary. Summing over north steps gives $\mathrm{inv}(w)$. □

5.4 Durfee squares and the central convolution

The largest square in the northwest corner of a partition diagram is its Durfee square.

Theorem 5.6 (Durfee-square decomposition)**.** *For* $m, n \ge 0$*,*

$$\begin{bmatrix} m+n \\ m \end{bmatrix}_q = \sum_{j=0}^{\min(m,n)} q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q .$$

Proof. This is Corollary 3.6; we give the partition proof. A partition in an $m \times n$ rectangle with Durfee square of side j consists of the $j \times j$ square, a partition to its right fitting in a $j \times (n-j)$ rectangle, and a partition below it fitting in an $(m-j) \times j$ rectangle. By Theorem 5.3, their generating functions are

$$q^{j^2} \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} m \\ j \end{bmatrix}_q .$$

The Durfee size is unique, so summing over j counts every partition exactly once. □

5.5 Words with several letters

Let $\mathcal{W}(n_1, \ldots, n_r)$ be the words containing n_i copies of the letter i, ordered by $1 < \cdots < r$. Let $\mathrm{inv}(w)$ count pairs $i < j$ of positions with $w_i > w_j$.

Theorem 5.7 (Multiset inversion enumerator)**.** *If $n = n_1 + \cdots + n_r$, then*

$$\begin{bmatrix} n \\ n_1, \ldots, n_r \end{bmatrix}_q = \sum_{w \in \mathcal{W}(n_1, \ldots, n_r)} q^{\mathrm{inv}(w)}.$$

Proof. Choose the positions of the largest letter r. If their binary indicator has inversion number s, then s counts precisely the inversions in which the larger letter is r. The remaining positions contain a word on $1, \ldots, r-1$, and its inversions are independent. Therefore the generating function factors as

$$\begin{bmatrix} n \\ n_r \end{bmatrix}_q \begin{bmatrix} n - n_r \\ n_1, \ldots, n_{r-1} \end{bmatrix}_q.$$

Iterating and using Proposition 2.8 proves the formula. □

5.6 Rogers–Szegő polynomials

Definition 5.8. The Rogers–Szegő polynomial is

$$H_n(z; q) = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k.$$

Proposition 5.9 (Basic recurrence)**.** *One has*

$$H_{n+1}(z; q) = zH_n(z; q) + H_n(qz; q), \qquad H_0(z; q) = 1.$$

Moreover, if $|q| < 1$, then

$$\sum_{n=0}^{\infty} \frac{H_n(z; q)}{(q; q)_n} t^n = \frac{1}{(t; q)_\infty (zt; q)_\infty}$$

for $|t| < 1$ and $|zt| < 1$.

Proof. Insert the first q-Pascal recurrence into H_{n+1}:

$$\sum_k \left(\begin{bmatrix} n \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n \\ k \end{bmatrix}_q \right) z^k = zH_n(z; q) + H_n(qz; q).$$

For the generating function, use

$$\frac{\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q}{(q; q)_n} = \frac{1}{(q; q)_k (q; q)_{n-k}}.$$

Writing $n = k + j$ and separating the double sum gives

$$\left(\sum_{j \geq 0} \frac{t^j}{(q; q)_j} \right) \left(\sum_{k \geq 0} \frac{(zt)^k}{(q; q)_k} \right).$$

Each factor equals the reciprocal infinite product by Euler's identity, proved later in Corollary 8.2; alternatively, the equality follows already by taking $n \to \infty$ in Theorem 3.4. This gives the stated product. □

Proposition 5.10 (Three-term recurrence)**.** *For $n \geq 1$,*

$$H_{n+1}(z;q) = (1+z)H_n(z;q) - z\,(1-q^n)H_{n-1}(z;q).$$

Proof. Both sides are polynomials in z, so it suffices to compare the coefficient of z^k for every $k \in \mathbb{Z}$, using the convention $\begin{bmatrix} n \\ k \end{bmatrix}_q = 0$ for $k < 0$ or $k > n$. On the left that coefficient is $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$; on the right it is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q - (1-q^n)\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q.$$

We claim that for every $k \in \mathbb{Z}$ and every $n \geq 1$,

$$(1-q^n)\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q = \left(1-q^{n+1-k}\right)\begin{bmatrix} n \\ k-1 \end{bmatrix}_q.$$

Granting the claim, the right-hand coefficient equals

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + \left(1-\left(1-q^{n+1-k}\right)\right)\begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n+1-k}\begin{bmatrix} n \\ k-1 \end{bmatrix}_q,$$

which is the second q-Pascal recurrence (2.2) with n replaced by $n+1$; by Theorem 2.3 that recurrence is valid for all k under the zero convention, and the right-hand coefficient is therefore $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$, as required.

To prove the claim, let first $1 \leq k \leq n$, so that $k-1 \geq 0$ and $n-k \geq 0$ and both Gaussian coefficients are given by factorial quotients. From $(q;q)_n = (1-q^n)(q;q)_{n-1}$ and $(q;q)_{n-k+1} = (1-q^{n-k+1})(q;q)_{n-k}$,

$$\frac{\begin{bmatrix} n \\ k-1 \end{bmatrix}_q}{\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q} = \frac{(q;q)_n\,(q;q)_{n-k}}{(q;q)_{n-1}\,(q;q)_{n-k+1}} = \frac{1-q^n}{1-q^{n-k+1}},$$

and cross-multiplication gives the claim. For $k \leq 0$ both sides vanish, since then $k-1 < 0$ and both Gaussian coefficients are zero. For $k = n+1$ the left side vanishes because $\begin{bmatrix} n-1 \\ n \end{bmatrix}_q = 0$ and the right side vanishes because $1 - q^{n+1-k} = 1 - q^0 = 0$. For $k > n+1$ every term is zero. This exhausts all k and completes the proof. □

Corollary 5.11 (Dilation formula)**.** *For $n \geq 1$,*

$$H_n(qz;q) = H_n(z;q) - z\,(1-q^n)H_{n-1}(z;q).$$

Proof. By Proposition 5.9, $H_n(qz;q) = H_{n+1}(z;q) - zH_n(z;q)$. Substituting Proposition 5.10 for $H_{n+1}(z;q)$ and cancelling $zH_n(z;q)$ gives the formula. □

Remark 5.12. The recurrence of Proposition 5.9 moves the argument, from z to qz; the one above does not. It is therefore a three-term recurrence in the index n alone, of the shape familiar from orthogonal polynomial sequences, and it evaluates $H_0, H_1, \dots, H_N$ at a fixed numerical z in $O(N)$ ring operations, without ever forming the Gaussian coefficients. At $q = 0$ every $\begin{bmatrix} n \\ k \end{bmatrix}_q$ equals 1 by Proposition 19.11, the recurrence becomes $H_{n+1} = (1+z)H_n - zH_{n-1}$, and its solution is the geometric sum $1 + z + \cdots + z^n$, as it must be.

Chapter 6

Finite fields, Grassmannians, and rank enumeration

6.1 Counting subspaces

Theorem 6.1 (Gaussian coefficients count subspaces)**.** *Let Q be a prime power. The number of k-dimensional linear subspaces of $\mathbb{F}_Q^n$ is*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

Proof. Count ordered linearly independent k-tuples in $\mathbb{F}_Q^n$. The first vector may be any nonzero vector, the second any vector outside the span of the first, and so on, giving

$$(Q^n - 1)(Q^n - Q)\cdots(Q^n - Q^{k-1}).$$

Each k-dimensional subspace has

$$(Q^k - 1)(Q^k - Q)\cdots(Q^k - Q^{k-1})$$

ordered bases. Dividing and cancelling Q^i in corresponding factors yields

$$\prod_{i=0}^{k-1} \frac{Q^{n-i} - 1}{Q^{k-i} - 1} = \prod_{j=1}^{k} \frac{1 - Q^{n-k+j}}{1 - Q^j} = \begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

□

Theorem 6.2 (Geometric Pascal recurrence)**.** *Let H be a fixed hyperplane in $\mathbb{F}_Q^n$. Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_Q + Q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_Q.$$

Proof. A k-subspace U is either contained in H or not. The contained subspaces contribute $\left[\begin{smallmatrix} n-1 \\ k \end{smallmatrix}\right]_Q$. If $U \not\subseteq H$, then $W = U \cap H$ has dimension $k-1$. Fixing W, the quotient $\mathbb{F}_Q^n/W$ has dimension $n-k+1$, and H/W is a hyperplane. The subspace U/W is a line not contained in H/W. The total number of lines in $\mathbb{F}_Q^{n-k+1}$ minus those in its hyperplane is

$$\frac{Q^{n-k+1} - 1}{Q - 1} - \frac{Q^{n-k} - 1}{Q - 1} = Q^{n-k}.$$

There are $\left[\begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix}\right]_Q$ choices for W, proving the formula. □

6.2 Flags and q-multinomial coefficients

Theorem 6.3 (Partial flags). *Let $d_i = n_1 + \cdots + n_i$. The number of flags*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_r = \mathbb{F}_Q^n, \qquad \dim(V_i/V_{i-1}) = n_i,$$

is

$$\begin{bmatrix} n \\ n_1, \ldots, n_r \end{bmatrix}_Q .$$

Proof. Choose V_1 in $\left[\begin{smallmatrix} n \\ n_1 \end{smallmatrix}\right]_Q$ ways. In the quotient $\mathbb{F}_Q^n/V_1$, choose V_2/V_1 in $\left[\begin{smallmatrix} n-n_1 \\ n_2 \end{smallmatrix}\right]_Q$ ways, and continue. The product is the factorization in Proposition 2.8. □

6.3 Matrices of fixed rank

Theorem 6.4 (Rank enumeration). *The number of $m \times n$ matrices of rank r over $\mathbb{F}_Q$ is*

$$N_{m,n,r}(Q) = \begin{bmatrix} m \\ r \end{bmatrix}_Q \begin{bmatrix} n \\ r \end{bmatrix}_Q |\mathrm{GL}_r(\mathbb{F}_Q)|,$$

where

$$|\mathrm{GL}_r(\mathbb{F}_Q)| = \prod_{i=0}^{r-1} (Q^r - Q^i).$$

Equivalently,

$$N_{m,n,r}(Q) = \frac{\prod_{i=0}^{r-1}(Q^m - Q^i)(Q^n - Q^i)}{\prod_{i=0}^{r-1}(Q^r - Q^i)}.$$

Proof. Every rank-r matrix A can be factored as $A = UV$, where U is an $m \times r$ matrix of full column rank and V is an $r \times n$ matrix of full row rank. There are $\prod_{i=0}^{r-1}(Q^m - Q^i)$ choices for U and $\prod_{i=0}^{r-1}(Q^n - Q^i)$ choices for V. Two pairs (U, V) and (U', V') give the same A exactly when

$$U' = UG, \qquad V' = G^{-1}V$$

for a unique $G \in \mathrm{GL}_r(\mathbb{F}_Q)$. Thus every matrix is counted $|\mathrm{GL}_r|$ times. Substituting the subspace-count formulas gives the first expression. □

6.4 Schubert cells and a topological meaning

Let $\mathrm{Gr}(k, n)$ be the complex Grassmannian of k-planes in $\mathbb{C}^n$. Its Schubert cell decomposition is indexed by partitions inside a $k \times (n-k)$ rectangle.

Theorem 6.5 (Poincaré polynomial of the Grassmannian). *The integral cohomology of* $\mathrm{Gr}(k, n)$ *is concentrated in even degrees, and its Poincaré polynomial is*

$$P_{\mathrm{Gr}(k,n)}(t) = \sum_{j \geq 0} \operatorname{rank} H^{2j}(\mathrm{Gr}(k, n); \mathbb{Z}) t^{2j} = \begin{bmatrix} n \\ k \end{bmatrix}_{t^2}.$$

Proof. Fix the standard flag in $\mathbb{C}^n$. Every k-plane has a unique row-reduced echelon matrix. Its pivot positions determine a binary word with k ones, hence a partition λ inside the $k \times (n-k)$ rectangle. Once the pivots are fixed, the entries allowed to vary freely occupy exactly the boxes of λ; the corresponding stratum is therefore an affine complex space $\mathbb{C}^{|\lambda|}$. Ordering the strata by inclusion of diagrams yields a CW decomposition with one real cell of dimension $2|\lambda|$ for each λ. Since all cells have even dimension, every cellular boundary map is zero. Thus the rank of H^{2j} is the number of such partitions of size j. Apply Theorem 5.3 with $q = t^2$. $\square$

Remark 6.6 (The "same polynomial" principle). For a prime power Q, evaluating $\begin{bmatrix} n \\ k \end{bmatrix}_q$ at $q = Q$ counts rational points of a Grassmannian over $\mathbb{F}_Q$. Evaluating it at $q = t^2$ records the even Betti numbers of the complex Grassmannian. This is an elementary instance of the broad principle that polynomial point counts over finite fields often shadow topological invariants over $\mathbb{C}$.

6.5 Growth over finite fields

Proposition 6.7 (Dimension-dominant bounds)**.** *For $Q > 1$ and $0 \le k \le n$,*

$$Q^{k(n-k)} \le \begin{bmatrix} n \\ k \end{bmatrix}_Q \le \frac{Q^{k(n-k)}}{(Q^{-1}; Q^{-1})_\infty}.$$

Proof. Rewrite the product as

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q = Q^{k(n-k)} \prod_{j=1}^{k} \frac{1 - Q^{-(n-k+j)}}{1 - Q^{-j}}.$$

Each ratio is at least 1 because $n - k + j \ge j$, proving the lower bound. For the upper bound, discard the numerator factors and note

$$\prod_{j=1}^{k} (1 - Q^{-j}) \ge \prod_{j=1}^{\infty} (1 - Q^{-j}) = (Q^{-1}; Q^{-1})_\infty.$$

$\square$

Chapter 7

Symmetry and unimodality

Palindromicity follows immediately from reciprocity. Unimodality is subtler: the coefficients rise toward the middle and then fall. A concise conceptual proof comes from the representation theory of $\mathfrak{sl}_2$.

7.1 A representation-theoretic lemma

Let $\mathfrak{sl}_2(\mathbb{C})$ have generators E, F, H with

$$[H,E] = 2E, \qquad [H,F] = -2F, \qquad [E,F] = H.$$

Lemma 7.1 (Structure of finite-dimensional $\mathfrak{sl}_2$-modules)**.** *Every finite-dimensional complex $\mathfrak{sl}_2$-module is a direct sum of irreducibles L_d indexed by $d \in \mathbb{N}_0$. The module L_d has dimension $d+1$ and H-weights*

$$d, d-2, \dots, -d,$$

each with multiplicity one.

Proof. We include the standard argument. Restrict a representation to the compact real form $\mathfrak{su}_2$. A finite-dimensional Lie-algebra representation integrates to the simply connected compact group $\mathrm{SU}(2)$ by exponentiating the representing matrices; the Lie relations ensure consistency of the local exponentials, and simple connectedness removes monodromy. Starting from any Hermitian inner product, average it over $\mathrm{SU}(2)$ with normalized Haar measure. The result is invariant. Hence the orthogonal complement of every invariant subspace is invariant, proving complete reducibility.

It remains to classify an irreducible module V. In the invariant Hermitian structure, a suitable scalar multiple of H is self-adjoint, so H is diagonalizable. Choose an eigenvector v whose eigenvalue d has maximal real part. Since E raises an H-weight by 2, maximality gives $Ev = 0$. The commutation relations imply by induction

$$HF^j v = (d-2j)F^j v, \qquad EF^j v = j(d-j+1)F^{j-1}v.$$

Because V is finite-dimensional, let m be maximal with $F^m v \neq 0$. Applying E to $F^{m+1}v = 0$ gives

$$(m+1)(d-m)F^m v = 0,$$

so $d = m \in \mathbb{N}_0$. The vectors $v, Fv, \dots, F^d v$ have distinct weights and are linearly independent. Their span is invariant under E, F, H and is nonzero; irreducibility makes it all of V. The displayed weights and multiplicities follow. □

7.2 Exterior powers and Gaussian coefficients

Let L_{n-1} have a weight basis $v_0, \ldots, v_{n-1}$ with $Hv_i = (n-1-2i)v_i$.

Proposition 7.2 (Exterior-power character)**.** *For* $0 \le k \le n$*,*

$$\operatorname{ch}\left(\bigwedge^k L_{n-1}\right)(x) = x^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{x^{-2}}.$$

Proof. A basis of the exterior power consists of $v_{i_1} \wedge \cdots \wedge v_{i_k}$ with $0 \le i_1 < \cdots < i_k \le n-1$. Its weight is

$$k(n-1) - 2(i_1 + \cdots + i_k).$$

Therefore

$$\operatorname{ch} = x^{k(n-1)} \sum_{i_1 < \cdots < i_k} (x^{-2})^{i_1 + \cdots + i_k}.$$

Use Corollary 3.3 with $q = x^{-2}$. The prefactor becomes

$$x^{k(n-1)} x^{-2\binom{k}{2}} = x^{k(n-k)}.$$

□

Theorem 7.3 (Unimodality of Gaussian coefficients)**.** *If*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = c_0 + c_1 q + \cdots + c_D q^D, \qquad D = k(n-k),$$

then

$$c_0 \le c_1 \le \cdots \le c_{\lfloor D/2 \rfloor}$$

and $c_j = c_{D-j}$*. Thus the coefficient sequence is symmetric and unimodal.*

Proof. Symmetry was proved in Theorem 2.5. By Lemma 7.1, decompose

$$\bigwedge^k L_{n-1} \cong \bigoplus_{d \ge 0} m_d L_d.$$

All weights in this exterior power have the same parity as D. The character of one summand L_d, placed inside the ambient sequence of weights

$$D, D-2, \ldots, -D,$$

contributes a consecutive block of ones centered at weight 0: it is zero until weight d is reached, then one at every step down to $-d$, then zero again. As one moves from weight D toward 0, such a block can turn on but cannot turn off. A nonnegative sum of these centered blocks therefore has weakly increasing multiplicities toward the middle.

By Proposition 7.2, the multiplicity of weight $D - 2j$ is exactly c_j. Hence $c_0 \le c_1 \le \cdots \le c_{\lfloor D/2 \rfloor}$. □

Remark 7.4. The proof explains more than the inequality: the successive differences $c_j - c_{j-1}$ record multiplicities of irreducible summands with prescribed highest weight. Stronger strict-unimodality results require additional arguments and have a small family of exceptional cases; ordinary unimodality is exactly what the $\mathfrak{sl}_2$ decomposition supplies for free.

Part II

Infinite series and basic hypergeometric summation

Chapter 8

The infinite q-binomial theorem

8.1 A functional-equation proof

Theorem 8.1 (Infinite q-binomial theorem)**.** *Let $|q| < 1$ and $|z| < 1$. Then*

$$\sum_{n=0}^{\infty} \frac{(a;q)_n}{(q;q)_n} z^n = \frac{(az;q)_\infty}{(z;q)_\infty}.$$

The convergence is locally uniform in (a, z) on compact subsets of $\mathbb{C} \times \{|z| < 1\}$.

Proof. Set

$$F(z) = \sum_{n\geq 0} c_n z^n, \qquad c_n = \frac{(a;q)_n}{(q;q)_n}.$$

Since $c_n/c_{n-1} = (1 - aq^{n-1})/(1 - q^n) \to 1$, the radius of convergence is 1 unless the series terminates; in every case it converges for $|z| < 1$. The coefficient recurrence is equivalent to

$$(1 - q^n)c_n = (1 - aq^{n-1})c_{n-1}.$$

Comparing coefficients shows that

$$(1 - z)F(z) = (1 - az)F(qz), \qquad F(0) = 1.$$

Now put $G(z) = (az;q)_\infty/(z;q)_\infty$. Using $(z;q)_\infty = (1 - z)(qz;q)_\infty$ gives the same functional equation and $G(0) = 1$.

To prove uniqueness, iterate the functional equation:

$$F(z) = \prod_{j=0}^{N-1} \frac{1 - azq^j}{1 - zq^j} F(q^N z).$$

As $N \to \infty$, $q^N z \to 0$, so $F(q^N z) \to 1$. The finite product tends to $G(z)$. Hence $F = G$. □

Corollary 8.2 (Euler's two infinite identities)**.** *For $|q| < 1$,*

$$(z;q)_\infty = \sum_{n=0}^{\infty} \frac{(-1)^n q^{\binom{n}{2}}}{(q;q)_n} z^n, \qquad z \in \mathbb{C}, \tag{8.1}$$

$$\frac{1}{(z;q)_\infty} = \sum_{n=0}^{\infty} \frac{z^n}{(q;q)_n}, \qquad |z| < 1. \tag{8.2}$$

Proof. Set $a = 0$ in Theorem 8.1 to obtain (8.2). For (8.1), start with the finite identity

$$(z;q)_N = \sum_{n=0}^{N} (-1)^n q^{\binom{n}{2}} \begin{bmatrix} N \\ n \end{bmatrix}_q z^n.$$

For fixed n, $\left[\begin{smallmatrix} N \\ n \end{smallmatrix}\right]_q \to 1/(q;q)_n$ as $N \to \infty$. The limiting series converges for every z because its successive-term ratio has magnitude asymptotic to $|z||q|^n$. To justify passage to the limit uniformly on $|z| \le R$, note that

$$\left| \begin{bmatrix} N \\ n \end{bmatrix}_q \right| \le \frac{\prod_{j\ge 1}(1+|q|^j)}{\left(\inf_{r\ge 0}|(q;q)_r|\right)^2},$$

where the denominator is positive because $(q;q)_r \to (q;q)_\infty \ne 0$. Thus the summands, extended by zero for $n > N$, are dominated by a constant multiple of $|q|^{\binom{n}{2}} R^n$. Dominated convergence, together with local uniform convergence of the products, now permits passage to the limit. □

Corollary 8.3 (A ratio expansion)**.** *For $|z| < 1$,*

$$\frac{(a_1 z;q)_\infty \cdots (a_r z;q)_\infty}{(b_1 z;q)_\infty \cdots (b_s z;q)_\infty}$$

can be expanded by multiplying r copies of (8.1) *and s copies of* (8.2)*. In particular, every coefficient is a finite convolution of q-Pochhammer ratios.*

Proof. Each numerator factor has an entire series expansion and each reciprocal denominator factor converges for $|b_j z| < 1$. Cauchy multiplication is valid in their common disk of absolute convergence. □

8.2 Formal and analytic interpretations

Proposition 8.4 (Formal version)**.** *If q is an indeterminate, then*

$$\frac{(az;q)_\infty}{(z;q)_\infty} = \sum_{n\ge 0} \frac{(a;q)_n}{(q;q)_n} z^n$$

holds in the q-adically complete ring $\mathbb{Q}(a)[[q]][[z]]$. More generally, it remains valid after any specialization into a complete topological ring for which the displayed products and series converge.

Proof. The products are well defined q-adically: apart from their initial factors, the jth factors differ from 1 by a multiple of q^j. For a formal power series $F(z) = \sum c_n z^n$, the equation

$$(1-z)F(z) = (1-az)F(qz), \qquad c_0 = 1,$$

determines c_n recursively by $(1-q^n)c_n = (1-aq^{n-1})c_{n-1}$, because $1-q^n$ is a unit in $\mathbb{Q}(a)[[q]]$. Both sides satisfy this equation, so their coefficients agree. The same uniqueness argument works in every stated complete specialization. □

8.3 Limits in the number of factors

Proposition 8.5. *For fixed k and $|q| < 1$,*

$$\lim_{n\to\infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q;q)_k}.$$

More generally,

$$\lim_{n\to\infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q = \frac{1}{(q;q)_k}.$$

Proof. Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1};q)_k}{(q;q)_k}.$$

Every numerator factor tends to 1. □

Remark 8.6. The two Euler identities are therefore the stable limits of the finite product and reciprocal-product formulas. This stability is one reason finite q-binomial identities so often serve as polynomial approximations to infinite q-series identities.

Chapter 9

Basic hypergeometric series

9.1 Definition and convergence

Definition 9.1. The basic hypergeometric series is

$${}_r\phi_s\left[\begin{matrix}a_1,\ldots,a_r\\ b_1,\ldots,b_s\end{matrix};q,z\right]=\sum_{n=0}^{\infty}\frac{(a_1,\ldots,a_r;q)_n}{(q,b_1,\ldots,b_s;q)_n}\left((-1)^n q^{\binom{n}{2}}\right)^{1+s-r}z^n,$$

provided no denominator factor vanishes. If one numerator parameter is q^{-N}, the series terminates at $n=N$.

Proposition 9.2 (Elementary convergence classification)**.** *Assume the series does not terminate and the denominator parameters avoid poles.*

(i) *If $r\le s$, the series converges for every z.*

(ii) *If $r=s+1$, its radius of convergence is* 1.

(iii) *If $r>s+1$, the terms generally fail to tend to zero for $z\neq 0$; such series are normally interpreted only when terminating or formally.*

Proof. Let T_n be the nth summand. Since $(1-aq^n)\to 1$ for each fixed parameter,

$$\frac{T_{n+1}}{T_n}=z\,\frac{\prod_{i=1}^{r}(1-a_iq^n)}{(1-q^{n+1})\prod_{j=1}^{s}(1-b_jq^n)}(-1)^{1+s-r}q^{n(1+s-r)}.$$

The ratio tends to 0 when $r\le s$, to z when $r=s+1$, and has unbounded magnitude when $r>s+1$ and $z\neq 0$. Apply the ratio test. $\square$

The infinite q-binomial theorem is the evaluation

$${}_1\phi_0\left[\begin{matrix}a\\ -\end{matrix};q,z\right]=\frac{(az;q)_\infty}{(z;q)_\infty}.$$

9.2 Heine's transformation

Theorem 9.3 (Heine transformation)**.** *Initially for $|z|<1$ and $|b|<1$,*

$${}_2\phi_1\left[\begin{matrix}a,b\\ c\end{matrix};q,z\right]=\frac{(b,az;q)_\infty}{(c,z;q)_\infty}\,{}_2\phi_1\left[\begin{matrix}c/b,z\\ az\end{matrix};q,b\right].$$

The identity extends wherever both sides are defined by analytic continuation.

Proof. Use the tail-factor identity to write

$$\frac{(b;q)_n}{(c;q)_n} = \frac{(b;q)_\infty}{(c;q)_\infty}\frac{(cq^n;q)_\infty}{(bq^n;q)_\infty}.$$

By the infinite q-binomial theorem,

$$\frac{(cq^n;q)_\infty}{(bq^n;q)_\infty} = \sum_{k\geq 0}\frac{(c/b;q)_k}{(q;q)_k}b^k q^{nk}.$$

Insert this into the ${}_2\phi_1$ series and interchange the absolutely convergent sums:

$$\frac{(b;q)_\infty}{(c;q)_\infty}\sum_{k\geq 0}\frac{(c/b;q)_k}{(q;q)_k}b^k\sum_{n\geq 0}\frac{(a;q)_n}{(q;q)_n}(zq^k)^n.$$

The inner sum is

$$\frac{(azq^k;q)_\infty}{(zq^k;q)_\infty} = \frac{(az;q)_\infty}{(z;q)_\infty}\frac{(z;q)_k}{(az;q)_k}.$$

Collecting factors yields the claimed transformation. Both sides are meromorphic functions of the parameters, so equality extends from the initial open domain by the identity theorem. □

9.3 The q-Gauss sum

Theorem 9.4 (q-Gauss summation)**.** *If* $|c/(ab)| < 1$, *then*

$${}_2\phi_1\left[\begin{matrix}a, b\\ c\end{matrix}; q, \frac{c}{ab}\right] = \frac{(c/a, c/b;q)_\infty}{(c, c/(ab);q)_\infty}.$$

Proof. Set $z = c/(ab)$ in Theorem 9.3. Then $az = c/b$, so one numerator parameter of the transformed series cancels its denominator parameter:

$${}_2\phi_1\left[\begin{matrix}c/b, c/(ab)\\ c/b\end{matrix}; q, b\right] = {}_1\phi_0\left[\begin{matrix}c/(ab)\\ -\end{matrix}; q, b\right] = \frac{(c/a;q)_\infty}{(b;q)_\infty}.$$

Multiplication by the prefactor in Heine's transformation gives the result. The initially stronger restrictions are removed by analytic continuation within the convergence domain. □

9.4 A second finite Cauchy identity

Lemma 9.5 (Cauchy convolution II)**.** *For* a, b, c *in a commutative ring,*

$$(ac;q)_n(bc;q)_n = \sum_{k=0}^{n}\begin{bmatrix}n\\ k\end{bmatrix}_q (a;q)_k(b;q)_k c^k(c;q)_{n-k}(abcq^k;q)_{n-k}.$$

Proof. Apply the q-Cauchy identity Theorem 3.10 with $u = bq^k$, $v = ac$, and length $n-k$:

$$(abcq^k;q)_{n-k} = \sum_{r=0}^{n-k}\begin{bmatrix}n-k\\ r\end{bmatrix}_q (bq^k;q)_r(ac)^r(ac;q)_{n-k-r}.$$

Insert this in the right side of the proposed identity. Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} n-k \\ r \end{bmatrix}_q = \begin{bmatrix} n \\ k+r \end{bmatrix}_q \begin{bmatrix} k+r \\ k \end{bmatrix}_q, \qquad (b;q)_k (bq^k;q)_r = (b;q)_{k+r}.$$

Put $m = k + r$ and use

$$(c;q)_{n-k} = (c;q)_{n-m}(cq^{n-m};q)_{m-k}.$$

The double sum becomes

$$\sum_{m=0}^{n} \begin{bmatrix} n \\ m \end{bmatrix}_q (b;q)_m c^m (c;q)_{n-m}(ac;q)_{n-m} \sum_{k=0}^{m} \begin{bmatrix} m \\ k \end{bmatrix}_q (a;q)_k a^{m-k} (cq^{n-m};q)_{m-k}.$$

In the inner sum replace $j = m-k$ and apply Theorem 3.10 with $u = cq^{n-m}$ and $v = a$; it equals $(acq^{n-m};q)_m$. Hence

$$(ac;q)_{n-m}(acq^{n-m};q)_m = (ac;q)_n$$

can be factored out. The remaining sum is

$$(ac;q)_n \sum_{m=0}^{n} \begin{bmatrix} n \\ m \end{bmatrix}_q (b;q)_m c^m (c;q)_{n-m} = (ac;q)_n (bc;q)_n$$

by one final application of the q-Cauchy identity. □

9.5 The q-Pfaff–Saalschütz sum

Theorem 9.6 (q-Pfaff–Saalschütz summation)**.** *For $n \in \mathbb{N}_0$,*

$${}_3\phi_2\left[\begin{matrix} a, b, q^{-n} \\ c, abq^{1-n}/c \end{matrix}; q, q\right] = \frac{(c/a, c/b;q)_n}{(c, c/(ab);q)_n}.$$

Proof. In Lemma 9.5, replace its parameters by

$$a \mapsto a, \qquad b \mapsto b, \qquad c \mapsto \frac{c}{ab}.$$

This gives

$$\frac{(c/a, c/b;q)_n}{(c, c/(ab);q)_n} = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q (a,b;q)_k \left(\frac{c}{ab}\right)^k \frac{(c/(ab);q)_{n-k}(cq^k;q)_{n-k}}{(c, c/(ab);q)_n}.$$

We show that the kth summand is the kth term of the stated ${}_3\phi_2$. The tail identities give

$$\frac{(cq^k;q)_{n-k}}{(c;q)_n} = \frac{1}{(c;q)_k}.$$

Also, reversal of the two finite products gives

$$(q^{-n};q)_k = (-1)^k q^{-nk+\binom{k}{2}} (q^{n-k+1};q)_k,$$

$$(abq^{1-n}/c;q)_k = \left(-\frac{ab}{c}\right)^k q^{k(1-n)+\binom{k}{2}} (cq^{n-k}/(ab);q)_k.$$

Therefore

$$\begin{aligned}
&\frac{(q^{-n};q)_k q^k}{(q;q)_k(abq^{1-n}/c;q)_k}\frac{(c/(ab);q)_n}{(c/(ab);q)_{n-k}} \\
&\quad= \left(\frac{c}{ab}\right)^k \frac{(q^{n-k+1};q)_k}{(q;q)_k}\frac{(c/(ab);q)_n}{(c/(ab);q)_{n-k}(cq^{n-k}/(ab);q)_k} \\
&\quad= \begin{bmatrix} n \\ k \end{bmatrix}_q \left(\frac{c}{ab}\right)^k.
\end{aligned}$$

Substitution yields exactly

$$\sum_{k=0}^{n} \frac{(a,b,q^{-n};q)_k}{(q,c,abq^{1-n}/c;q)_k} q^k.$$

□

9.6 The two q-Chu–Vandermonde sums

Corollary 9.7 (q-Chu–Vandermonde)**.** *For $n \in \mathbb{N}_0$,*

$$ {}_2\phi_1\left[\begin{matrix} q^{-n}, a \\ c \end{matrix}; q, \frac{cq^n}{a}\right] = \frac{(c/a;q)_n}{(c;q)_n}, \tag{9.1}$$

$$ {}_2\phi_1\left[\begin{matrix} q^{-n}, a \\ c \end{matrix}; q, q\right] = a^n \frac{(c/a;q)_n}{(c;q)_n}. \tag{9.2}$$

Proof. For (9.1), put $b = q^{-n}$ in the q-Gauss sum. The infinite products telescope:

$$\frac{(cq^n, c/a;q)_\infty}{(c, cq^n/a;q)_\infty} = \frac{(c/a;q)_n}{(c;q)_n}.$$

The substitution is legitimate because $\{|b| > |c/a|\}$ is connected and meets Heine's initial region $|b| < 1$, so Theorem 9.4 continues analytically to $b = q^{-n}$. Both sides of (9.1) are now rational in (a, c), and they agree on the nonempty open set $|cq^n/a| < 1$ where the q-Gauss hypothesis holds; a rational function vanishing on a nonempty open subset of $\mathbb{C}^2$ vanishes identically, so the identity holds wherever both sides are defined. Termination alone would not give this: it makes the two sides defined, not equal.

For (9.2), let $b \to 0$ in Theorem 9.6, with its other numerator parameter equal to a. Then $(b;q)_k \to 1$ and $(abq^{1-n}/c;q)_k \to 1$, so the left side becomes the displayed ${}_2\phi_1$. On the right,

$$\frac{(c/b;q)_n}{(c/(ab);q)_n} \longrightarrow a^n$$

by comparing the leading factors of the two degree-n polynomials in $1/b$. □

9.7 A hierarchy of mechanisms

The proofs in this chapter form a hierarchy:

finite q-Pascal $\Longrightarrow$ q-Cauchy $\Longrightarrow$ q-Pfaff–Saalschütz,

while

infinite q-binomial $\Longrightarrow$ Heine $\Longrightarrow$ q-Gauss $\Longrightarrow$ first q-Chu–Vandermonde.

The finite chain is valid in a commutative ring after denominators are cleared. The infinite chain requires convergence or a suitable complete formal topology. This distinction becomes important in computer formalization; see Chapter 20.

Chapter 10

Bilateral series and Ramanujan's ${}_1\psi_1$ sum

10.1 Definition

For integral n, negative q-shifted factorials are understood as in Proposition 1.8. Define

$$ {}_1\psi_1\left[\begin{matrix}a\\b\end{matrix};q,z\right]=\sum_{n\in\mathbb{Z}}\frac{(a;q)_n}{(b;q)_n}z^n. $$

The positive tail behaves like a geometric series in z, while the negative tail behaves like a geometric series in $b/(az)$.

Proposition 10.1 (Convergence annulus)**.** *If denominator factors do not vanish, the ${}_1\psi_1$ series converges absolutely in*

$$ |b/a|<|z|<1. $$

Proof. For $n\to+\infty$, the ratio of consecutive terms tends to z. For the negative tail put $n=-m$ and use the negative-index formula. The ratio of successive terms in m tends to $b/(az)$. The ratio test gives the annulus. □

Theorem 10.2 (Ramanujan's ${}_1\psi_1$ summation)**.** *For $|b/a|<|z|<1$,*

$$ {}_1\psi_1\left[\begin{matrix}a\\b\end{matrix};q,z\right]=\frac{(q,b/a,az,q/(az);q)_\infty}{(b,q/a,z,b/(az);q)_\infty}. $$

Proof. We first prove the formula for $b=q^{m+1}$, where $m\in\mathbb{N}_0$. In that case $1/(b;q)_n=0$ for $n<-m$ in the limiting sense, so shift $n=k-m$:

$$ \begin{aligned}\sum_{n\in\mathbb{Z}}\frac{(a;q)_n}{(q^{m+1};q)_n}z^n&=z^{-m}\frac{(a;q)_{-m}}{(q^{m+1};q)_{-m}}\sum_{k=0}^{\infty}\frac{(aq^{-m};q)_k}{(q;q)_k}z^k\\&=z^{-m}\frac{(q;q)_m}{(aq^{-m};q)_m}\frac{(aq^{-m}z;q)_\infty}{(z;q)_\infty}.\end{aligned} $$

The second equality is the infinite q-binomial theorem. Reverse the finite products:

$$ (aq^{-m};q)_m=(-a)^mq^{-m(m+1)/2}(q/a;q)_m, $$

$$ (aq^{-m}z;q)_m=(-az)^mq^{-m(m+1)/2}(q/(az);q)_m. $$

Since $(aq^{-m}z;q)_\infty = (aq^{-m}z;q)_m(az;q)_\infty$, we obtain

$$\frac{(q;q)_m}{(q/a;q)_m}\frac{(q/(az);q)_m(az;q)_\infty}{(z;q)_\infty}.$$

On the other hand, substituting $b = q^{m+1}$ in the proposed product and cancelling tails gives exactly the same expression:

$$\frac{(q;q)_\infty}{(q^{m+1};q)_\infty}\frac{(q^{m+1}/a;q)_\infty}{(q/a;q)_\infty}\frac{(q/(az);q)_\infty}{(q^{m+1}/(az);q)_\infty}\frac{(az;q)_\infty}{(z;q)_\infty}.$$

For fixed a, z with $|z| < 1$, both sides are analytic functions of b in a neighborhood of 0 contained in $|b| < |az|$. Absolute convergence is locally uniform there by Proposition 10.1. For all sufficiently large m, the points $b = q^{m+1}$ lie in that neighborhood, and they accumulate at 0. The identity theorem therefore proves equality near 0. Both sides are meromorphic functions of b throughout the disk $|b| < |az|$: the series gives the meromorphic continuation termwise on compacta avoiding denominator zeros, and the product quotient is visibly meromorphic. Uniqueness of meromorphic continuation proves the identity everywhere in that disk where both sides are defined, which is precisely the asserted annulus for fixed a and z. □

Remark 10.3. Ramanujan's sum contains the infinite q-binomial theorem and several theta-product identities as limiting or specialized cases. It is a model example of a bilateral identity: the second inequality in the convergence annulus is controlled by the positive tail, while the first is controlled by the negative tail.

Chapter 11

q-difference calculus and the quantum binomial

11.1 The q-derivative

Assume $q \neq 1$. For $x \neq 0$ define

$$D_q f(x) = \frac{f(x) - f(qx)}{(1-q)x}.$$

At $x = 0$ one uses the continuous extension when it exists.

Proposition 11.1 (Elementary rules)**.** *For $n \geq 1$,*

$$D_q x^n = [n]_q x^{n-1}.$$

The product rules are

$$\begin{aligned} D_q(fg)(x) &= g(x) D_q f(x) + f(qx) D_q g(x), \\ &= f(x) D_q g(x) + g(qx) D_q f(x). \end{aligned}$$

Moreover,

$$D_q\big(f(q^r x)\big) = q^r (D_q f)(q^r x)$$

for integral r.

Proof. The monomial identity follows from $(x^n - q^n x^n)/((1-q)x) = [n]_q x^{n-1}$. For the first product rule, insert and subtract $f(qx)g(x)$ in $f(x)g(x) - f(qx)g(qx)$. For the second, insert and subtract $f(x)g(qx)$. The scaling rule follows directly from the definition. □

Theorem 11.2 (q-Leibniz rule)**.** *For $n \in \mathbb{N}_0$,*

$$D_q^n(fg)(x) = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k}x)(D_q^{n-k} g)(x).$$

Proof. Induct on n. The assertion is the ordinary product at $n = 0$. Apply D_q to the kth term at level n. By the product and scaling rules, it contributes

$$q^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^{k+1} f)(q^{n-k}x)(D_q^{n-k} g)(x)$$

to index $k+1$, and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k+1}x)(D_q^{n-k+1} g)(x)$$

to index k. The total coefficient of index k is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the second q-Pascal recurrence. □

11.2 q-Taylor and Newton interpolation

Define the q-falling power based at a by

$$(x-a)_q^m = \prod_{j=0}^{m-1} (x - q^j a), \qquad (x-a)_q^0 = 1.$$

Lemma 11.3. *For $m \geq 1$,*

$$D_q (x-a)_q^m = [m]_q (x-a)_q^{m-1}.$$

Consequently,

$$D_q^j (x-a)_q^m = \frac{[m]_q!}{[m-j]_q!} (x-a)_q^{m-j}$$

for $0 \leq j \leq m$.

Proof. Let $P_m(x) = (x-a)_q^m$. Then

$$P_m(qx) = q^m (x - q^{-1}a) \prod_{j=0}^{m-2} (x - q^j a).$$

Subtracting from $P_m(x) = (x - q^{m-1}a) \prod_{j=0}^{m-2} (x - q^j a)$ gives

$$P_m(x) - P_m(qx) = (1 - q^m) x P_{m-1}(x).$$

Divide by $(1-q)x$. Iteration proves the second formula. □

Theorem 11.4 (q-Taylor formula)**.** *Assume $[j]_q \neq 0$ for $1 \leq j \leq N$. If f is a polynomial of degree at most N, then*

$$f(x) = \sum_{k=0}^{N} \frac{(D_q^k f)(a)}{[k]_q!} (x-a)_q^k.$$

Proof. The polynomials $(x-a)_q^k$, $0 \leq k \leq N$, are monic of distinct degrees and hence form a basis. Write $f = \sum c_k (x-a)_q^k$. Apply D_q^j and set $x = a$. By Lemma 11.3, every term with $k > j$ still contains the factor $x - a$, terms with $k < j$ vanish after too many derivatives, and the $k = j$ term equals $c_j [j]_q!$. Thus $c_j = (D_q^j f)(a)/[j]_q!$. □

Remark 11.5. When the nodes $a, aq, aq^2, \ldots$ are distinct, this is a Newton interpolation formula on that geometric grid. In every case covered by the theorem, its basis zeros follow a multiplicative rather than an additive progression.

11.3 The quantum plane

Theorem 11.6 (Noncommutative q-binomial theorem)**.** *Let X, Y belong to an associative algebra and satisfy*

$$YX = qXY.$$

Then

$$(X+Y)^n = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q X^{n-k} Y^k.$$

Proof. Induct on n and multiply on the right by $X+Y$. Since $Y^k X = q^k X Y^k$, the coefficient of $X^{n+1-k} Y^k$ becomes

$$q^k \begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the first q-Pascal recurrence. □

Corollary 11.7 (Noncommutative q-multinomial theorem)**.** *Let $r \geq 1$ and let $x_1, \ldots, x_r$ be elements of an associative algebra over a commutative ring in which q is a central element, and suppose that*

$$x_j x_i = q x_i x_j \qquad (1 \leq i < j \leq r).$$

Then, for every $n \geq 0$,

$$(x_1 + \cdots + x_r)^n = \sum_{\substack{n_1, \ldots, n_r \geq 0 \\ n_1 + \cdots + n_r = n}} \begin{bmatrix} n \\ n_1, \ldots, n_r \end{bmatrix}_q x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r},$$

where on the right the factors are written in increasing order of index. Equivalently, the q-multinomial coefficients are exactly the structure constants that expand powers of $x_1 + \cdots + x_r$ in the ordered monomial basis of the quantum affine r-space.

Proof. Induct on r. For $r = 1$ there is nothing to prove, and $r = 2$ is Theorem 11.6.

Let $r \geq 2$ and set $X = x_1$ and $Y = x_2 + \cdots + x_r$. Because $x_j x_1 = q x_1 x_j$ for every $j \geq 2$ and q is central,

$$YX = \sum_{j=2}^{r} x_j x_1 = q \sum_{j=2}^{r} x_1 x_j = qXY,$$

so Theorem 11.6 applies to the pair (X, Y) and gives

$$(x_1 + \cdots + x_r)^n = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q x_1^{n-k} (x_2 + \cdots + x_r)^k.$$

Substituting $n_1 = n - k$ and using the symmetry $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q$ of Theorem 2.5, this reads

$$(x_1 + \cdots + x_r)^n = \sum_{n_1=0}^{n} \begin{bmatrix} n \\ n_1 \end{bmatrix}_q x_1^{n_1} (x_2 + \cdots + x_r)^{n-n_1}.$$

The elements $x_2, \ldots, x_r$ satisfy the same commutation relations among themselves, so the inductive hypothesis applies to them and gives, with $m = n - n_1$,

$$(x_2 + \cdots + x_r)^m = \sum_{\substack{n_2, \ldots, n_r \geq 0 \\ n_2 + \cdots + n_r = m}} \begin{bmatrix} m \\ n_2, \ldots, n_r \end{bmatrix}_q x_2^{n_2} \cdots x_r^{n_r}.$$

Since the coefficients are central, substitution produces precisely the ordered monomials $x_1^{n_1}x_2^{n_2}\cdots x_r^{n_r}$, the coefficient of $x_1^{n_1}\cdots x_r^{n_r}$ being

$$\begin{bmatrix} n \\ n_1 \end{bmatrix}_q \begin{bmatrix} n-n_1 \\ n_2,\dots,n_r \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_{n_1}(q;q)_{n-n_1}} \cdot \frac{(q;q)_{n-n_1}}{(q;q)_{n_2}\cdots(q;q)_{n_r}} = \begin{bmatrix} n \\ n_1,\dots,n_r \end{bmatrix}_q$$

by the definition of the q-multinomial coefficient preceding Proposition 2.8; the factor $(q;q)_{n-n_1}$ telescopes away, exactly as in the proof of that proposition. No factor was transposed at any stage after the expansion, so the monomials are produced in increasing order of index, as asserted. □

Remark 11.8 (The same coefficient read combinatorially). Expanding $(x_1+\cdots+x_r)^n$ without using any relation produces the r^n words $x_{w_1}x_{w_2}\cdots x_{w_n}$ with $w\in\{1,\dots,r\}^n$. Sorting such a word into increasing order uses one adjacent transposition for each inversion of w, and each transposition contributes exactly one factor q; hence

$$x_{w_1}x_{w_2}\cdots x_{w_n} = q^{\mathrm{inv}(w)}x_1^{n_1}\cdots x_r^{n_r},$$

where n_i is the multiplicity of the letter i in w. Comparing coefficients with Corollary 11.7 recovers the multiset inversion enumerator of Theorem 5.7, and conversely that theorem proves Corollary 11.7 again. The quantum affine space is therefore a purely algebraic encoding of the inversion statistic, just as Theorem 11.6 encodes Theorem 5.1.

Corollary 11.9 (Factorization of q-exponentials)**.** *Assume $|q|<1$. Define*

$$e_q(x) = \sum_{n=0}^{\infty} \frac{x^n}{[n]_q!} = \frac{1}{((1-q)x;q)_\infty}, \qquad |(1-q)x|<1,$$

and

$$E_q(x) = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}}x^n}{[n]_q!} = (-(1-q)x;q)_\infty, \qquad x\in\mathbb{C}.$$

Then $e_q(x)E_q(-x)=1$ wherever the first series is defined. If $YX=qXY$, then, as an identity of formal series in total degree,

$$e_q(X+Y)=e_q(X)e_q(Y).$$

Proof. The product formulas and the stated convergence domains follow from Euler's identities after replacing z by $(1-q)x$. Their product is 1. For the noncommutative factorization, apply Theorem 11.6 degree by degree and use

$$\frac{1}{[n]_q!}\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{[n-k]_q![k]_q!}.$$

Reindex with $r=n-k$ to obtain the Cauchy product of $e_q(X)$ and $e_q(Y)$, with all X factors preceding all Y factors. Since every total degree receives only finitely many contributions, the formal rearrangement is valid. □

Proposition 11.10 (Eigenfunction properties)**.** *Under the hypotheses and within the convergence domains of Corollary 11.9, for scalar λ,*

$$D_q e_q(\lambda x) = \lambda e_q(\lambda x), \qquad D_q E_q(\lambda x) = \lambda E_q(q\lambda x).$$

Proof. Differentiate the series term by term and use $D_q x^n = [n]_q x^{n-1}$. □

11.4 Jackson's q-integral

Assume $0 < q < 1$. Define

$$\int_0^a f(t)\, d_q t = (1-q)a \sum_{n=0}^{\infty} q^n f(aq^n),$$

whenever the series converges.

Theorem 11.11 (Fundamental theorem of q-calculus)**.** *Let*

$$F(x) = \int_0^x f(t)\, d_q t.$$

Whenever the defining series converges locally uniformly,

$$D_q F(x) = f(x).$$

Conversely, if $G(aq^n) \to G(0)$, then

$$\int_0^a D_q G(t)\, d_q t = G(a) - G(0).$$

Proof. From the definition,

$$F(x) = (1-q)x \sum_{n \geq 0} q^n f(xq^n),$$

while

$$F(qx) = (1-q)x \sum_{n \geq 1} q^n f(xq^n).$$

Their difference is $(1-q)xf(x)$, proving the first identity. For the second,

$$\begin{aligned}\int_0^a D_q G(t)\, d_q t &= (1-q)a \sum_{n \geq 0} q^n \frac{G(aq^n) - G(aq^{n+1})}{(1-q)aq^n} \\ &= \sum_{n \geq 0} \big(G(aq^n) - G(aq^{n+1})\big),\end{aligned}$$

which telescopes to $G(a) - G(0)$. □

Corollary 11.12 (Jackson integration by parts)**.** *Under convergence assumptions,*

$$\int_0^a f(t) D_q g(t)\, d_q t = f(a)g(a) - f(0)g(0) - \int_0^a g(qt) D_q f(t)\, d_q t.$$

Proof. Use the product rule $D_q(fg) = fD_q g + g(q\,\cdot)D_q f$ and integrate with Theorem 11.11. □

Example 11.13 (Monomials)**.** For $\Re(s) > -1$,

$$\int_0^a t^s\, d_q t = (1-q)a^{s+1} \sum_{n \geq 0} q^{n(s+1)} = \frac{a^{s+1}}{[s+1]_q},$$

where $[u]_q = (1-q^u)/(1-q)$ for complex u.

Chapter 12

The q-gamma and q-beta functions

Throughout this chapter $0 < q < 1$ unless stated otherwise.

12.1 Definition and functional equation

Definition 12.1. For $x > 0$, define

$$\Gamma_q(x) = (1-q)^{1-x} \frac{(q;q)_\infty}{(q^x;q)_\infty}.$$

With $q^z = \exp(z \log q)$, the same product quotient defines a meromorphic function of $z \in \mathbb{C}$; its possible poles occur where $q^{z+j} = 1$ for some $j \in \mathbb{N}_0$. Whenever complex arguments of Γ_q occur below, this meromorphic continuation is intended.

Theorem 12.2 (Basic properties)**.** *For $x > 0$,*

$$\Gamma_q(1) = 1, \qquad \Gamma_q(x+1) = [x]_q \Gamma_q(x).$$

Consequently, for $n \in \mathbb{N}$,

$$\Gamma_q(n) = [n-1]_q!.$$

The recurrence extends as an identity of meromorphic functions of a complex argument.

Proof. The value at 1 is immediate. The tail identity $(q^x;q)_\infty = (1-q^x)(q^{x+1};q)_\infty$ gives

$$\frac{\Gamma_q(x+1)}{\Gamma_q(x)} = (1-q)^{-1}(1-q^x) = [x]_q.$$

Iteration gives the integral values. The same tail identity is valid for complex x, and hence proves the meromorphic recurrence wherever both sides are finite; equality elsewhere follows in the sense of meromorphic continuation. □

Proposition 12.3 (Logarithmic derivatives and convexity)**.** *For $x > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}x} \log \Gamma_q(x) = -\log(1-q) + (\log q) \sum_{n=0}^{\infty} \frac{q^{n+x}}{1-q^{n+x}},$$

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} \log \Gamma_q(x) = (\log q)^2 \sum_{n=0}^{\infty} \frac{q^{n+x}}{(1-q^{n+x})^2} > 0.$$

Thus Γ_q is strictly log-convex.

Proof. Differentiate the locally uniformly convergent logarithm of the defining infinite product. A second differentiation gives the positive series. □

12.2 A q-Bohr–Mollerup theorem

Theorem 12.4 (q-Bohr–Mollerup characterization)**.** *The function Γ_q is the unique function $f : (0,\infty) \to (0,\infty)$ satisfying*

$$f(1) = 1, \qquad f(x+1) = [x]_q f(x),$$

and such that $\log f$ *is convex.*

Proof. Existence follows from Theorem 12.2 and proposition 12.3. For uniqueness let f have the stated properties and fix $0 < x \le 1$. Convexity on the interval from n to $n+1$ gives

$$f(n+x) \le f(n)^{1-x} f(n+1)^x.$$

A comparison of the adjacent secant slopes on $[n+x, n+1]$ and $[n+1, n+2]$ gives

$$f(n+x) \ge \frac{f(n+1)}{[n+1]_q^{1-x}}.$$

Using the recurrence,

$$f(n+x) = f(x)\prod_{j=0}^{n-1}[x+j]_q, \quad f(n) = [n-1]_q!, \quad f(n+1) = [n]_q!,$$

we obtain

$$\frac{[n]_q!}{[n+1]_q^{1-x}\prod_{j=0}^{n-1}[x+j]_q} \le f(x) \le \frac{[n-1]_q![n]_q^x}{\prod_{j=0}^{n-1}[x+j]_q}.$$

The ratio of the upper bound to the lower bound is

$$\left(\frac{[n+1]_q}{[n]_q}\right)^{1-x} \longrightarrow 1.$$

Hence the bounds determine $f(x)$ uniquely for $0 < x \le 1$, and the recurrence determines it on all of $(0,\infty)$. Since Γ_q satisfies the same conditions, $f = \Gamma_q$. □

12.3 Classical limit

Theorem 12.5 (Limit to Euler's gamma function)**.** *For every $x > 0$,*

$$\lim_{q\to 1^-} \Gamma_q(x) = \Gamma(x).$$

The convergence is locally uniform on $(0,\infty)$.

Proof. It suffices by the functional equations to consider $0 < x \le 1$. The bounds in the proof of Theorem 12.4 hold for $\Gamma_q(x)$. Fix n and let $q \to 1^-$. Since $[u]_q \to u$, they become

$$\frac{n!}{(n+1)^{1-x}x(x+1)\cdots(x+n-1)} \le \liminf_{q\to 1^-}\Gamma_q(x)$$

$$\le \limsup_{q\to 1^-}\Gamma_q(x) \le \frac{(n-1)!n^x}{x(x+1)\cdots(x+n-1)}.$$

Both outer expressions tend to $\Gamma(x)$ as $n \to \infty$. To see this directly, use Euler's beta integral:

$$B(x,n) = \int_0^1 t^{x-1}(1-t)^{n-1}\,dt = \frac{(n-1)!}{x(x+1)\cdots(x+n-1)}.$$

After $t = u/n$, dominated convergence gives $n^x B(x,n) \to \int_0^\infty u^{x-1}\mathrm{e}^{-u}\,du = \Gamma(x)$. The lower expression differs from the upper by a factor tending to 1. This proves pointwise convergence. We record the uniformity argument. On a compact interval $x \in [\delta, 1]$, the upper outer bound can be written as

$$n^x B(x,n) = \int_0^n u^{x-1}\left(1 - \frac{u}{n}\right)^{n-1}\,\mathrm{d}u.$$

For $n \geq 2$, its integrand is dominated uniformly in x by $u^{\delta-1}$ on $(0,1]$ and by $e^{-u/2}$ on $[1,\infty)$. More explicitly, after extending the kernel by 0 for $u > n$,

$$\sup_{\delta \leq x \leq 1}\left| u^{x-1}\left(1 - \frac{u}{n}\right)_+^{n-1} - u^{x-1}e^{-u}\right| \longrightarrow 0$$

for every $u > 0$, while this supremum is bounded by an integrable multiple of $u^{\delta-1}\mathbf{1}_{(0,1]}(u) + e^{-u/2}\mathbf{1}_{[1,\infty)}(u)$. The dominated convergence theorem therefore gives

$$\sup_{\delta \leq x \leq 1} |n^x B(x,n) - \Gamma(x)| \longrightarrow 0.$$

The lower outer bound differs by the factor $(n/(n+1))^{1-x}$, which also tends uniformly to 1. For fixed n, the original q-bounds converge uniformly as $q \to 1^-$ because they contain only finitely many continuous factors. A uniform squeeze proves local uniform convergence on $(0,1]$, and the recurrence extends it to arbitrary compact subsets of $(0,\infty)$. □

12.4 The Jackson q-beta integral

Definition 12.6. For $x, y > 0$, define

$$B_q(x,y) = \int_0^1 t^{x-1}\frac{(qt;q)_\infty}{(q^y t;q)_\infty}\,d_q t.$$

Theorem 12.7 (q-beta evaluation)**.** *For $x, y > 0$,*

$$B_q(x,y) = \frac{\Gamma_q(x)\Gamma_q(y)}{\Gamma_q(x+y)}.$$

Proof. Expand the Jackson integral:

$$\begin{aligned} B_q(x,y) &= (1-q)\sum_{n\geq 0} q^{nx}\frac{(q^{n+1};q)_\infty}{(q^{n+y};q)_\infty} \\ &= (1-q)\frac{(q;q)_\infty}{(q^y;q)_\infty}\sum_{n\geq 0}\frac{(q^y;q)_n}{(q;q)_n}q^{nx}. \end{aligned}$$

The infinite q-binomial theorem evaluates the sum as $(q^{x+y};q)_\infty/(q^x;q)_\infty$. Thus

$$B_q(x,y) = (1-q)\frac{(q;q)_\infty(q^{x+y};q)_\infty}{(q^x;q)_\infty(q^y;q)_\infty},$$

which is exactly the gamma quotient after substituting the definition of Γ_q. □

Corollary 12.8 (Recurrences)**.** *For $x, y > 0$,*

$$B_q(x+1, y) = \frac{[x]_q}{[x+y]_q} B_q(x, y), \qquad B_q(x, y+1) = \frac{[y]_q}{[x+y]_q} B_q(x, y).$$

Proof. Apply the functional equation for Γ_q to the quotient in Theorem 12.7. □

12.5 Multiplication and reflection-type formulas

Theorem 12.9 (q-multiplication formula)**.** *For an integer $m \geq 1$ and $x > 0$,*

$$\Gamma_q(mx) \prod_{j=1}^{m-1} \Gamma_{q^m}\left(\frac{j}{m}\right) = [m]_q^{mx-1} \prod_{j=0}^{m-1} \Gamma_{q^m}\left(x + \frac{j}{m}\right).$$

Proof. Multiply the definitions on the right. By dissection,

$$\prod_{j=0}^{m-1} (q^{mx+j}; q^m)_\infty = (q^{mx}; q)_\infty.$$

Hence

$$\prod_{j=0}^{m-1} \Gamma_{q^m}\left(x + \frac{j}{m}\right) = (1-q^m)^{(m+1)/2-mx} \frac{(q^m; q^m)_\infty^m}{(q^{mx}; q)_\infty}.$$

Similarly,

$$\prod_{j=1}^{m-1} \Gamma_{q^m}\left(\frac{j}{m}\right) = (1-q^m)^{(m-1)/2} \frac{(q^m; q^m)_\infty^m}{(q; q)_\infty}.$$

Taking the ratio and using $[m]_q = (1-q^m)/(1-q)$ produces the formula. □

Proposition 12.10 (Reflection-type product)**.** *For $0 < x < 1$,*

$$\Gamma_q(x)\Gamma_q(1-x) = (1-q) \frac{(q; q)_\infty^2}{(q^x, q^{1-x}; q)_\infty}.$$

Proof. Multiply the two defining products; the powers of $1-q$ combine to one. □

Remark 12.11. Unlike Euler's reflection formula, the right side is naturally a theta quotient rather than an elementary trigonometric function. Jacobi's triple product in the next part makes that theta structure explicit.

Part III

Theta products and partition identities

Chapter 13

Jacobi's triple product

13.1 A bilateral theta series

For $z \neq 0$ and $|q| < 1$, define

$$\Theta(z;q) = \sum_{n\in\mathbb{Z}} q^{\binom{n}{2}} z^n.$$

The series converges locally uniformly on $\mathbb{C}^*$ because its terms decay quadratically in both directions.

Theorem 13.1 (Jacobi triple product)**.** *For $z \neq 0$ and $|q| < 1$,*

$$\sum_{n\in\mathbb{Z}} q^{\binom{n}{2}} z^n = (-z, -q/z, q; q)_\infty, \tag{13.1}$$

$$\sum_{n\in\mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n = (z, q/z, q; q)_\infty. \tag{13.2}$$

Proof. It suffices to prove (13.1); replace z by $-z$ for (13.2). Put

$$F(z) = (-z;q)_\infty(-q/z;q)_\infty(q;q)_\infty.$$

The products converge locally uniformly on $\mathbb{C}^*$, so F has a Laurent expansion $F(z) = \sum_{n\in\mathbb{Z}} c_n z^n$. The shift identities give

$$F(qz) = \frac{1+z^{-1}}{1+z} F(z) = z^{-1}F(z).$$

Comparing Laurent coefficients yields

$$c_{n+1} = q^n c_n, \qquad c_n = q^{\binom{n}{2}} c_0$$

for every integer n.

It remains to find c_0. Euler's product expansions give

$$(-z;q)_\infty = \sum_{r\geq 0} \frac{q^{\binom{r}{2}} z^r}{(q;q)_r}, \qquad (-q/z;q)_\infty = \sum_{s\geq 0} \frac{q^{\binom{s+1}{2}} z^{-s}}{(q;q)_s}.$$

Hence

$$c_0 = (q;q)_\infty \sum_{r\geq 0} \frac{q^{r^2}}{(q;q)_r^2}.$$

The sum is the generating function obtained by decomposing an arbitrary partition according to the side r of its Durfee square: the square contributes q^{r^2}, while the diagrams to its right and below are independent partitions with at most r parts, each having generating function $1/(q;q)_r$. Therefore

$$\sum_{r\ge 0}\frac{q^{r^2}}{(q;q)_r^2}=\frac{1}{(q;q)_\infty},$$

so $c_0 = 1$. Thus $F(z)=\sum_{n\in\mathbb{Z}} q^{\binom{n}{2}} z^n$. □

Corollary 13.2 (Quasi-periodicity and zeros). *The theta function satisfies*

$$\Theta(qz;q)=z^{-1}\Theta(z;q).$$

Its zeros are the simple points $z=-q^m$, $m\in\mathbb{Z}$.

Proof. The functional equation follows either from reindexing the bilateral series or from the proof above. The product in (13.1) vanishes exactly when $1+zq^j=0$ or $1+q^{j+1}/z=0$, which together give $z=-q^m$ for $m\in\mathbb{Z}$. Exactly one factor vanishes at each such point. □

13.2 A finite triple product

Jacobi's identity has an exact polynomial refinement. The truncated product $(z;q)_N(q/z;q)_N$ is again a bilateral theta-like sum, with the factor $1/(q;q)_\infty$ replaced by a central Gaussian coefficient; letting $N\to\infty$ then reproves Theorem 13.1 from finite algebra plus a single application of dominated convergence.

Theorem 13.3 (Finite triple product). *For $N\in\mathbb{N}_0$ and $z\neq 0$,*

$$(z;q)_N\,(q/z;q)_N=\sum_{k=-N}^{N}(-1)^k q^{\binom{k}{2}}\begin{bmatrix}2N\\ N+k\end{bmatrix}_q z^k. \tag{13.3}$$

Here $\binom{k}{2}$ denotes $k(k-1)/2$ for every integer k, negative k included; equivalently $\binom{-m}{2}=\binom{m+1}{2}$ for $m\ge 0$. Both sides lie in $\mathbb{Z}[q][z,z^{-1}]$ and (13.3) is an identity there; in particular it carries no convergence hypothesis and holds with q an indeterminate.

Proof. By Theorem 3.1, (3.1),

$$(z;q)_N=\sum_{i=0}^{N}(-1)^i q^{\binom{i}{2}}\begin{bmatrix}N\\ i\end{bmatrix}_q z^i,$$

and, replacing z by q/z in the same identity and using $\binom{j}{2}+j=\binom{j+1}{2}$,

$$(q/z;q)_N=\sum_{j=0}^{N}(-1)^j q^{\binom{j+1}{2}}\begin{bmatrix}N\\ j\end{bmatrix}_q z^{-j}.$$

Multiply the two expansions and collect the pairs contributing to z^k, namely those with $i=j+k$. Since $(-1)^{j+k}(-1)^j=(-1)^k$, and since $\begin{bmatrix}N\\ j+k\end{bmatrix}_q=0$ unless $0\le j+k\le N$, the inner sum may be extended over all $j\in\mathbb{Z}$:

$$(z;q)_N(q/z;q)_N=\sum_{k\in\mathbb{Z}}(-1)^k z^k\sum_{j\in\mathbb{Z}} q^{\binom{j+k}{2}+\binom{j+1}{2}}\begin{bmatrix}N\\ j+k\end{bmatrix}_q\begin{bmatrix}N\\ j\end{bmatrix}_q.$$

The two binomial exponents combine cleanly. As an identity between polynomials in j and k,

$$\binom{j+k}{2}+\binom{j+1}{2}-\binom{k}{2}=\tfrac{1}{2}\big[(j+k)(j+k-1)+j(j+1)-k(k-1)\big]=j^2+jk,$$

and being a polynomial identity it is valid for negative k as well, with the convention above. On a term with $\left[\begin{smallmatrix}N\\ j\end{smallmatrix}\right]_q\left[\begin{smallmatrix}N\\ j+k\end{smallmatrix}\right]_q\neq 0$ one has $0\le j\le N$ and $0\le j+k\le N$, so $j^2+jk=j(j+k)\ge 0$ and no negative power of q is created; terms with $j(j+k)<0$ carry a vanishing Gaussian coefficient. Hence the inner sum equals

$$q^{\binom{k}{2}}\sum_{j\in\mathbb{Z}}q^{j^2+jk}\begin{bmatrix}N\\ j\end{bmatrix}_q\begin{bmatrix}N\\ j+k\end{bmatrix}_q=q^{\binom{k}{2}}\begin{bmatrix}2N\\ N+k\end{bmatrix}_q$$

by Corollary 3.7. Therefore

$$(z;q)_N(q/z;q)_N=\sum_{k\in\mathbb{Z}}(-1)^kq^{\binom{k}{2}}\begin{bmatrix}2N\\ N+k\end{bmatrix}_q z^k,$$

and the summand vanishes for $|k|>N$, which is (13.3). Both sides are Laurent polynomials in z with coefficients in $\mathbb{Z}[q]$ because each Gaussian coefficient is such a polynomial by Theorem 2.5. □

Example 13.4. At $N=1$ the coefficients are $\left[\begin{smallmatrix}2\\ 0\end{smallmatrix}\right]_q=\left[\begin{smallmatrix}2\\ 2\end{smallmatrix}\right]_q=1$ and $\left[\begin{smallmatrix}2\\ 1\end{smallmatrix}\right]_q=1+q$, so (13.3) reads

$$(1-z)(1-q/z)=-qz^{-1}+(1+q)-z,$$

which is correct. The coefficient $q^{\binom{-1}{2}}=q$ of z^{-1} shows the convention for negative k at work. At $N=2$ one gets

$$(z;q)_2(q/z;q)_2=q^3z^{-2}-q(1+q+q^2+q^3)z^{-1}+(1+q+2q^2+q^3+q^4)-(1+q+q^2+q^3)z+qz^2,$$

the coefficients being $\left[\begin{smallmatrix}4\\ 0\end{smallmatrix}\right]_q=\left[\begin{smallmatrix}4\\ 4\end{smallmatrix}\right]_q=1$, $\left[\begin{smallmatrix}4\\ 1\end{smallmatrix}\right]_q=\left[\begin{smallmatrix}4\\ 3\end{smallmatrix}\right]_q=1+q+q^2+q^3$ and $\left[\begin{smallmatrix}4\\ 2\end{smallmatrix}\right]_q=1+q+2q^2+q^3+q^4$, with the outer q-powers $q^{\binom{-2}{2}}=q^3$, $q^{\binom{-1}{2}}=q$ and $q^{\binom{2}{2}}=q$.

Corollary 13.5 (Second proof of Jacobi's triple product)**.** *Let $|q|<1$ and $z\neq 0$. Letting $N\to\infty$ in (13.3) and multiplying by $(q;q)_\infty$ yields (13.2). The passage to the limit is uniform on every closed annulus $r\le|z|\le R$ with $0<r\le R$.*

Proof. Extend the summand of (13.3) by zero, that is, read $\left[\begin{smallmatrix}2N\\ N+k\end{smallmatrix}\right]_q=0$ for $|k|>N$, so that the right side is a sum over all $k\in\mathbb{Z}$.

A bound uniform in both N and k. By Proposition 2.2,

$$\begin{bmatrix}2N\\ N+k\end{bmatrix}_q=\frac{(q;q)_{2N}}{(q;q)_{N+k}\,(q;q)_{N-k}}\qquad(|k|\le N).$$

Put

$$\mu=\inf_{m\ge 0}|(q;q)_m|,\qquad M=\prod_{j\ge 1}\big(1+|q|^j\big)<\infty.$$

No factor of $(q;q)_m$ vanishes, because $|q|<1$ forces $q^j\neq 1$ for $j\ge 1$, and $(q;q)_m\to(q;q)_\infty$, which is nonzero by Theorem 1.12. A convergent sequence of nonzero numbers with nonzero limit is bounded away from 0, so $\mu>0$; and $|(q;q)_m|\le M$ for every m. Hence

$$\left|\begin{bmatrix}2N\\ N+k\end{bmatrix}_q\right|\le\frac{M}{\mu^2}\qquad\text{for all } N\ge 0 \text{ and all } k\in\mathbb{Z}. \tag{13.4}$$

A summable dominating function. For every integer k one has $\binom{k}{2} = \frac{1}{2}(k^2 - k) \geq \frac{1}{2}(k^2 - |k|)$, with equality for $k \geq 0$, so $|q|^{\binom{k}{2}} \leq |q|^{(k^2-|k|)/2}$. On the annulus $r \leq |z| \leq R$ we have $|z^k| \leq R^{|k|} + r^{-|k|}$. Therefore, for all N and all z in the annulus,

$$\left|(-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k\right| \leq g(k) := \frac{M}{\mu^2} |q|^{(k^2-|k|)/2}\left(R^{|k|} + r^{-|k|}\right),$$

and $\sum_{k\in\mathbb{Z}} g(k) < \infty$, since $|q|^{k^2/2}$ decays faster than any geometric sequence grows.

The termwise limit. For each fixed k, using $(q;q)_m \to (q;q)_\infty \neq 0$ again,

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \frac{(q;q)_{2N}}{(q;q)_{N+k}\,(q;q)_{N-k}} \xrightarrow[N\to\infty]{} \frac{(q;q)_\infty}{(q;q)_\infty^2} = \frac{1}{(q;q)_\infty}.$$

This is Theorem 19.7 when $k = 0$; the argument is unchanged for fixed $k \neq 0$, since $N \pm k \to \infty$ with N. Dominated convergence for series, applied on the counting measure of $\mathbb{Z}$ with dominating function g, therefore gives

$$\lim_{N\to\infty} \sum_{k\in\mathbb{Z}} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k = \frac{1}{(q;q)_\infty} \sum_{k\in\mathbb{Z}} (-1)^k q^{\binom{k}{2}} z^k,$$

The convergence is moreover uniform on the annulus: given $\varepsilon > 0$, choose K with $\sum_{|k|>K} g(k) < \varepsilon/3$, which bounds the two tails simultaneously for every N and every z in the annulus because g depends on neither; the remaining $2K+1$ terms converge because $\left[\begin{smallmatrix} 2N \\ N+k \end{smallmatrix}\right]_q \to 1/(q;q)_\infty$ is a limit of numbers, and $|z^k| \leq R^K + r^{-K}$ there. On the left of (13.3), $(z;q)_N(q/z;q)_N \to (z;q)_\infty(q/z;q)_\infty$ locally uniformly on $\mathbb{C}^*$ by Theorem 1.12. Multiplying by $(q;q)_\infty$ we obtain

$$(q, z, q/z; q)_\infty = \sum_{k\in\mathbb{Z}} (-1)^k q^{\binom{k}{2}} z^k,$$

which is (13.2). □

Remark 13.6 (Two mechanisms for one identity). Theorem 13.1 was proved above from a functional equation together with the Durfee-square evaluation of the constant Laurent coefficient. Corollary 13.5 proves it instead from the polynomial identity (13.3), whose only inputs are Theorem 3.1 and Theorem 3.5. The second route separates algebra from analysis completely: (13.3) holds in $\mathbb{Z}[q][z, z^{-1}]$, and the entire analytic content is the single limit $\left[\begin{smallmatrix} 2N \\ N+k \end{smallmatrix}\right]_q \to 1/(q;q)_\infty$, which is the same finite-to-infinite stabilization recorded coefficientwise in Proposition 14.5 and columnwise in Theorem 19.6. That separation is exactly the architecture recommended in Chapter 20. Note also that (13.3) is genuinely finer than (13.2): its coefficients are the central Gaussian coefficients $\left[\begin{smallmatrix} 2N \\ N+k \end{smallmatrix}\right]_q$, which by Theorem 2.5 are palindromic polynomials of degree $(N+k)(N-k)$, whereas in the limit every one of them collapses to the single k-independent factor $1/(q;q)_\infty$; what survives of the polynomial is not its constant term but its stabilised coefficient sequence, namely the partition numbers of Proposition 14.5.

13.3 Euler's pentagonal number theorem

Theorem 13.7 (Pentagonal number theorem)**.** *For* $|q| < 1$*,*

$$(q;q)_\infty = \sum_{n\in\mathbb{Z}} (-1)^n q^{n(3n-1)/2} = 1 + \sum_{n=1}^{\infty} (-1)^n \left(q^{n(3n-1)/2} + q^{n(3n+1)/2}\right).$$

Proof. In (13.2), replace the base q by q^3 and set $z = q$. The product side is

$$(q;q^3)_\infty(q^2;q^3)_\infty(q^3;q^3)_\infty = (q;q)_\infty$$

by dissection. The series exponent is

$$3\binom{n}{2} + n = \frac{n(3n-1)}{2}.$$

Pair the terms n and $-n$ to obtain the second form. □

Corollary 13.8 (Euler's partition recurrence)**.** *Let $p(0) = 1$ and $p(n) = 0$ for $n < 0$. Then for $n \geq 1$,*

$$p(n) = \sum_{j=1}^{\infty}(-1)^{j-1}\left(p\left(n - \frac{j(3j-1)}{2}\right) + p\left(n - \frac{j(3j+1)}{2}\right)\right).$$

Proof. The generating function for partitions is $\sum_{n\geq 0} p(n)q^n = 1/(q;q)_\infty$. Multiply this by the pentagonal expansion and compare the coefficient of q^n. □

13.4 Jacobi's cubic identity

Theorem 13.9 (Jacobi identity)**.** *For $|q| < 1$,*

$$(q;q)_\infty^3 = \sum_{n=0}^{\infty}(-1)^n(2n+1)q^{n(n+1)/2}.$$

Proof. Differentiate (13.2) with respect to z and set $z = 1$. Since

$$(z;q)_\infty = (1-z)(qz;q)_\infty,$$

the derivative of the product side at $z = 1$ is $-(q;q)_\infty^3$. On the series side it is

$$\sum_{m\in\mathbb{Z}} m(-1)^m q^{\binom{m}{2}}.$$

Pair $m = -n$ with $m = n+1$ for $n \geq 0$. Their sum is

$$-(2n+1)(-1)^n q^{n(n+1)/2}.$$

Cancelling the common minus sign proves the identity. □

13.5 A theta form of the q-gamma reflection product

Corollary 13.10. *For $0 < x < 1$,*

$$\Gamma_q(x)\Gamma_q(1-x) = (1-q)\frac{(q;q)_\infty^3}{\sum_{n\in\mathbb{Z}}(-1)^n q^{\binom{n}{2}+xn}}.$$

Proof. Set $z = q^x$ in (13.2):

$$\sum_{n\in\mathbb{Z}}(-1)^n q^{\binom{n}{2}+xn} = (q^x, q^{1-x}, q;q)_\infty.$$

Substitute this in Proposition 12.10. □

Chapter 14

Partitions and product–sum identities

14.1 Unrestricted, distinct, and odd parts

Theorem 14.1 (Basic partition products)**.** *Let $p(n)$ count unrestricted partitions, $d(n)$ partitions into distinct parts, and $o(n)$ partitions into odd parts. Then*

$$\begin{aligned}\sum_{n\geq 0} p(n)q^n &= \frac{1}{(q;q)_\infty},\\ \sum_{n\geq 0} d(n)q^n &= (-q;q)_\infty,\\ \sum_{n\geq 0} o(n)q^n &= \frac{1}{(q;q^2)_\infty}.\end{aligned}$$

Moreover, $d(n) = o(n)$ for every n.

Proof. For unrestricted partitions, the part j may occur $0, 1, 2, \ldots$ times and contributes the geometric factor $(1 - q^j)^{-1}$. For distinct partitions, the part j occurs zero or one time and contributes $1 + q^j$. For odd parts, retain only the odd geometric factors. Finally,

$$(-q;q)_\infty = \prod_{j\geq 1} \frac{1-q^{2j}}{1-q^j} = \frac{(q^2;q^2)_\infty}{(q;q)_\infty} = \frac{1}{(q;q^2)_\infty},$$

so the two generating functions agree coefficientwise. □

Remark 14.2 (A bijection). The equality $d(n) = o(n)$ also has a direct proof. In a partition into distinct parts, write each part uniquely as $2^r u$ with u odd. Replace that part by 2^r copies of u. Distinct powers of two ensure that the process is reversible by binary expansion of the multiplicity of every odd part.

14.2 Bounded partitions

Proposition 14.3. *The following generating functions are equal:*

$$\frac{1}{(q;q)_r} = \sum_{n\geq 0} p_{\leq r}(n)q^n,$$

where $p_{\leq r}(n)$ counts partitions with largest part at most r, or equivalently partitions with at most r parts.

Proof. The product allows parts $1, \ldots, r$ with arbitrary multiplicity. Conjugation of Ferrers diagrams exchanges the condition "largest part at most r" with "at most r parts." □

Corollary 14.4 (Durfee decomposition of all partitions)**.**

$$\frac{1}{(q;q)_\infty} = \sum_{r=0}^{\infty} \frac{q^{r^2}}{(q;q)_r^2}.$$

Proof. Decompose every partition by the side r of its Durfee square. The diagrams to the right and below are independent partitions with at most r parts. This is the identity used in the proof of Jacobi's triple product. □

14.3 Finite approximations and stabilization

Proposition 14.5. *For fixed N, the coefficient of q^N in $\begin{bmatrix} m+k \\ k \end{bmatrix}_q$ equals $p(N)$ whenever $m, k \geq N$.*

Proof. The Gaussian coefficient counts partitions inside a $k \times m$ rectangle. Any partition of N has at most N parts and largest part at most N, so for $m, k \geq N$ the rectangle restriction is vacuous. □

Remark 14.6. Thus $\begin{bmatrix} m+k \\ k \end{bmatrix}_q$ converges coefficientwise to $1/(q;q)_\infty$ as both sides of the rectangle tend to infinity. This finite-to-infinite passage is a recurring bridge between Gaussian polynomials and partition identities.

Chapter 15

Bailey pairs and the Rogers–Ramanujan identities

15.1 Bailey pairs

For brevity, write $(a)_n = (a;q)_n$ in this chapter.

Definition 15.1. Sequences $(\alpha_n)_{n\geq 0}$ and $(\beta_n)_{n\geq 0}$ form a *Bailey pair relative to a* if

$$\beta_n = \sum_{r=0}^{n} \frac{\alpha_r}{(q)_{n-r}(aq)_{n+r}} \qquad (n \geq 0).$$

15.2 Bailey inversion

Lemma 15.2 (q-difference annihilation)**.** *If $P(x)$ is a polynomial of degree less than N, then*

$$\sum_{s=0}^{N}(-1)^{N-s}q^{\binom{N-s}{2}}\begin{bmatrix}N\\ s\end{bmatrix}_q P(q^s) = 0.$$

Proof. By linearity it suffices to take $P(x) = x^m$ with $0 \leq m < N$. Put $t = N - s$ and use the finite q-binomial theorem:

$$\begin{aligned}\sum_{s=0}^{N}(-1)^{N-s}q^{\binom{N-s}{2}}\begin{bmatrix}N\\ s\end{bmatrix}_q q^{ms} &= q^{mN}\sum_{t=0}^{N}(-1)^t q^{\binom{t}{2}}\begin{bmatrix}N\\ t\end{bmatrix}_q q^{-mt}\\ &= q^{mN}(q^{-m};q)_N = 0,\end{aligned}$$

for the product contains the factor $1 - q^0$. □

Theorem 15.3 (Bailey inversion)**.** *Assume $a \neq 1$ initially. The Bailey-pair relation is equivalent to*

$$\alpha_n = \frac{1 - aq^{2n}}{1 - a}\sum_{j=0}^{n}\frac{(a)_{n+j}}{(q)_{n-j}}(-1)^{n-j}q^{\binom{n-j}{2}}\beta_j.$$

At $a = 1$, the right side is interpreted by its removable limit.

Proof. Substitute the definition of β_j into the proposed inverse. The coefficient of α_r is

$$\frac{1-aq^{2n}}{1-a}\sum_{j=r}^{n}\frac{(a)_{n+j}}{(q)_{n-j}(q)_{j-r}(aq)_{j+r}}(-1)^{n-j}q^{\binom{n-j}{2}}.$$

If $n=r$, only $j=n$ occurs, and

$$\frac{1-aq^{2n}}{1-a}\frac{(a)_{2n}}{(aq)_{2n}}=1.$$

Suppose $N=n-r>0$ and put $j=r+s$. Then

$$\frac{(a)_{n+r+s}}{(aq)_{2r+s}}=(1-a)(aq^{2r+s+1};q)_{N-1}.$$

The coefficient becomes

$$\frac{1-aq^{2n}}{(q)_N}\sum_{s=0}^{N}(-1)^{N-s}q^{\binom{N-s}{2}}\begin{bmatrix}N\\ s\end{bmatrix}_q P(q^s),$$

where $P(x)=(aq^{2r+1}x;q)_{N-1}$ has degree $N-1$. It vanishes by Lemma 15.2. Thus the two triangular matrices are inverse. Since every expression is rational in a and the singularity at $a=1$ is removable, the limiting statement follows. □

Corollary 15.4 (Two unit Bailey pairs)**.** *Let* $\beta_n=\delta_{n,0}$.

(i) *Relative to* $a=1$,

$$\alpha_0=1,\qquad \alpha_n=(-1)^nq^{\binom{n}{2}}(1+q^n)\quad (n>0).$$

(ii) *Relative to* $a=q$,

$$\alpha_n=(-1)^nq^{\binom{n}{2}}[2n+1]_q\quad (n\ge 0).$$

Proof. With $\beta_j=\delta_{j,0}$, Bailey inversion gives

$$\alpha_n=\frac{1-aq^{2n}}{1-a}\frac{(a)_n}{(q)_n}(-1)^nq^{\binom{n}{2}}.$$

For $a\to 1$ and $n>0$, $(a)_n/(1-a)\to (q)_{n-1}$, so the prefactor reduces to $(1-q^{2n})/(1-q^n)=1+q^n$. For $a=q$, the ratio $(a)_n/(q)_n$ is 1 and the remaining factor is $[2n+1]_q$. □

15.3 The limiting Bailey lemma

Lemma 15.5 (Auxiliary finite identity)**.** *For* $N\ge 0$,

$$\sum_{k=0}^{N} z^kq^{k(k-1)}\begin{bmatrix}N\\ k\end{bmatrix}_q (zq^k;q)_{N-k}=1.$$

Proof. Call the left side $S_N(z)$. Using the second q-Pascal recurrence and splitting the last factor of $(zq^k;q)_{N+1-k}$ gives

$$S_{N+1}(z)=(1-zq^N)S_N(z)+zq^NS_N(qz).$$

The identity is immediate for $N=0$. If S_N is identically 1, the recurrence gives $S_{N+1}=1-zq^N+zq^N=1$. Induction completes the proof. □

Theorem 15.6 (Limiting Bailey lemma, finite form)**.** *If (α_n, β_n) is a Bailey pair relative to a, define*

$$\alpha'_n = a^n q^{n^2} \alpha_n, \qquad \beta'_n = \sum_{j=0}^{n} \frac{a^j q^{j^2}}{(q)_{n-j}} \beta_j.$$

Then (α'_n, β'_n) is also a Bailey pair relative to a.

Proof. Substitute the Bailey relation for β_j and interchange the finite sums. The coefficient of α_r in β'_n is

$$a^r q^{r^2} \sum_{s=0}^{N} \frac{a^s q^{s^2+2rs}}{(q)_{N-s}(q)_s(aq)_{2r+s}}, \qquad N = n - r.$$

Let $A = aq^{2r+1}$. Since

$$a^s q^{s^2+2rs} = A^s q^{s(s-1)}, \qquad (aq)_{2r+s} = (aq)_{2r}(A)_s,$$

the desired Bailey coefficient $a^r q^{r^2}/((q)_N(aq)_{n+r})$ is equivalent, after clearing factors, to

$$\sum_{s=0}^{N} \begin{bmatrix} N \\ s \end{bmatrix}_q A^s q^{s(s-1)} (Aq^s)_{N-s} = 1.$$

This is Lemma 15.5. □

Corollary 15.7 (Limiting Bailey lemma, infinite form)**.** *If the series converge absolutely, then every Bailey pair relative to a satisfies*

$$\sum_{n=0}^{\infty} a^n q^{n^2} \beta_n = \frac{1}{(aq;q)_\infty} \sum_{n=0}^{\infty} a^n q^{n^2} \alpha_n.$$

Proof. The transformed pair in Theorem 15.6 satisfies

$$\beta'_N = \sum_{r=0}^{N} \frac{\alpha'_r}{(q)_{N-r}(aq)_{N+r}}.$$

Let $N \to \infty$. On the left,

$$\beta'_N \longrightarrow \frac{1}{(q)_\infty} \sum_{j \geq 0} a^j q^{j^2} \beta_j.$$

On the right,

$$\beta'_N \longrightarrow \frac{1}{(q)_\infty (aq)_\infty} \sum_{r \geq 0} a^r q^{r^2} \alpha_r.$$

Absolute convergence permits dominated convergence. Cancel $(q)_\infty^{-1}$. □

15.4 The Rogers–Ramanujan identities

Theorem 15.8 (Rogers–Ramanujan)**.** *For $|q| < 1$,*

$$\sum_{n=0}^{\infty} \frac{q^{n^2}}{(q;q)_n} = \frac{1}{(q;q^5)_\infty (q^4;q^5)_\infty}, \tag{15.1}$$

$$\sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(q;q)_n} = \frac{1}{(q^2;q^5)_\infty (q^3;q^5)_\infty}. \tag{15.2}$$

Proof. For the first identity, begin with the unit pair relative to $a = 1$ in Corollary 15.4. Apply the finite limiting lemma once. The new pair is

$$\alpha'_n = q^{n^2}\alpha_n, \qquad \beta'_n = \frac{1}{(q)_n}.$$

Applying the infinite limiting lemma to this new pair gives

$$\begin{aligned}\sum_{n\geq 0}\frac{q^{n^2}}{(q)_n} &= \frac{1}{(q)_\infty}\left(1+\sum_{r\geq 1}(-1)^r q^{2r^2+\binom{r}{2}}(1+q^r)\right)\\ &= \frac{1}{(q)_\infty}\sum_{r\in\mathbb{Z}}(-1)^r q^{(5r^2-r)/2}.\end{aligned}$$

By Jacobi's triple product with base q^5 and $z = q^2$,

$$\sum_{r\in\mathbb{Z}}(-1)^r q^{(5r^2-r)/2} = (q^2, q^3, q^5; q^5)_\infty.$$

Dissection gives

$$(q;q)_\infty = (q, q^2, q^3, q^4, q^5; q^5)_\infty,$$

which proves (15.1).

For the second identity use the unit pair relative to $a = q$. After one finite limiting step,

$$\alpha'_n = (-1)^n q^{n^2+n+\binom{n}{2}}[2n+1]_q, \qquad \beta'_n = \frac{1}{(q)_n}.$$

The infinite limiting lemma gives

$$\begin{aligned}\sum_{n\geq 0}\frac{q^{n^2+n}}{(q)_n} &= \frac{1}{(q^2;q)_\infty}\sum_{r\geq 0}q^{r^2+r}\alpha'_r\\ &= \frac{1}{(q;q)_\infty}\sum_{r\geq 0}(-1)^r q^{(5r^2+3r)/2}(1-q^{2r+1})\\ &= \frac{1}{(q;q)_\infty}\sum_{r\in\mathbb{Z}}(-1)^r q^{(5r^2+3r)/2}.\end{aligned}$$

In the last step, the subtracted term indexed by $r \geq 0$ is the term with bilateral index $-r-1$. Jacobi's triple product with base q^5 and $z = q^4$ gives

$$\sum_{r\in\mathbb{Z}}(-1)^r q^{(5r^2+3r)/2} = (q, q^4, q^5; q^5)_\infty.$$

Divide by the fivefold dissection of $(q;q)_\infty$ to obtain (15.2). □

Remark 15.9 (Partition interpretation). The product side of (15.1) counts partitions into parts congruent to 1 or 4 modulo 5; the product side of (15.2) counts parts congruent to 2 or 3 modulo 5. The sum sides count partitions with successive parts differing by at least 2, with an additional smallest-part condition in the second identity. Establishing the latter combinatorial interpretations requires a separate bijective or finite-polynomial argument; the proof above instead reveals the hypergeometric engine.

Part IV

Cyclotomic arithmetic and extended indices

Chapter 16

Cyclotomic factorization and congruences

The arithmetic of Gaussian coefficients is governed by cyclotomic polynomials. This is the precise sense in which a primitive root of unity plays the role of a prime. The point of this chapter is that the essential congruences do not require mysterious cancellation: they follow from product factorization, coefficient extraction, and a careful accounting of blocks.

16.1 Cyclotomic valuations and carries

Let $\Phi_d(q)$ denote the dth cyclotomic polynomial, normalized by

$$q^n - 1 = \prod_{d|n} \Phi_d(q).$$

For a nonzero polynomial $F \in \mathbb{Z}[q]$, write $\nu_d(F)$ for the exponent of $\Phi_d(q)$ in its factorization.

Theorem 16.1 (Cyclotomic factorization of a shifted factorial)**.** *For every $n \geq 0$,*

$$(q;q)_n = (-1)^n \prod_{d=1}^{n} \Phi_d(q)^{\lfloor n/d \rfloor}.$$

Consequently,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{d=1}^{n} \Phi_d(q)^{e_d(n,k)}, \qquad e_d(n,k) = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor. \tag{16.1}$$

Moreover $e_d(n,k)$ is always either 0 *or* 1.

Proof. Since $1 - q^j = -(q^j - 1)$ and $q^j - 1 = \prod_{d|j} \Phi_d(q)$,

$$(q;q)_n = \prod_{j=1}^{n} (1 - q^j) = (-1)^n \prod_{j=1}^{n} \prod_{d|j} \Phi_d(q).$$

For fixed d, the multiples of d among $1, \ldots, n$ are $d, 2d, \ldots, \lfloor n/d \rfloor d$, so the exponent of Φ_d is $\lfloor n/d \rfloor$. Dividing the factorizations of $(q;q)_n$, $(q;q)_k$, and $(q;q)_{n-k}$ gives (16.1); the signs cancel.

Write $k = ad + r$ and $n - k = bd + s$ with $0 \leq r, s < d$. Then

$$e_d(n,k) = \lfloor (r+s)/d \rfloor,$$

which is 0 or 1. □

Corollary 16.2 (Square-free cyclotomic structure)**.** *Every Gaussian polynomial is a product of distinct cyclotomic polynomials. More precisely,*

$$\Phi_d(q) \mid \begin{bmatrix} n \\ k \end{bmatrix}_q \quad \Longleftrightarrow \quad (k \bmod d) > (n \bmod d). \tag{16.2}$$

Equivalently, Φ_d occurs exactly when adding k and $n-k$ produces a carry in the units place in base d.

Proof. The first assertion follows because every exponent in (16.1) belongs to $\{0,1\}$. If $n = cd + t$ and $k = ad + r$ with $0 \le r, t < d$, then the residue of $n-k$ is $t-r$ when $r \le t$ and $t - r + d$ when $r > t$. Thus the two residues add to t in the first case and to $t + d$ in the second. The latter is precisely the carry case, and by the proof of Theorem 16.1 it is equivalent to $e_d(n,k) = 1$. □

Example 16.3. For $n = 10$ and $k = 4$, the values of d for which $4 \bmod d > 10 \bmod d$ are $d = 5, 7, 8, 9$. Hence

$$\begin{bmatrix} 10 \\ 4 \end{bmatrix}_q = \Phi_5(q)\Phi_7(q)\Phi_8(q)\Phi_9(q).$$

The degrees add to $4 + 6 + 4 + 6 = 20 = 4(10-4)$, as required by reciprocity.

Corollary 16.4 (MacMahon's q-Catalan polynomial)**.** *For $n \ge 0$ put*

$$C_n(q) := \frac{1}{[n+1]_q} \begin{bmatrix} 2n \\ n \end{bmatrix}_q.$$

Then $C_n(q) \in \mathbb{Z}[q]$; it has degree $n(n-1)$, and $C_n(1) = \frac{1}{n+1}\binom{2n}{n}$.

Proof. Since $q^N - 1 = \prod_{d \mid N} \Phi_d(q)$ and $\Phi_1(q) = q - 1$,

$$[n+1]_q = \frac{q^{n+1}-1}{q-1} = \prod_{\substack{d \mid n+1 \\ d>1}} \Phi_d(q).$$

Hence $\nu_d([n+1]_q) = 1$ exactly for those $d > 1$ dividing $n+1$, and $\nu_d([n+1]_q) = 0$ for every other d, the case $d = 1$ included: indeed $[n+1]_q$ takes the value $n + 1 \ne 0$ at $q = 1$, so $q - 1$ does not divide it.

Let $d > 1$ divide $n + 1$. Then $n \equiv -1 \pmod d$, so $n \bmod d = d - 1$, while $2n \equiv -2 \pmod d$ together with $0 \le d - 2 < d$ gives $2n \bmod d = d - 2$. Therefore

$$n \bmod d = d - 1 > d - 2 = 2n \bmod d,$$

and the carry criterion (16.2) of Corollary 16.2, applied with upper index $2n$ and lower index n, gives $\Phi_d(q) \mid \left[\begin{smallmatrix} 2n \\ n \end{smallmatrix}\right]_q$, that is, $\nu_d\big(\left[\begin{smallmatrix} 2n \\ n \end{smallmatrix}\right]_q\big) = 1$. Combining the two paragraphs,

$$\nu_d\Big(\begin{bmatrix} 2n \\ n \end{bmatrix}_q\Big) \ge \nu_d\big([n+1]_q\big) \qquad \text{for every } d \ge 1,$$

so $[n+1]_q$ divides $\left[\begin{smallmatrix} 2n \\ n \end{smallmatrix}\right]_q$ in $\mathbb{Q}[q]$ and $C_n \in \mathbb{Q}[q]$. Moreover $[n+1]_q = 1 + q + \cdots + q^n$ is monic and $\left[\begin{smallmatrix} 2n \\ n \end{smallmatrix}\right]_q \in \mathbb{Z}[q]$ by Corollary 2.4, so division with remainder in $\mathbb{Z}[q]$ produces a quotient in $\mathbb{Z}[q]$; by uniqueness of the quotient in $\mathbb{Q}[q]$ that quotient is C_n, whence $C_n \in \mathbb{Z}[q]$.

Finally, $\deg \left[\begin{smallmatrix} 2n \\ n \end{smallmatrix}\right]_q = n \cdot n = n^2$ by Theorem 2.5(iii) and $\deg [n+1]_q = n$, so $\deg C_n = n^2 - n$. Evaluating the polynomial identity $[n+1]_q\, C_n(q) = \left[\begin{smallmatrix} 2n \\ n \end{smallmatrix}\right]_q$ at $q = 1$ and using Corollary 2.6 gives $(n+1)C_n(1) = \binom{2n}{n}$. □

Remark 16.5. The argument above establishes integrality and says nothing about signs. The coefficients of $C_n(q)$ are in fact nonnegative — C_n enumerates the Dyck paths of semilength n by a suitable major-index statistic, for which see [30] — but that is a combinatorial theorem and not a consequence of cyclotomic divisibility. Compare Corollary 5.5, where the corresponding statistic for unrestricted monotone lattice paths is the area and the enumeration is the unconstrained $\left[\begin{smallmatrix} 2n \\ n \end{smallmatrix}\right]_q$.

16.2 The q-Lucas congruence

The following theorem is often called the q-Lucas theorem. It is naturally proved by grouping a finite product into complete root-of-unity blocks.

Lemma 16.6 (A complete root-of-unity block)**.** *If ζ is a primitive dth root of unity, then*

$$\prod_{j=0}^{d-1}(1+x\zeta^j) = 1-(-x)^d. \tag{16.3}$$

Proof. The roots in x of the left side are $-\zeta^{-j}$, which are exactly the roots of $1-(-x)^d$. Both polynomials have constant term 1, so they are equal. ◻

Theorem 16.7 (q-Lucas)**.** *Let $a, r \geq 0$, let $d \geq 1$, and let $0 \leq b, s < d$. Then*

$$\begin{bmatrix} ad+b \\ rd+s \end{bmatrix}_q \equiv \binom{a}{r}\begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}. \tag{16.4}$$

Here the right side is understood as 0 when $r > a$ or $s > b$.

Proof. It suffices to prove equality after substituting an arbitrary primitive dth root ζ for q, because Φ_d is the minimal polynomial of ζ over $\mathbb{Q}$.

The finite q-binomial theorem gives

$$\prod_{j=0}^{ad+b-1}(1+x\zeta^j) = \sum_{h=0}^{ad+b}\zeta^{\binom{h}{2}}\begin{bmatrix} ad+b \\ h \end{bmatrix}_\zeta x^h. \tag{16.5}$$

Group the product into a complete blocks and one incomplete block. By Lemma 16.6,

$$\prod_{j=0}^{ad+b-1}(1+x\zeta^j) = \left(1-(-x)^d\right)^a\prod_{j=0}^{b-1}(1+x\zeta^j).$$

The coefficient of x^{rd+s} on the right is

$$\binom{a}{r}(-1)^{r(d+1)}\zeta^{\binom{s}{2}}\begin{bmatrix} b \\ s \end{bmatrix}_\zeta. \tag{16.6}$$

Comparing (16.5) and (16.6) leaves only the phase. We claim that

$$(-1)^{r(d+1)}\zeta^{\binom{s}{2}-\binom{rd+s}{2}} = 1. \tag{16.7}$$

Indeed,

$$2\left(\binom{rd+s}{2}-\binom{s}{2}\right) = rd(rd+2s-1).$$

If d is odd, $r(rd+2s-1)$ is even, so the exponent difference is a multiple of d, while $r(d+1)$ is even. If d is even, then $\zeta^{d/2} = -1$ and $rd+2s-1$ is odd; hence the second factor in (16.7) equals $(-1)^r$, as does the first. Their product is 1. The coefficient comparison therefore gives (16.4) at $q = \zeta$, completing the proof. ◻

Corollary 16.8 (Evaluation at $q = -1$)**.** *For* $0 \le k \le n$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Bigg|_{q=-1} = \begin{cases} \dbinom{n/2}{k/2}, & n \text{ even and } k \text{ even}, \\ 0, & n \text{ even and } k \text{ odd}, \\ \dbinom{(n-1)/2}{\lfloor k/2 \rfloor}, & n \text{ odd}. \end{cases}$$

Proof. Apply Theorem 16.7 with $d = 2$. Write $n = 2a + b$ and $k = 2r + s$, where $b, s \in \{0, 1\}$. Since $\left[\begin{smallmatrix} b \\ s \end{smallmatrix}\right]_{-1}$ is 1 when $s \le b$ and 0 otherwise, the three cases follow. □

16.3 Cyclic sieving for subsets

A triple $(X, C, F(q))$, where a finite cyclic group $C = \langle c \rangle$ acts on a finite set X and $F(q) \in \mathbb{Z}[q]$, exhibits the *cyclic sieving phenomenon* when

$$\#\,\mathrm{Fix}(c^j) = F(\omega^j)$$

for every j, where ω is a primitive $|C|$th root of unity.

Theorem 16.9 (Cyclic sieving for k-subsets)**.** *Let X be the set of k-element subsets of $\mathbb{Z}/n\mathbb{Z}$, and let C_n act by translation. Then*

$$\left(X, C_n, \begin{bmatrix} n \\ k \end{bmatrix}_q\right)$$

exhibits the cyclic sieving phenomenon.

Proof. Fix j, and let d be the order of translation by j. Its orbits on $\mathbb{Z}/n\mathbb{Z}$ all have size d, and there are n/d such orbits. A subset is fixed precisely when it is a union of these orbits. Thus

$$\#\,\mathrm{Fix}(c^j) = \begin{cases} \dbinom{n/d}{k/d}, & d \mid k, \\ 0, & d \nmid k. \end{cases} \tag{16.8}$$

The number ω^j is a primitive dth root. Write $n = (n/d)d$ and $k = rd + s$ with $0 \le s < d$. By q-Lucas,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Bigg|_{q=\omega^j} = \binom{n/d}{r} \begin{bmatrix} 0 \\ s \end{bmatrix}_{\omega^j}.$$

This is 0 if $s > 0$, and it is $\binom{n/d}{k/d}$ if $s = 0$, exactly as in (16.8). □

16.4 A q-Babbage congruence modulo a cyclotomic square

The modulus $\Phi_n(q)^2$ records not only the value at a primitive nth root, but also first-order contact there. The next congruence is a particularly transparent example. Its block proof is also a useful template for stronger q-congruences; see Andrews [3] and Straub [31].

Theorem 16.10 (q-Babbage congruence)**.** *For integers $a \ge b \ge 0$ and $n \ge 1$,*

$$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}} \pmod{\Phi_n(q)^2}. \tag{16.9}$$

Proof. Split the finite product into a consecutive blocks of length n:

$$\prod_{j=0}^{an-1}(1+xq^j)=\prod_{i=0}^{a-1}\prod_{j=0}^{n-1}(1+xq^{in+j}).$$

Apply the finite q-binomial theorem to each block. Comparing the coefficient of x^{bn} gives

$$q^{\binom{bn}{2}}\begin{bmatrix}an\\bn\end{bmatrix}_q=\sum_{\substack{0\le k_0,\dots,k_{a-1}\le n\\k_0+\cdots+k_{a-1}=bn}}q^{\sum_i\binom{k_i}{2}+n\sum_i ik_i}\prod_{i=0}^{a-1}\begin{bmatrix}n\\k_i\end{bmatrix}_q. \tag{16.10}$$

If $0<k_i<n$, then $\Phi_n(q)$ divides $\left[\begin{smallmatrix}n\\k_i\end{smallmatrix}\right]_q$ by Corollary 16.2. A tuple contributing to (16.10) cannot have exactly one such intermediate entry: all other entries would be 0 or n, making the sum of the k_i nonzero modulo n. Therefore every tuple containing an intermediate entry contributes a multiple of $\Phi_n(q)^2$.

Modulo $\Phi_n(q)^2$, only tuples with each $k_i\in\{0,n\}$ survive. Such a tuple is specified by a b-element subset $S\subseteq\{0,1,\dots,a-1\}$. After division by $q^{\binom{bn}{2}}$, its exponent is

$$n^2\sum_{i\in S}i+b\binom{n}{2}-\binom{bn}{2}=n^2\left(\sum_{i\in S}i-\binom{b}{2}\right).$$

Hence

$$\begin{bmatrix}an\\bn\end{bmatrix}_q\equiv\sum_{\substack{S\subseteq\{0,\dots,a-1\}\\|S|=b}}(q^{n^2})^{\sum_{i\in S}i-\binom{b}{2}}\pmod{\Phi_n(q)^2}.$$

The subset form of the finite q-binomial theorem identifies the sum as $\left[\begin{smallmatrix}a\\b\end{smallmatrix}\right]_{q^{n^2}}$. □

Remark 16.11 (Why the square appears). The decisive observation in the proof is combinatorial: because the target degree is a multiple of the block length, nontrivial block choices occur in pairs. Each nontrivial block supplies one factor $\Phi_n(q)$, so two of them supply the square. Congruences modulo Φ_n^3 require genuinely finer first- and second-order information; the q-Ljunggren congruences of Straub and subsequent refinements belong to that deeper layer.

Corollary 16.12 (Primitive-root value and derivative)**.** *Let ζ be a primitive nth root of unity. Then*

$$\begin{bmatrix}an\\bn\end{bmatrix}_\zeta=\binom{a}{b},$$

and the derivatives at $q=\zeta$ of the two sides of (16.9) *are equal.*

Proof. The first assertion follows either from q-Lucas or from (16.9), because $\zeta^{n^2}=1$. If a polynomial is divisible by Φ_n^2, it has a zero of order at least two at every primitive nth root, so both its value and first derivative vanish there. □

Chapter 17

Negative upper indices and geometric Newton interpolation

Ordinary binomial coefficients acquire a useful extension to negative upper arguments. Gaussian coefficients do as well, but the powers of q are essential: omitting them destroys the Pascal recurrences. The same algebra leads naturally to interpolation on a geometric grid.

17.1 Generalized Gaussian coefficients

Definition 17.1. For an integer N and $k \geq 0$, define

$$\begin{bmatrix} N \\ k \end{bmatrix}_q := \frac{(q^{N-k+1}; q)_k}{(q; q)_k}. \tag{17.1}$$

For $N \geq 0$ this agrees with the usual Gaussian coefficient, including the value 0 when $k > N$.

Proposition 17.2 (Negative-index reflection)**.** *For $m \geq 1$ and $k \geq 0$,*

$$\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk-\binom{k}{2}} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q. \tag{17.2}$$

Proof. From the definition,

$$\begin{aligned}
(q^{-m-k+1}; q)_k &= \prod_{j=0}^{k-1} \left(1 - q^{-(m+k-1-j)}\right) \\
&= (-1)^k q^{-\sum_{j=0}^{k-1}(m+k-1-j)} \prod_{j=0}^{k-1} (1 - q^{m+k-1-j}) \\
&= (-1)^k q^{-mk-\binom{k}{2}} (q^m; q)_k.
\end{aligned}$$

After division by $(q; q)_k$, the final ratio is $\left[\begin{smallmatrix} m+k-1 \\ k \end{smallmatrix}\right]_q$. □

Theorem 17.3 (Pascal recurrences for all integral upper indices)**.** *For every integer N and every $k \geq 1$,*

$$\begin{aligned}
\begin{bmatrix} N \\ k \end{bmatrix}_q &= \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} N-1 \\ k \end{bmatrix}_q, &(17.3) \\
&= q^{N-k} \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} N-1 \\ k \end{bmatrix}_q. &(17.4)
\end{aligned}$$

Both formulas are identities in $\mathbb{Q}(q)$.

Proof. Put the two terms on the right of (17.3) over the common denominator $(q;q)_k$. Since

$$\begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q = \frac{(q^{N-k+1};q)_{k-1}(1-q^k)}{(q;q)_k}$$

the numerator of the right side is

$$\begin{aligned}(q^{N-k+1};q)_{k-1}&\left[(1-q^k)+q^k(1-q^{N-k})\right]\\ &=(q^{N-k+1};q)_{k-1}(1-q^N)=(q^{N-k+1};q)_k,\end{aligned}$$

which proves (17.3). For (17.4), the corresponding bracket is

$$q^{N-k}(1-q^k)+(1-q^{N-k})=1-q^N.$$

The same common-denominator calculation applies. □

Corollary 17.4 (Reciprocal product series)**.** *For* $m \geq 1$ *and* $|z| < 1$*,*

$$\frac{1}{(z;q)_m}=\sum_{k=0}^{\infty}\begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q z^k=\sum_{k=0}^{\infty}(-1)^k q^{mk+\binom{k}{2}}\begin{bmatrix} -m \\ k \end{bmatrix}_q z^k. \tag{17.5}$$

The first equality also holds in the formal power-series ring $\mathbb{Z}[q][[z]]$.

Proof. The reciprocal finite q-binomial theorem, already proved in Theorem 3.4, gives the first expansion. Substitute (17.2) to obtain the second. Formal validity follows because each coefficient of z^k is a polynomial and multiplication by the finite product $(z;q)_m$ can be checked coefficientwise. □

17.2 An arbitrary upper parameter

When a branch of q^α has been fixed, it is often convenient to define

$$\begin{bmatrix} \alpha \\ k \end{bmatrix}_q = \frac{(q^{\alpha-k+1};q)_k}{(q;q)_k}. \tag{17.6}$$

The proofs of the Pascal recurrences are purely algebraic in q^α and therefore remain valid. A related, and usually more analytic, expansion is the following.

Theorem 17.5 (Generalized upper-parameter series)**.** *For* $|q| < 1$*,* $|z| < 1$*, and any parameter* α *for which the factors are defined,*

$$\frac{(q^{\alpha+1}z;q)_\infty}{(z;q)_\infty}=\sum_{k=0}^{\infty}\begin{bmatrix} \alpha+k \\ k \end{bmatrix}_q z^k. \tag{17.7}$$

Proof. Apply the infinite q-binomial theorem with $a = q^{\alpha+1}$:

$$\frac{(q^{\alpha+1}z;q)_\infty}{(z;q)_\infty}=\sum_{k\geq 0}\frac{(q^{\alpha+1};q)_k}{(q;q)_k}z^k.$$

The coefficient is exactly $\left[\begin{smallmatrix} \alpha+k \\ k \end{smallmatrix}\right]_q$ by (17.6). □

17.3 Two complex arguments and generalized binomial series

The extension in Definition 1.13 also permits both arguments of a Gaussian coefficient to be complex. Throughout this section take $0 < q < 1$ and set $q^z = \exp(z \log q)$, using the real logarithm of q.

Definition 17.6. For $\alpha, \beta \in \mathbb{C}$, define

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q := \frac{(q^{\beta+1}, q^{\alpha-\beta+1}; q)_\infty}{(q, q^{\alpha+1}; q)_\infty}, \tag{17.8}$$

whenever the denominator is nonzero.

Theorem 17.7 (Gamma representation and Pascal laws)**.** *Wherever the expressions are defined,*

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q = \frac{\Gamma_q(\alpha+1)}{\Gamma_q(\beta+1)\Gamma_q(\alpha-\beta+1)}, \tag{17.9}$$

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q = \begin{bmatrix} \alpha \\ \alpha-\beta \end{bmatrix}_q, \tag{17.10}$$

$$\begin{bmatrix} \alpha+1 \\ \beta \end{bmatrix}_q = q^\beta \begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q + \begin{bmatrix} \alpha \\ \beta-1 \end{bmatrix}_q, \tag{17.11}$$

$$= \begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q + q^{\alpha+1-\beta} \begin{bmatrix} \alpha \\ \beta-1 \end{bmatrix}_q. \tag{17.12}$$

If $\beta = k \in \mathbb{N}_0$, this definition agrees with (17.6).

Proof. Substitution of the product definition of Γ_q proves (17.9); all powers of $1-q$ cancel. Symmetry is immediate either from that quotient or directly from (17.8). For $\beta = k \in \mathbb{N}_0$, tail cancellation gives

$$\frac{(q^{k+1}; q)_\infty}{(q; q)_\infty} = \frac{1}{(q; q)_k}, \qquad \frac{(q^{\alpha-k+1}; q)_\infty}{(q^{\alpha+1}; q)_\infty} = (q^{\alpha-k+1}; q)_k,$$

which is exactly (17.6).

For the first Pascal law, divide every term by $\left[\begin{smallmatrix} \alpha \\ \beta \end{smallmatrix}\right]_q$. The gamma functional equation gives

$$\frac{\left[\begin{smallmatrix} \alpha+1 \\ \beta \end{smallmatrix}\right]_q}{\left[\begin{smallmatrix} \alpha \\ \beta \end{smallmatrix}\right]_q} = \frac{[\alpha+1]_q}{[\alpha-\beta+1]_q}, \qquad \frac{\left[\begin{smallmatrix} \alpha \\ \beta-1 \end{smallmatrix}\right]_q}{\left[\begin{smallmatrix} \alpha \\ \beta \end{smallmatrix}\right]_q} = \frac{[\beta]_q}{[\alpha-\beta+1]_q}.$$

The identity $[\alpha+1]_q = q^\beta[\alpha-\beta+1]_q + [\beta]_q$ proves (17.11). The alternative identity $[\alpha+1]_q = [\alpha-\beta+1]_q + q^{\alpha+1-\beta}[\beta]_q$ proves (17.12). □

Corollary 17.8 (Classical generalized binomial)**.** *If $\alpha > \beta > -1$ are real, then*

$$\lim_{q\to1^-} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q = \frac{\Gamma(\alpha+1)}{\Gamma(\beta+1)\Gamma(\alpha-\beta+1)}.$$

Proof. Apply the locally uniform q-gamma limit in Theorem 12.5 to (17.9). □

Theorem 17.9 (Generalized finite and reciprocal q-binomial theorems)**.** *Let $0 < q < 1$. If $|zq^\alpha| < 1$, then*

$$(z; q)_\alpha = \sum_{k=0}^{\infty} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} \alpha \\ k \end{bmatrix}_q z^k. \tag{17.13}$$

If $|z| < 1$, *then*

$$\frac{1}{(z;q)_\alpha} = \sum_{k=0}^{\infty} \begin{bmatrix} \alpha + k - 1 \\ k \end{bmatrix}_q z^k. \tag{17.14}$$

For a nonnegative integral α, *the first series terminates and becomes Euler's finite theorem; in that case the restriction* $|zq^\alpha| < 1$ *can be removed because both sides are polynomials. For a positive integral* α, *the second becomes the reciprocal finite theorem.*

Proof. Apply the infinite q-binomial theorem with $a = q^{-\alpha}$ and argument zq^α:

$$\frac{(z;q)_\infty}{(zq^\alpha;q)_\infty} = \sum_{k\geq 0} \frac{(q^{-\alpha};q)_k}{(q;q)_k}(zq^\alpha)^k.$$

The left side is $(z;q)_\alpha$. Reversing the finite numerator product gives

$$\frac{(q^{-\alpha};q)_k}{(q;q)_k} q^{\alpha k} = (-1)^k q^{\binom{k}{2}} \begin{bmatrix} \alpha \\ k \end{bmatrix}_q,$$

which proves (17.13).

For the reciprocal formula, apply the infinite theorem with $a = q^\alpha$ and argument z:

$$\frac{(q^\alpha z;q)_\infty}{(z;q)_\infty} = \sum_{k\geq 0} \frac{(q^\alpha;q)_k}{(q;q)_k} z^k.$$

The left side is $(z;q)_\alpha^{-1}$ and the coefficient is $\begin{bmatrix} \alpha+k-1 \\ k \end{bmatrix}_q$. The integral specializations follow from the consistency results already proved. □

17.4 Newton interpolation on the grid $1, q, q^2, \ldots$

The usual Newton basis is adapted to an arbitrary sequence of distinct nodes. At geometric nodes its basis polynomials are q-Pochhammer symbols.

Theorem 17.10 (Geometric Newton interpolation)**.** *Assume* $q^i \neq q^j$ *whenever* $0 \leq i < j \leq n$. *For arbitrary values* $y_0, \ldots, y_n$ *there is a unique polynomial* P *of degree at most* n *satisfying*

$$P(q^j) = y_j \qquad (0 \leq j \leq n).$$

It has the Newton expansion

$$P(x) = \sum_{k=0}^{n} c_k \prod_{r=0}^{k-1} (x - q^r) = \sum_{k=0}^{n} c_k (-1)^k q^{\binom{k}{2}} (x;q^{-1})_k, \tag{17.15}$$

where

$$c_k = \sum_{j=0}^{k} \frac{y_j}{\prod_{\substack{0\leq r\leq k \\ r\neq j}} (q^j - q^r)} \tag{17.16}$$

$$= \sum_{j=0}^{k} \frac{(-1)^j q^{-jk+\binom{j+1}{2}}}{(q;q)_j (q;q)_{k-j}} y_j. \tag{17.17}$$

Proof. For distinct nodes $x_0, \dots, x_n$, Lagrange interpolation gives existence and uniqueness. The coefficient of the Newton basis polynomial $\prod_{r=0}^{k-1}(x - x_r)$ is the divided difference

$$y[x_0, \dots, x_k] = \sum_{j=0}^{k} \frac{y_j}{\prod_{r \neq j}(x_j - x_r)};$$

this follows by comparing the coefficient of x^k in the Lagrange interpolant through the first $k + 1$ nodes. Taking $x_j = q^j$ proves (17.16).

It remains to simplify the denominator. Splitting the factors below and above j gives

$$\begin{aligned}\prod_{\substack{0 \le r \le k \\ r \neq j}} (q^j - q^r) &= \left[\prod_{r=0}^{j-1} -q^r(1 - q^{j-r})\right]\left[\prod_{r=j+1}^{k} q^j(1 - q^{r-j})\right] \\ &= (-1)^j q^{\binom{j}{2} + j(k-j)} (q;q)_j (q;q)_{k-j} \\ &= (-1)^j q^{jk - \binom{j+1}{2}} (q;q)_j (q;q)_{k-j}.\end{aligned}$$

Taking reciprocals yields (17.17). Finally,

$$\prod_{r=0}^{k-1}(x - q^r) = (-1)^k q^{\binom{k}{2}} \prod_{r=0}^{k-1}(1 - xq^{-r}) = (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k,$$

which proves the second form of (17.15). □

Corollary 17.11 (Triangular reconstruction)**.** *The coefficients in* (17.15) *can be recovered successively by*

$$c_0 = y_0, \qquad c_k = \frac{y_k - \sum_{r=0}^{k-1} c_r \prod_{j=0}^{r-1}(q^k - q^j)}{\prod_{j=0}^{k-1}(q^k - q^j)}. \tag{17.18}$$

Proof. Evaluate (17.15) at $x = q^k$. Every term with index greater than k vanishes, while the coefficient multiplying c_k is nonzero by the distinct-node hypothesis. Solving for c_k gives the formula. □

Remark 17.12. Kupershmidt's q-Newton viewpoint emphasizes that the classical binomial theorem, Newton interpolation, and q-difference calculus are not separate constructions: each is controlled by a basis whose successive factors form a geometric progression. The finite formula above is algebraic and needs no convergence hypothesis; infinite Newton series require additional growth conditions on the data.

Part V

Central sums, limits, and effective methods

Chapter 18

Central q-binomial sums and Lambert series

Central Gaussian coefficients frequently become tractable only after a seemingly extraneous factor is inserted. The elementary identity in the next lemma shows why the factor $(-q;q)_{2k}$ is natural: it converts a central coefficient at base q^2 into a single quotient of shifted factorials. This chapter develops both a regular q-beta evaluation and a deleted-singularity identity proved recently by Chern, Dilcher, and Jiu [7].

18.1 The central-factor reduction

Lemma 18.1. *For every $k \geq 0$,*

$$\begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{1}{(-q;q)_{2k}} = \frac{(q;q^2)_k}{(q^2;q^2)_k}. \tag{18.1}$$

Proof. The elementary product identity

$$(q;q)_{2k}(-q;q)_{2k} = (q^2;q^2)_{2k}$$

gives

$$\begin{aligned} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{1}{(-q;q)_{2k}} &= \frac{(q^2;q^2)_{2k}}{(q^2;q^2)_k^2} \frac{(q;q)_{2k}}{(q^2;q^2)_{2k}} \\ &= \frac{(q;q^2)_k (q^2;q^2)_k}{(q^2;q^2)_k^2}, \end{aligned}$$

which is the claimed quotient. □

For a complex parameter z and $0 < q < 1$, we use the analytic extension

$$[z]_q = \frac{1-q^z}{1-q},$$

with the real logarithm in $q^z = \exp(z \log q)$.

18.2 A q-beta evaluation

Theorem 18.2 (A regular central q-binomial sum)**.** *Let $0 < q < 1$, and let $\alpha \in \mathbb{C}$ be such that none of the denominators below vanishes. Then*

$$\sum_{k=0}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q;q)_{2k}[2k+1+\alpha]_q} = \frac{\Gamma_{q^2}(3/2)\,\Gamma_{q^2}((\alpha+1)/2)}{\Gamma_{q^2}((\alpha+2)/2)}. \tag{18.2}$$

In particular, for real α it is enough to exclude $\alpha = -1, -3, -5, \ldots$.

Proof. By Lemma 18.1 and the telescoping quotient

$$\frac{1}{[2k+1+\alpha]_q} = \frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+1};q^2)_k}{(q^{\alpha+3};q^2)_k},$$

the left side is

$$\frac{1}{[\alpha+1]_q}\, {}_2\phi_1\left[\begin{matrix} q, q^{\alpha+1} \\ q^{\alpha+3} \end{matrix}; q^2, q\right].$$

The q-Gauss sum from Theorem 9.4, with base q^2, gives

$$\frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+2}, q^2; q^2)_\infty}{(q^{\alpha+3}, q; q^2)_\infty}. \tag{18.3}$$

Now

$$(q^{\alpha+1};q^2)_\infty = (1-q^{\alpha+1})(q^{\alpha+3};q^2)_\infty, \qquad (q;q^2)_\infty = (1-q)(q^3;q^2)_\infty.$$

Using these two relations to simplify (18.3), we obtain

$$\frac{(q^2;q^2)_\infty (q^{\alpha+2};q^2)_\infty}{(q^3;q^2)_\infty (q^{\alpha+1};q^2)_\infty}. \tag{18.4}$$

Finally insert

$$\Gamma_{q^2}(z) = (1-q^2)^{1-z} \frac{(q^2;q^2)_\infty}{(q^{2z};q^2)_\infty}.$$

In the quotient on the right of (18.2), all powers of $1-q^2$ cancel, and the remaining products are exactly (18.4). ◻

Corollary 18.3 (Classical limit of the regular sum)**.** *For $\alpha \notin \{-1, -3, -5, \ldots\}$,*

$$\sum_{k=0}^{\infty} \binom{2k}{k} \frac{1}{4^k(2k+1+\alpha)} = \frac{\Gamma(3/2)\Gamma((\alpha+1)/2)}{\Gamma((\alpha+2)/2)}. \tag{18.5}$$

Proof. The series is

$$\frac{1}{\alpha+1}\, {}_2F_1\left(\begin{matrix} 1/2, (\alpha+1)/2 \\ (\alpha+3)/2 \end{matrix}; 1\right).$$

Gauss's classical summation gives

$$\frac{1}{\alpha+1} \frac{\Gamma((\alpha+3)/2)\Gamma(1/2)}{\Gamma((\alpha+2)/2)}.$$

Since $\Gamma((\alpha+3)/2) = ((\alpha+1)/2)\Gamma((\alpha+1)/2)$ and $\Gamma(1/2)/2 = \Gamma(3/2)$, this is (18.5). The same identity is also the $q \to 1^-$ limit of Theorem 18.2 by the q-gamma limit in Theorem 12.5. ◻

18.3 Deleting a singular term

The regular formula becomes singular when $\alpha = -2m - 1$. Removing the offending $k = m$ term does not merely produce a finite part of the same gamma quotient; it yields an alternating Lambert-series tail.

Theorem 18.4 (Deleted-singularity central identity)**.** *Let* $m \geq 0$ *and* $0 < q < 1$. *Then*

$$\sum_{\substack{k\geq 0\\ k\neq m}} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q;q)_{2k}[k-m]_{q^2}} = \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1+q)q^m}{(-q;q)_{2m}} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1}q^r}{[r]_q}. \tag{18.6}$$

Proof. By Lemma 18.1, the left side is

$$(1-q^2)\sum_{\substack{k\geq 0\\ k\neq m}} \frac{(q;q^2)_k q^k}{(q^2;q^2)_k(1-q^{2k-2m})}. \tag{18.7}$$

Introduce an auxiliary parameter a and define

$$F(a,q) := \frac{1}{1-aq^{-2m}}\left({}_2\phi_1\left[\begin{matrix} q, aq^{-2m} \\ aq^{-2m+2}\end{matrix}; q^2, q\right] - \frac{(q, aq^{-2m};q^2)_m q^m}{(q^2, aq^{-2m+2};q^2)_m}\right). \tag{18.8}$$

The elementary cancellation

$$\frac{(aq^{-2m};q^2)_k}{(aq^{-2m+2};q^2)_k} = \frac{1-aq^{-2m}}{1-aq^{2k-2m}}$$

shows that

$$F(a,q) = \sum_{\substack{k\geq 0\\ k\neq m}} \frac{(q;q^2)_k q^k}{(q^2;q^2)_k(1-aq^{2k-2m})}. \tag{18.9}$$

Thus (18.7) equals $(1-q^2)\lim_{a\to 1} F(a,q)$.

Apply q-Gauss with base q^2 to the ${}_2\phi_1$ in (18.8). We obtain

$$\begin{aligned} F(a,q) &= \frac{1}{1-aq^{-2m}}\left(\frac{(q^2, aq^{-2m+1};q^2)_\infty}{(q, aq^{-2m+2};q^2)_\infty} - \frac{(q, aq^{-2m};q^2)_m q^m}{(q^2, aq^{-2m+2};q^2)_m}\right) \\ &= \frac{H(a) - C_m}{1-a}, \end{aligned} \tag{18.10}$$

where

$$H(a) = \frac{(q^2, aq;q^2)_\infty (aq^{-2m+1};q^2)_m}{(q, aq^2;q^2)_\infty (aq^{-2m};q^2)_m},$$
$$C_m = \frac{(q;q^2)_m q^m}{(q^2;q^2)_m}.$$

The reversal identity for finite shifted factorials gives

$$\frac{(q^{-2m+1};q^2)_m}{(q^{-2m};q^2)_m} = q^m \frac{(q;q^2)_m}{(q^2;q^2)_m},$$

so $H(1) = C_m$. Hence L'Hospital's rule yields

$$\lim_{a\to 1} F(a,q) = -H'(1) = -H(1)\,\frac{\partial}{\partial a}\log H(a)\bigg|_{a=1}. \tag{18.11}$$

Differentiating the logarithms of the product factors gives

$$\lim_{a\to 1} F(a,q) = C_m\Bigg(\sum_{r=1}^{\infty}\frac{(-1)^{r-1}q^r}{1-q^r} + \sum_{r=1}^{2m}\frac{(-1)^{r-1}q^{-r}}{1-q^{-r}}\Bigg). \tag{18.12}$$

For $r \ge 1$,

$$\frac{q^{-r}}{1-q^{-r}} = -\frac{1}{1-q^r}.$$

Consequently, the two summands in (18.12) combine, for $1 \le r \le 2m$, to $(-1)^r$. Their sum is 0 because $2m$ is even. Therefore

$$\lim_{a\to 1} F(a,q) = C_m \sum_{r=2m+1}^{\infty}\frac{(-1)^{r-1}q^r}{1-q^r}. \tag{18.13}$$

Multiply by $1-q^2 = (1+q)(1-q)$, use $(1-q)/(1-q^r) = 1/[r]_q$, and finally use (18.1) at $k = m$ to rewrite

$$C_m = q^m\frac{(q;q^2)_m}{(q^2;q^2)_m} = q^m\begin{bmatrix}2m\\ m\end{bmatrix}_{q^2}\frac{1}{(-q;q)_{2m}}.$$

The result is exactly (18.6). □

Remark 18.5 (The mechanism). The deleted term is encoded by the auxiliary parameter a. The q-Gauss product has a removable zero-over-zero singularity at $a = 1$, and its logarithmic derivative turns products into Lambert series. This parameter-differentiation mechanism recurs throughout the theory of harmonic-number and Lambert-series refinements of hypergeometric summations.

18.4 Separate finite and infinite components

The deleted sum in Theorem 18.4 naturally splits at the missing index. Each component has a simpler one-sided Lambert-series evaluation.

Theorem 18.6 (Infinite and finite components)**.** *Let $m \ge 0$ and $0 < q < 1$, and put*

$$P_m(q) = \begin{bmatrix}2m\\ m\end{bmatrix}_{q^2}\frac{(1+q)q^m}{(-q;q)_{2m}}.$$

Then

$$\sum_{k=m+1}^{\infty}\begin{bmatrix}2k\\ k\end{bmatrix}_{q^2}\frac{q^k}{(-q;q)_{2k}[k-m]_{q^2}} = P_m(q)\Bigg(\sum_{r=1}^{\infty}\frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty}\frac{q^{2r+2m}}{[2r+2m]_q}\Bigg), \tag{18.14}$$

and

$$\sum_{k=0}^{m-1}\begin{bmatrix}2k\\ k\end{bmatrix}_{q^2}\frac{q^k}{(-q;q)_{2k}[k-m]_{q^2}} = -P_m(q)\sum_{r=1}^{m}\frac{q^{2r-1}}{[2r-1]_q}. \tag{18.15}$$

For $m = 0$, the finite sum and the sum on its right are both empty.

Proof. By Lemma 18.1, shifting $k \mapsto k+m$ in the infinite component gives

$$\sum_{k=m+1}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q;q)_{2k}[k-m]_{q^2}}$$
$$= (1-q^2)q^m \frac{(q;q^2)_m}{(q^2;q^2)_m} \sum_{k=1}^{\infty} \frac{(q^{2m+1};q^2)_k q^k}{(q^{2m+2};q^2)_k(1-q^{2k})}.$$

Introduce

$$U(a,q) = \frac{1}{1-a}\left({}_2\phi_1\left[\begin{matrix} a, q^{2m+1} \\ q^{2m+2} \end{matrix}; q^2, a^{-1}q\right] - 1\right).$$

For a near 1, absolute convergence permits termwise passage to the limit. Since

$$\frac{(a;q^2)_k}{1-a} = (aq^2;q^2)_{k-1} \qquad (k \geq 1),$$

we obtain

$$\lim_{a\to 1} U(a,q) = \sum_{k=1}^{\infty} \frac{(q^{2m+1};q^2)_k q^k}{(q^{2m+2};q^2)_k(1-q^{2k})}.$$

The q-Gauss summation, at base q^2, evaluates the auxiliary series as

$$U(a,q) = \frac{1}{1-a}\left(\frac{(q, a^{-1}q^{2m+2};q^2)_\infty}{(a^{-1}q, q^{2m+2};q^2)_\infty} - 1\right).$$

The product quotient tends to 1 as $a \to 1$. L'Hospital's rule and termwise logarithmic differentiation, justified locally uniformly, give

$$\lim_{a\to 1} U(a,q) = \sum_{r=1}^{\infty} \frac{q^{2r-1}}{1-q^{2r-1}} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{1-q^{2r+2m}}.$$

Multiplication by the prefactor above, followed by $(1-q)/(1-q^j) = 1/[j]_q$ and Lemma 18.1 at $k=m$, proves (18.14).

To obtain the finite component, subtract (18.14) from the full deleted sum (18.6). Indeed,

$$\sum_{j=2m+1}^{\infty} \frac{(-1)^{j-1}q^j}{[j]_q} = \sum_{r=m+1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=m+1}^{\infty} \frac{q^{2r}}{[2r]_q}$$
$$= \sum_{r=1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{[2r+2m]_q} - \sum_{r=1}^{m} \frac{q^{2r-1}}{[2r-1]_q}.$$

The first two sums constitute the infinite component, leaving exactly (18.15). □

18.5 The classical finite part

Define the alternating harmonic number

$$\overline{H}_N = \sum_{r=1}^{N} \frac{(-1)^{r-1}}{r}.$$

Theorem 18.7 (Classical deleted-singularity identity)**.** *For every $m \geq 0$,*

$$\sum_{\substack{k\geq 0\\ k\neq m}} \binom{2k}{k} \frac{1}{4^k(k-m)} = \binom{2m}{m} \frac{\log 2 - \overline{H}_{2m}}{2^{2m-1}}. \tag{18.16}$$

Proof. Put $c_k = (1/2)_k/k! = \binom{2k}{k}/4^k$. For a away from the nonpositive integers, the series

$$T(a) = \sum_{k=0}^{\infty} \frac{c_k}{k+a}$$

converges locally uniformly and defines a meromorphic function. For $\Re a > 0$, termwise integration of $(1-x)^{-1/2} = \sum_{k\geq 0} c_k x^k$ gives

$$T(a) = \int_0^1 x^{a-1}(1-x)^{-1/2}\, \mathrm{d}x = B(a, 1/2) = \frac{\Gamma(a)\Gamma(1/2)}{\Gamma(a+1/2)}. \tag{18.17}$$

Analytic continuation makes (18.17) valid wherever both sides are defined. The desired deleted sum is the finite part at $a = -m$:

$$\lim_{a\to -m} \left(T(a) - \frac{c_m}{a+m} \right). \tag{18.18}$$

Write $a = -m + \varepsilon$. The standard expansions of the gamma function at a negative integer and at a regular point are

$$\Gamma(-m+\varepsilon) = \frac{(-1)^m}{m!\,\varepsilon} \Big(1 + (H_m - \gamma)\varepsilon + O(\varepsilon^2) \Big),$$
$$\Gamma(1/2 - m + \varepsilon) = \Gamma(1/2-m) \Big(1 + \psi(1/2-m)\varepsilon + O(\varepsilon^2) \Big).$$

Moreover,

$$\frac{(-1)^m \Gamma(1/2)}{m!\Gamma(1/2-m)} = c_m.$$

Thus the finite part in (18.18) is

$$c_m\big(H_m - \gamma - \psi(1/2-m)\big). \tag{18.19}$$

Using $\psi(1/2) = -\gamma - 2\log 2$ and the recurrence $\psi(z+1) = \psi(z) + 1/z$, we find

$$\psi(1/2-m) = -\gamma - 2\log 2 + 2\sum_{r=1}^{m} \frac{1}{2r-1}.$$

Since

$$\overline{H}_{2m} = \sum_{r=1}^{m} \frac{1}{2r-1} - \frac{1}{2} H_m,$$

the bracket in (18.19) is $2(\log 2 - \overline{H}_{2m})$. Hence the finite part equals

$$2\frac{\binom{2m}{m}}{4^m}(\log 2 - \overline{H}_{2m}),$$

which is (18.16). ◻

Proposition 18.8 (Limit of the Lambert tail)**.** *For fixed $m \geq 0$,*

$$\lim_{q\to 1^-} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1}q^r}{[r]_q} = \log 2 - \overline{H}_{2m}. \tag{18.20}$$

Proof. For $0 < q < 1$, set $a_r(q) = q^r/[r]_q$. These numbers are positive and decrease with r, since

$$\frac{a_{r+1}(q)}{a_r(q)} = q\frac{[r]_q}{[r+1]_q} < 1.$$

Moreover,

$$a_r(q) = \frac{q^r}{1 + q + \cdots + q^{r-1}} \leq \frac{1}{r},$$

because every summand in the denominator is at least q^{r-1}. The alternating-series remainder after N terms therefore has absolute value at most $a_{N+1}(q) \leq 1/(N+1)$, uniformly in q. For each fixed r, $a_r(q) \to 1/r$ as $q \to 1^-$. First pass to the limit in a finite partial sum and then let $N \to \infty$. The result is the tail of the alternating harmonic series,

$$\sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1}}{r} = \log 2 - \overline{H}_{2m}.$$

□

Chapter 19

Asymptotic regimes and Bernoulli expansions

A q-analogue should explain both the deformation near $q = 1$ and behavior far from it. Gaussian coefficients have especially clean asymptotic descriptions because their logarithms are finite sums of a single elementary function.

19.1 A complete expansion as $q \to 1$

Let the Bernoulli numbers be defined by

$$\frac{x}{e^x - 1} = \sum_{r=0}^{\infty} B_r \frac{x^r}{r!}.$$

Lemma 19.1 (The basic logarithmic expansion)**.** *As $x \to 0$,*

$$\log \frac{1 - e^{-x}}{x} = -\frac{x}{2} + \sum_{r=1}^{\infty} \frac{B_{2r}}{2r(2r)!} x^{2r}. \tag{19.1}$$

The series converges for $|x| < 2\pi$.

Proof. Differentiate the left side:

$$\frac{\mathrm{d}}{\mathrm{d}x} \log \frac{1 - e^{-x}}{x} = \frac{1}{e^x - 1} - \frac{1}{x}.$$

By the Bernoulli generating function,

$$\frac{1}{e^x - 1} - \frac{1}{x} = -\frac{1}{2} + \sum_{r=1}^{\infty} B_{2r} \frac{x^{2r-1}}{(2r)!}.$$

Integrate term by term and use the value 0 at $x = 0$. The nearest nonzero singularities are at $x = \pm 2\pi i$, giving radius 2π. □

Theorem 19.2 (Bernoulli expansion of a Gaussian coefficient)**.** *Let $0 \le k \le n$, put $D = k(n-k)$, and set $q = e^{-t}$. As $t \to 0$,*

$$\log \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}}}{\binom{n}{k}} = -\frac{D}{2} t + \sum_{r=1}^{\infty} \frac{B_{2r} t^{2r}}{2r(2r)!} \sum_{j=1}^{k} \left((n-k+j)^{2r} - j^{2r} \right). \tag{19.2}$$

For fixed n, k, the displayed series converges whenever $|t| < 2\pi/n$.

Proof. From the product formula,

$$\begin{aligned}\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} &= \prod_{j=1}^{k} \frac{1-e^{-(n-k+j)t}}{1-e^{-jt}} \\ &= \binom{n}{k} \prod_{j=1}^{k} \frac{(1-e^{-(n-k+j)t})/((n-k+j)t)}{(1-e^{-jt})/(jt)}.\end{aligned}$$

Take logarithms and apply Lemma 19.1 to every factor. The linear term is

$$-\frac{t}{2}\sum_{j=1}^{k}(n-k) = -\frac{1}{2}k(n-k)t = -\frac{D}{2}t,$$

and the even terms give the sum in (19.2). Every argument of the basic series has modulus below 2π when $|t| < 2\pi/n$, so the finite sum converges there. □

Remark 19.3 (Power-sum form of the coefficients). Let $S_m(N) = \sum_{j=1}^{N} j^m$ denote the mth power sum. For every integer $m \geq 1$, reindexing the first half of an inner sum by $i = n - k + j$ gives

$$\sum_{j=1}^{k}((n-k+j)^m - j^m) = \sum_{i=n-k+1}^{n} i^m - \sum_{j=1}^{k} j^m = S_m(n) - S_m(k) - S_m(n-k).$$

Applied with $m = 2r$ to the inner sum of (19.2), and with $m = 1$ to the sum $\sum_{j=1}^{k}\big((n-k+j)-j\big) = \sum_{j=1}^{k}(n-k)$ that produced the linear term in the proof of Theorem 19.2, this turns the expansion into

$$\log \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}}}{\binom{n}{k}} = -\frac{t}{2}\big(S_1(n) - S_1(k) - S_1(n-k)\big) + \sum_{r=1}^{\infty} \frac{B_{2r}t^{2r}}{2r(2r)!}\big(S_{2r}(n) - S_{2r}(k) - S_{2r}(n-k)\big),$$

still for $|t| < 2\pi/n$, the factor $-\frac{1}{2}$ being the coefficient of x in (19.1). The case $m = 1$ of the display evaluates the linear bracket outright: $S_1(n) - S_1(k) - S_1(n-k) = \sum_{j=1}^{k}(n-k) = k(n-k) = D$.

Three features become visible in this form. First, each coefficient is a difference of Faulhaber polynomials, hence itself a polynomial in n and k with rational coefficients, of total degree exactly $2r+1$: since $S_{2r}(N) = N^{2r+1}/(2r+1) + O(N^{2r})$, the homogeneous part of top degree is $\big(n^{2r+1} - k^{2r+1} - (n-k)^{2r+1}\big)/(2r+1)$, whose coefficient of $n^{2r}k$ is $-\frac{1}{2r+1}\binom{2r+1}{1}(-1) = 1$, so no cancellation lowers the degree. Second, the expression is manifestly invariant under $k \mapsto n-k$, as it must be by the symmetry recorded in Theorem 2.5. Third, at $r = 1$ the identity $S_2(N) = N(N+1)(2N+1)/6$ gives

$$\begin{aligned}S_2(n) - S_2(k) - S_2(n-k) &= \frac{n(n+1)(2n+1) - k(k+1)(2k+1)}{6} \\ &\quad - \frac{(n-k)(n-k+1)(2n-2k+1)}{6} = k(n-k)(n+1) = D(n+1),\end{aligned}$$

the value used in Corollary 19.4 below.

[The second "

Corollary 19.4 (First terms)**.** *With $D = k(n-k)$,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} = \binom{n}{k}\exp\left(-\frac{D}{2}t + \frac{D(n+1)}{24}t^2 + O(t^4)\right), \tag{19.3}$$

$$= \binom{n}{k}\left(1 - \frac{D}{2}t + \left(\frac{D^2}{8} + \frac{D(n+1)}{24}\right)t^2 + O(t^3)\right). \tag{19.4}$$

Proof. For $r = 1$ in (19.2),

$$\sum_{j=1}^{k}\Big((n-k+j)^2 - j^2\Big) = k(n-k)^2 + (n-k)k(k+1)$$
$$= k(n-k)(n+1) = D(n+1).$$

Since $B_2/(2\cdot 2!) = 1/24$, the logarithmic formula follows. Expanding the exponential gives the second formula. □

Remark 19.5 (Reciprocity and centering). The factor $e^{-Dt/2}$ is forced by reciprocity:

$$\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} = e^{-Dt}\begin{bmatrix} n \\ k \end{bmatrix}_{e^{t}}.$$

Therefore the centered quantity $e^{Dt/2}\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{e^{-t}}$ is an even analytic function of t. This explains, before any calculation, why all odd powers beyond the linear term vanish from the logarithm.

19.2 Fixed q and growing parameters

Theorem 19.6 (Fixed-column limit)**.** *If* $0 < |q| < 1$ *and* k *is fixed, then*

$$\lim_{n\to\infty}\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q;q)_k}. \tag{19.5}$$

More precisely,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q;q)_k}\Big(1 + O(|q|^{n-k+1})\Big). \tag{19.6}$$

Proof. The product formula gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1};q)_k}{(q;q)_k}.$$

The numerator tends to 1. Since the number of factors is fixed,

$$\prod_{j=0}^{k-1}(1 - q^{n-k+1+j}) = 1 + O(|q|^{n-k+1}),$$

which proves both assertions. □

Theorem 19.7 (Fixed-q limit with both indices growing)**.** *Let* $0 < |q| < 1$*. For all integers* $0 \le k \le n$*,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{k+1};q)_\infty\,(q^{n-k+1};q)_\infty}{(q;q)_\infty\,(q^{n+1};q)_\infty},$$

and consequently

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q;q)_\infty}\Big(1 + O\big(|q|^{\min(k,n-k)+1}\big)\Big),$$

the implied constant depending only on q*. In particular,*

$$\lim_{\min(k,n-k)\to\infty}\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q;q)_\infty}, \tag{19.7}$$

and for every fixed $r \in \mathbb{Z}$,

$$\lim_{m\to\infty} \begin{bmatrix} 2m \\ m+r \end{bmatrix}_q = \frac{1}{(q;q)_\infty},$$

the coefficient being read for $m \ge |r|$, so that $m + r$ and $m - r$ are admissible indices.

Proof. By Theorem 1.12 applied at $a = q$, the product $(q;q)_\infty$ converges, and it is nonzero: the zeros of $a \mapsto (a;q)_\infty$ are the points q^{-j} with $j \ge 0$, and $q = q^{-j}$ would give $q^{j+1} = 1$, hence $|q| = 1$. The same theorem at $a = q^{N+1}$ gives $(q^{N+1};q)_\infty \ne 0$ for every $N \ge 0$. The tail factorization of Proposition 1.2,

$$(q;q)_\infty = (q;q)_N\,(q^{N+1};q)_\infty,$$

therefore yields $(q;q)_N = (q;q)_\infty/(q^{N+1};q)_\infty$. Substituting this at $N = n$, $N = k$, and $N = n - k$ in the factorial form $\begin{bmatrix} n \\ k \end{bmatrix}_q = (q;q)_n/\big((q;q)_k(q;q)_{n-k}\big)$ gives the first display.

For the error term, let $M \ge 0$. By the elementary product estimate,

$$|(q^{M+1};q)_\infty - 1| \le \prod_{j\ge M+1} \big(1 + |q|^j\big) - 1 \le \exp\Big(\frac{|q|^{M+1}}{1-|q|}\Big) - 1 \le C_q |q|^{M+1}, \qquad C_q = \frac{e^{1/(1-|q|)}}{1-|q|},$$

since $|q|^{M+1}/(1-|q|) \le 1/(1-|q|)$ and $e^x - 1 \le x e^{x_0}$ for $0 \le x \le x_0$. Moreover

$$|(q^{M+1};q)_\infty| \ge \prod_{j\ge 1}\big(1 - |q|^j\big) = (|q|;|q|)_\infty > 0$$

uniformly in M. As $\min(k, n-k) \le n$, applying both bounds at $M = k$, $M = n - k$, and $M = n$ to the first display gives the second.

The two limits follow at once: in the first, $\min(k, n-k) \to \infty$ by hypothesis; in the second, $n = 2m$ and $k = m + r$ give $\min(k, n-k) = m - |r| \to \infty$, which is also what makes both indices admissible once $m \ge |r|$. □

Remark 19.8. The off-centre column in the regime $q > 1$ needs no new argument. By (19.9) at upper index $2m$ and lower index $m + r$, whose exponent is $k(n-k) = (m+r)(m-r) = m^2 - r^2$,

$$\begin{bmatrix} 2m \\ m+r \end{bmatrix}_q = q^{m^2-r^2} \begin{bmatrix} 2m \\ m+r \end{bmatrix}_{q^{-1}},$$

and Theorem 19.7 applied with q^{-1} in place of q, which is legitimate because $0 < q^{-1} < 1$, gives $\begin{bmatrix} 2m \\ m+r \end{bmatrix}_{q^{-1}} \to (q^{-1};q^{-1})_\infty^{-1}$. Hence

$$\begin{bmatrix} 2m \\ m+r \end{bmatrix}_q \sim \frac{q^{m^2-r^2}}{(q^{-1};q^{-1})_\infty},$$

which is (19.11) at $r = 0$.

Proposition 19.9 (Uniform elementary bound)**.** *For $0 < q < 1$ and $0 \le k \le n$,*

$$1 \le \begin{bmatrix} n \\ k \end{bmatrix}_q \le \frac{1}{(q;q)_\infty}. \tag{19.8}$$

Proof. The lower bound follows from the partition interpretation: the constant term is 1 and every coefficient is nonnegative. The same interpretation says that $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is the generating function of partitions fitting in a k by $(n-k)$ rectangle. Dropping the rectangle restriction embeds this set into all partitions, whose generating function is $(q;q)_\infty^{-1}$. Evaluating at a positive $q < 1$ gives the upper bound. □

19.3 The regime $q > 1$

Corollary 19.10 (Transfer through reciprocity)**.** *Let $q > 1$ and $D = k(n-k)$. Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^D \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}. \tag{19.9}$$

Consequently, for fixed k,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{q^{k(n-k)}}{(q^{-1};q^{-1})_k}\Big(1 + O(q^{-n+k-1})\Big), \tag{19.10}$$

and

$$\begin{bmatrix} 2m \\ m \end{bmatrix}_q \sim \frac{q^{m^2}}{(q^{-1};q^{-1})_\infty}. \tag{19.11}$$

Proof. The reciprocity identity was proved in Theorem 2.5. Apply Theorems 19.6 and 19.7 with q^{-1} in place of q. □

19.4 Extreme specializations

Proposition 19.11. *For $0 \le k \le n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Bigg|_{q=0} = 1, \qquad \lim_{q \to 1} \begin{bmatrix} n \\ k \end{bmatrix}_q = \binom{n}{k},$$

and the leading term as $q \to \infty$ is $q^{k(n-k)}$ with coefficient 1.

Proof. The partition generating polynomial has constant term 1, proving the first assertion. The second follows by replacing every ratio $(1-q^a)/(1-q^b)$ in the product formula by its limit a/b. Palindromicity and degree $k(n-k)$ show that the leading coefficient equals the constant coefficient, namely 1. □

Chapter 20

Computation, certification, and formal proof architecture

The identities in this book are well suited to exact computation because their natural objects are polynomials, rational functions, and formal power series. They are also well suited to proof assistants, provided algebraic identities are separated cleanly from analytic convergence. Lau, Lee, and Ono's 2026 formalization project [20] demonstrates this architecture for q-Pochhammer symbols, Bailey pairs, Jacobi's triple product, and the Rogers–Ramanujan identities.

20.1 Dynamic programming for Gaussian polynomials

A polynomial is represented here by its coefficient vector. Multiplication by q^h is therefore a shift by h places.

Algorithm 20.1 (Pascal dynamic program)**.** Given $n \geq 0$ and $0 \leq k \leq n$:

1. Replace k by $\min(k, n-k)$, using symmetry.
2. Initialize $G_0(q) = 1$ and $G_s(q) = 0$ for $1 \leq s \leq k$.
3. For $r = 1, 2, \ldots, n$, update in descending order

$$G_s(q) \leftarrow G_s(q) + q^{r-s} G_{s-1}(q) \qquad \big(s = \min(r,k), \ldots, 1\big).$$

4. Return $G_k(q)$.

Theorem 20.2 (Correctness and complexity)**.** *Algorithm 20.1 returns $\begin{bmatrix} n \\ k \end{bmatrix}_q$. If $D = k(n-k)$ after the symmetry reduction, its coefficient-level arithmetic cost is $O(nkD)$ and its storage cost is $O(kD)$.*

Proof. Let $G_s^{(r)}$ denote the value after row r. We prove by induction on r that

$$G_s^{(r)} = \begin{bmatrix} r \\ s \end{bmatrix}_q \qquad (0 \leq s \leq \min(r,k)).$$

The assertion is true at $r = 0$. Descending updates ensure that both G_s and G_{s-1} on the right are still values from row $r-1$. Hence the update is

$$G_s^{(r)} = \begin{bmatrix} r-1 \\ s \end{bmatrix}_q + q^{r-s} \begin{bmatrix} r-1 \\ s-1 \end{bmatrix}_q = \begin{bmatrix} r \\ s \end{bmatrix}_q$$

by the second q-Pascal recurrence. At $r = n$ this gives the desired output.

After replacing k by $\min(k, n-k)$, we have $k \le n/2$. Every intermediate polynomial $\begin{bmatrix} r \\ s \end{bmatrix}_q$ with $s \le k$ has degree $s(r-s) \le k(n-k) = D$. There are at most nk shift-and-add updates, each touching at most $D+1$ coefficients. Storing $k+1$ such vectors uses $O(kD)$ coefficients. □

Remark 20.3 (Numerical specialization). When only a numerical value is required, one should not first expand the polynomial. For positive q near 1, the stable formula is

$$\log \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=1}^{k} \left[\log\bigl(-\operatorname{expm1}((n-k+j)\log q)\bigr) - \log\bigl(-\operatorname{expm1}(j \log q)\bigr) \right],$$

where `expm1` avoids cancellation in $e^x - 1$. For exact symbolic work, the Pascal algorithm or the cyclotomic factorization is preferable.

20.2 A factorization algorithm

Algorithm 20.4 (Cyclotomic factor list)**.** For each $1 \le d \le n$, compute

$$e_d = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor .$$

Return the list of d for which $e_d = 1$.

Theorem 20.5 (Correctness of the factor list)**.** *The output of Algorithm 20.4 is exactly the set of cyclotomic factors of* $\begin{bmatrix} n \\ k \end{bmatrix}_q$*, each with multiplicity one.*

Proof. This is Theorem 16.1: the exponent of $\Phi_d(q)$ is e_d, and the floor calculation in that theorem proves $e_d \in \{0, 1\}$. □

Remark 20.6. The factor-list algorithm takes only $O(n)$ integer floor operations. It is therefore far cheaper than expanding $\begin{bmatrix} n \\ k \end{bmatrix}_q$ when the purpose is root-of-unity evaluation, divisibility testing, or congruence work. Polynomial expansion may be postponed until the selected cyclotomic factors are actually multiplied.

20.3 Telescoping certificates for finite sums

A term $T(n, k)$ is called *q-hypergeometric* if the ratios

$$\frac{T(n, k+1)}{T(n, k)}, \qquad \frac{T(n+1, k)}{T(n, k)}$$

are rational functions of q^n and q^k (and of any fixed parameters). Shifted-factorial terms have this property because adjacent factors cancel.

Theorem 20.7 (Finite telescoping certificate)**.** *Let*

$$S_n = \sum_{k=L(n)}^{U(n)} T(n, k).$$

Suppose there are functions $A_0(n), \ldots, A_r(n)$ *and a certificate* $G(n, k)$ *such that*

$$\sum_{j=0}^{r} A_j(n) T(n+j, k) = G(n, k+1) - G(n, k) \tag{20.1}$$

for all relevant k, after extending T by zero outside its natural finite range. If $G(n,k)$ vanishes at both ends of the resulting finite summation interval, then

$$\sum_{j=0}^{r} A_j(n) S_{n+j} = 0. \tag{20.2}$$

Proof. Sum (20.1) over an interval containing the support of every $T(n+j,k)$. The left side becomes $\sum_j A_j(n)S_{n+j}$. The right side telescopes to the terminal value of G minus its initial value, which is 0 by hypothesis. ◻

Corollary 20.8 (Identity certification). *If two finite sums satisfy the same recurrence of order r and agree for r consecutive initial values at every nonsingular parameter choice, then they are equal wherever the recurrence determines subsequent values.*

Proof. Subtract the two sequences. The difference satisfies the homogeneous recurrence and has r consecutive zero initial values. Induction through the recurrence gives zero thereafter. ◻

Remark 20.9 (Creative telescoping). The q-Zeilberger method [23] searches algorithmically for a rational multiple $G(n,k) = R(q^n, q^k)T(n,k)$ satisfying (20.1). Once found, the certificate is independently checkable by rational-function arithmetic. The search may be sophisticated, but verification is elementary: divide by $T(n,k)$ and clear denominators.

20.4 Denominator clearing and specialization

Lemma 20.10 (Polynomial identity principle). *Let $R(q), S(q) \in \mathbb{Q}(q)$. If $R(q) = S(q)$ at infinitely many complex values of q where both sides are defined, then $R = S$ in $\mathbb{Q}(q)$.*

Proof. Write $R - S = A/B$ with $A, B \in \mathbb{Q}[q]$ and $B \neq 0$. The hypothesis says that A has infinitely many zeros. A nonzero polynomial has only finitely many zeros, so $A = 0$. ◻

Corollary 20.11 (Safe generic specialization). *A rational q-identity proved under generic nonvanishing assumptions remains valid after any specialization at which the denominator-cleared identity is defined. Apparent excluded values may therefore be recovered by cancellation or by taking limits.*

Proof. Clear all denominators to obtain a polynomial identity. Polynomial evaluation is valid at every specialization. If the original rational expressions have removable singularities, divide out common factors or take their uniquely determined limits. ◻

This principle is particularly important at roots of unity. One should first rewrite a Gaussian coefficient as its polynomial, not substitute a root into the factorial quotient and accept an indeterminate $0/0$.

20.5 Integer-exponent translation and automated identity discovery

A large class of Gaussian identities can be generated by starting from a terminating basic hypergeometric summation and replacing every shifted factorial whose parameter is an integral power of q by factorial blocks. This old method is now amenable to systematic symbolic search; the identity finder of Zhong and Zhao [40] organizes the necessary side conditions and backtracking choices. The proof certificate, however, remains elementary substitution.

Lemma 20.12 (Two translation rules)**.** *For nonnegative integers a, n, m, k with $0 \le k \le m$,*

$$\frac{(q^{a+1};q)_n}{(q;q)_n} = \begin{bmatrix} a+n \\ n \end{bmatrix}_q, \tag{20.1}$$

$$\frac{(q^{-m};q)_k}{(q;q)_k} = (-1)^k q^{-mk+\binom{k}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q. \tag{20.2}$$

More generally,

$$(q^{a+1};q)_n = \frac{(q;q)_{a+n}}{(q;q)_a}. \tag{20.3}$$

Proof. The factors in (20.3) are precisely $1-q^{a+1}, \dots, 1-q^{a+n}$. Division by $(q;q)_n$ gives (20.1). For the negative rule, factor $-q^{-m+j}$ from the jth factor:

$$\begin{aligned}(q^{-m};q)_k &= \prod_{j=0}^{k-1}(1-q^{-m+j}) \\ &= (-1)^k q^{-mk+\binom{k}{2}} \prod_{j=0}^{k-1}(1-q^{m-j}) \\ &= (-1)^k q^{-mk+\binom{k}{2}} \frac{(q;q)_m}{(q;q)_{m-k}}.\end{aligned}$$

Divide by $(q;q)_k$ to obtain (20.2). □

Theorem 20.13 (Certified integral-parameter translation)**.** *Consider a terminating identity in which every summand and the closed form are products and quotients of monomials in q and shifted factorials of the forms*

$$(q^{a+1};q)_n, \qquad (q^{-m};q)_k, \qquad (q;q)_r,$$

with integer-valued parameters satisfying side conditions that keep all displayed finite indices nonnegative and all original denominators nonzero. Replacing the factors by (20.1)–(20.3)*, cancelling common factorial blocks, and collecting monomial powers of q produces an algebraically equivalent identity in Gaussian coefficients and residual q-factorials. If all residual factorial multiplicities cancel, the output is a pure q-binomial identity.*

Proof. Each replacement is an equality by Lemma 20.12. Products of equal rational functions are equal, cancellation of a common nonzero factor preserves equality under the stated generic assumptions, and collecting monomials is ordinary exponent arithmetic. Hence the transformed summand agrees term by term with the original summand, and the transformed closed form agrees with the original closed form. Summing the finitely many terms preserves equality. Finally, clearing denominators turns the result into a polynomial identity, so removable exceptional specializations are recovered by Corollary 20.11. □

Remark 20.14 (Constraint generation). The nontrivial search problem is not verification but representation: the same factorial block can participate in several Gaussian coefficients, and the integer parameters must satisfy inequalities such as $0 \le k \le m$. An automated finder records the multiplicity of each $(q;q)_r$, searches for compatible triples

$$\frac{(q;q)_u}{(q;q)_v(q;q)_{u-v}} = \begin{bmatrix} u \\ v \end{bmatrix}_q,$$

and carries the corresponding linear inequalities as side conditions. Once a candidate decomposition is printed, Theorem 20.13 reduces its proof to exact algebra.

Corollary 20.15 (Mechanically checkable certificates)**.** *A computer-generated integral-parameter translation can be certified without trusting the search program: expand every Gaussian coefficient into factorial quotients, apply base reversal to every negative parameter, clear denominators, and compare the resulting Laurent monomials and finite products.*

Proof. The prescribed steps reverse the equalities in Lemma 20.12. After denominator clearing, equality is decidable by polynomial arithmetic. Therefore the search code may be treated as a conjecture generator while the emitted certificate is checked independently. □

Historical note 20.16. Recent formal libraries reinforce the same separation of concerns. Eberl's Isabelle development [36] formalizes Gaussian polynomials, finite and infinite q-Pochhammer symbols, Euler expansions, the q-binomial theorem, and q-Vandermonde. The Lean developments cited earlier and the Fabius-function repository [38] similarly distinguish denominator-free polynomial recurrences from analytic infinite products. This architecture is exactly what makes automated discovery compatible with proof-level reliability.

20.6 Formal series versus analytic series

Theorem 20.17 (Formal-to-analytic transfer)**.** *Suppose*

$$\sum_{n\geq 0} a_n z^n = \sum_{n\geq 0} b_n z^n$$

coefficientwise in $\mathbb{C}[[z]]$. *If both series converge for* $|z| < R$*, then they define the same analytic function on that disk. Conversely, if two power series converge in a disk and define the same function on a subset with an accumulation point in the disk, then their coefficients are equal.*

Proof. The forward implication is immediate because the two partial sums agree term by term. For the converse, the identity theorem makes the analytic functions equal throughout the disk; differentiating n times at 0 gives $n!a_n = n!b_n$. □

Principle 20.18 (A scalable proof architecture)**.** *For a mixed algebraic-analytic q-identity, the following order minimizes hidden assumptions:*

1. *define finite shifted factorials and prove their algebraic laws;*
2. *prove terminating identities after denominator clearing;*
3. *define formal infinite products by coefficient stabilization when possible;*
4. *separately establish analytic convergence for* $|q| < 1$*;*
5. *transfer the formal coefficient identity to analytic functions;*
6. *only then take limits, differentiate parameters, or specialize at roots of unity.*

Proof. Each step is justified by results already proved: finite product algebra in Chapter 1; polynomial identity and specialization by Lemma 20.10 and corollary 20.11; formal-to-analytic transfer by Theorem 20.17; and the convergence theorems in Chapters 8 to 10. The ordering prevents an analytic operation from being used before its convergence hypothesis has been supplied. □

Historical note 20.19. Formal verification forces distinctions that handwritten mathematics often suppresses: a finite product indexed by a natural number is not the same object as a bilateral product; a polynomial

Gaussian coefficient is not definitionally the same as a rational factorial quotient; and a formal power-series identity does not automatically contain a proof of analytic convergence. The Lean development of Lau, Lee, and Ono [20] turns these distinctions into reusable interfaces rather than obstacles.

Part VI

Geometric filters, dyadic structure, and the Fabius bridge

Chapter 21

Geometric interpolation and exact q-filters

The geometric grid

$$1, q, q^2, \dots, q^n$$

is the natural interpolation grid for q-calculus. The Newton basis on this grid was developed in Chapter 17. The complementary Lagrange viewpoint is equally useful: its weights are signed Gaussian coefficients, and every moment beyond the interpolation range is again a Gaussian coefficient. This gives exact coefficient filters, closed error formulas for geometric Richardson extrapolation, and the algebraic mechanism behind several constructions in the Fabius-function literature.

21.1 A reversed Gaussian row

For $0 \le k \le n$ define

$$N_{n,k}(q) := (-1)^{n-k} q^{\binom{n-k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q, \tag{21.1}$$

and put $N_{n,k}(q) = 0$ outside this range. Although the definition is written with a quotient notation, $N_{n,k}(q)$ is a polynomial in q with integral coefficients up to the displayed monomial and sign.

Theorem 21.1 (Reversed finite q-binomial theorem)**.** *For every $n \ge 0$,*

$$\sum_{k=0}^{n} N_{n,k}(q) z^k = \prod_{j=1}^{n} (z - q^j). \tag{21.2}$$

Consequently, for every integer d with $0 \le d \le n$,

$$\sum_{k=0}^{n} N_{n,k}(q) q^{kd} = \begin{cases} (q;q)_n, & d = 0, \\ 0, & 1 \le d \le n. \end{cases} \tag{21.3}$$

Proof. Write the product on the right as

$$\prod_{j=1}^{n} (z - q^j) = z^n \prod_{j=1}^{n} \left(1 - \frac{q^j}{z}\right) = z^n (q/z; q)_n.$$

The finite q-binomial theorem, Theorem 3.1, gives

$$z^n (q/z; q)_n = \sum_{r=0}^{n} (-1)^r q^{\binom{r}{2}+r} \begin{bmatrix} n \\ r \end{bmatrix}_q z^{n-r}.$$

Put $k = n - r$. Since $\begin{bmatrix} n \\ r \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q$ by symmetry and $\binom{r}{2} + r = \binom{r+1}{2} = \binom{n-k+1}{2}$, this is exactly (21.2).

Now substitute $z = q^d$. At $d = 0$ the product is $\prod_{j=1}^{n}(1 - q^j) = (q;q)_n$. If $1 \le d \le n$, the factor with $j = d$ vanishes. This proves (21.3). □

Remark 21.2 (Denominator-free meaning). The identity (21.2) belongs to $\mathbb{Z}[q, z]$ and therefore remains valid after mapping q into any commutative ring. No division by $1 - q^j$ is needed. This is the robust way to use Gaussian coefficients when q is a root of unity or when the coefficient ring has zero divisors.

21.2 Lagrange weights at zero

Assume henceforth that q lies in a field K and that

$$q^i \neq q^j \qquad (0 \le i < j \le n).$$

Equivalently, the nodes $1, q, \dots, q^n$ are distinct. Define

$$\lambda_{n,k}^{(q)} := \frac{N_{n,k}(q)}{(q;q)_n} = \frac{(-1)^{n-k} q^{\binom{n-k+1}{2}}}{(q;q)_n} \begin{bmatrix} n \\ k \end{bmatrix}_q. \tag{21.4}$$

Theorem 21.3 (Evaluation at zero on a geometric grid)**.** *For every polynomial $p \in K[X]$ of degree at most n,*

$$p(0) = \sum_{k=0}^{n} \lambda_{n,k}^{(q)} p(q^k). \tag{21.5}$$

Thus $\lambda_{n,k}^{(q)}$ are precisely the Lagrange interpolation weights for evaluation at 0 on the nodes $1, q, \dots, q^n$.

Proof. It is enough to verify (21.5) on the basis $1, X, \dots, X^n$. For $p(X) = 1$, (21.3) divided by $(q;q)_n$ gives 1. For $p(X) = X^d$ with $1 \le d \le n$, the same identity gives $0 = p(0)$. Linearity proves the formula for every polynomial of degree at most n.

The Lagrange interpolation functional at 0 is the unique linear combination of the point evaluations $p \mapsto p(q^k)$ that agrees with $p \mapsto p(0)$ on all polynomials of degree at most n: uniqueness follows from invertibility of the Vandermonde matrix of the distinct nodes. Hence the displayed coefficients are the Lagrange weights. □

Corollary 21.4 (Explicit product form)**.** *For $0 \le k \le n$,*

$$\lambda_{n,k}^{(q)} = \prod_{\substack{0 \le j \le n \\ j \neq k}} \frac{-q^j}{q^k - q^j}. \tag{21.6}$$

Proof. The right side is the usual Lagrange basis polynomial associated with the node q^k, evaluated at 0. By uniqueness in Theorem 21.3, it equals (21.4). One can also simplify the product directly by splitting the factors with $j < k$ and $j > k$. □

Example 21.5. For $n = 2$ the weights are

$$\lambda_{2,0}^{(q)} = \frac{q^3}{(1-q)(1-q^2)}, \qquad \lambda_{2,1}^{(q)} = -\frac{q(1+q)}{(1-q)(1-q^2)}, \qquad \lambda_{2,2}^{(q)} = \frac{1}{(1-q)(1-q^2)}.$$

They satisfy

$$\sum_{k=0}^{2} \lambda_{2,k}^{(q)} = 1, \qquad \sum_{k=0}^{2} \lambda_{2,k}^{(q)} q^k = 0, \qquad \sum_{k=0}^{2} \lambda_{2,k}^{(q)} q^{2k} = 0.$$

21.3 All residual moments

The vanishing moments in (21.3) are only the beginning. Every later moment has a closed form. The key fact is a universal remainder formula involving complete homogeneous symmetric functions.

Theorem 21.6 (Universal Lagrange residual formula)**.** *Let $v_1, \dots, v_N$ be distinct elements of a field, let*

$$\ell_i(x) = \prod_{\substack{1\le j\le N\\ j\ne i}} \frac{x-v_j}{v_i-v_j},$$

and let h_r denote the complete homogeneous symmetric polynomial, with $h_0 = 1$. For every $r \ge 0$,

$$\sum_{i=1}^{N} \ell_i(x) v_i^{N+r} = x^{N+r} - \prod_{i=1}^{N}(x-v_i)\, h_r(x, v_1, \dots, v_N). \tag{21.7}$$

For exponents $0 \le d < N$, the left side with $N + r$ replaced by d is simply x^d.

Proof. The assertion for $d < N$ is ordinary Lagrange interpolation. For $r \ge 0$, put

$$P(T) = \prod_{i=1}^{N}(T - v_i), \qquad H_r(T) = T^{N+r} - P(T)h_r(T, v_1, \dots, v_N).$$

We first show that $\deg H_r < N$. Write $e_j = e_j(v_1, \dots, v_N)$ and $h_j = h_j(v_1, \dots, v_N)$. Then

$$P(T) = \sum_{j=0}^{N}(-1)^j e_j T^{N-j}, \qquad h_r(T, v_1, \dots, v_N) = \sum_{s=0}^{r} T^s h_{r-s}.$$

For $0 \le \ell \le r$, the coefficient of $T^{N+r-\ell}$ in their product is

$$\sum_{j=0}^{\ell}(-1)^j e_j h_{\ell-j}.$$

By elementary–complete orthogonality, Corollary 4.4, this is 1 when $\ell = 0$ and 0 for $1 \le \ell \le r$. Hence the terms of degrees $N, N+1, \dots, N+r$ in $P(T)h_r(T, v_1, \dots, v_N)$ agree with those of T^{N+r}, and therefore $\deg H_r < N$.

At every node v_i we have $P(v_i) = 0$, so $H_r(v_i) = v_i^{N+r}$. Since H_r has degree less than N, Lagrange interpolation gives

$$H_r(x) = \sum_{i=1}^{N} \ell_i(x) v_i^{N+r}.$$

Substituting the definition of H_r proves (21.7). □

Theorem 21.7 (Complete geometric moment formula)**.** *Under the distinct-node hypothesis, for every integer $d \ge 0$,*

$$\sum_{k=0}^{n} \lambda_{n,k}^{(q)} q^{kd} = \begin{cases} 1, & d = 0, \\ 0, & 1 \le d \le n, \\ (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q, & d \ge n+1. \end{cases} \tag{21.8}$$

Equivalently, with the convention that a Gaussian coefficient is zero outside its natural range,

$$\sum_{k=0}^{n} \lambda_{n,k}^{(q)} q^{kd} = (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q \qquad (d > 0). \tag{21.9}$$

Proof. The cases $0 \le d \le n$ were proved in Theorem 21.3. Let $d = n + 1 + r$ with $r \ge 0$. Apply Theorem 21.6 with target $x = 0$ and nodes $v_k = q^k$ for $0 \le k \le n$. There are $N = n + 1$ nodes, and the weights $\ell_k(0)$ are $\lambda^{(q)}_{n,k}$. Since

$$\prod_{k=0}^{n}(0 - q^k) = (-1)^{n+1} q^{\binom{n+1}{2}},$$

we obtain

$$\sum_{k=0}^{n} \lambda^{(q)}_{n,k} q^{k(n+1+r)} = (-1)^n q^{\binom{n+1}{2}} h_r(1, q, \dots, q^n).$$

The geometric specialization in Theorem 4.5 gives

$$h_r(1, q, \dots, q^n) = \begin{bmatrix} n + r \\ n \end{bmatrix}_q .$$

Because $n + r = d - 1$, this is the desired formula. □

Corollary 21.8 (Scaled geometric nodes)**.** *Let $c \ne 0$ and use the same weights $\lambda^{(q)}_{n,k}$ on the nodes cq^k. Then for $d > 0$,*

$$\sum_{k=0}^{n} \lambda^{(q)}_{n,k} (cq^k)^d = c^d (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d - 1 \\ n \end{bmatrix}_q .$$

In particular, these weights evaluate every polynomial of degree at most n at 0 independently of the nonzero scale c.

Proof. Factor c^d from the moment sum and apply Theorem 21.7. Exactness on polynomials follows by linearity. □

21.4 Formal coefficient filters

For a formal power series $f(X) = \sum_{m\ge 0} a_m X^m$, define

$$(\mathcal{G}_{n,q} f)(X) := \sum_{k=0}^{n} \lambda^{(q)}_{n,k} f(q^k X). \tag{21.10}$$

The sum is finite, so this is well defined over any coefficient field in which the weights exist.

Theorem 21.9 (Diagonal action and exact tail)**.** *For every formal power series $f(X) = \sum_{m\ge 0} a_m X^m$,*

$$[X^m]\mathcal{G}_{n,q} f = \left(\sum_{k=0}^{n} \lambda^{(q)}_{n,k} q^{km} \right) a_m, \tag{21.11}$$

$$\mathcal{G}_{n,q} f(X) = a_0 + (-1)^n q^{\binom{n+1}{2}} \sum_{r \ge 0} \begin{bmatrix} n + r \\ n \end{bmatrix}_q a_{n+1+r} X^{n+1+r}. \tag{21.12}$$

Thus the filter deletes the coefficients of degrees $1, \dots, n$ and multiplies every later coefficient by an explicit Gaussian polynomial.

Proof. Substitution gives

$$\mathcal{G}_{n,q} f(X) = \sum_{k=0}^{n} \lambda_{n,k}^{(q)} \sum_{m\geq 0} a_m q^{km} X^m.$$

The outer sum is finite, so coefficient extraction may be interchanged with it, proving (21.11). Apply Theorem 21.7: the multiplier is 1 at $m = 0$, is 0 for $1 \leq m \leq n$, and at $m = n + 1 + r$ is $(-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} n+r \\ n \end{bmatrix}_q$. This gives (21.12). □

Corollary 21.10 (Geometric Richardson extrapolation)**.** *Let f be analytic at 0. The values*

$$f(z),\ f(qz),\ \ldots,\ f(q^n z)$$

produce the extrapolation formula

$$\sum_{k=0}^{n} \lambda_{n,k}^{(q)} f(q^k z) = f(0) + O(z^{n+1}) \qquad (z \to 0),$$

and the complete asymptotic error is the convergent specialization of (21.12).

Proof. Apply Theorem 21.9 to the Taylor series of f. The first nonconstant surviving term has degree $n + 1$. □

Theorem 21.11 (A quantitative analytic bound)**.** *Assume $0 < q < 1$ and $f(z) = \sum_{m\geq 0} a_m z^m$ with $|a_m| \leq M R^{-m}$. If $|z| < R$, then*

$$\left| \sum_{k=0}^{n} \lambda_{n,k}^{(q)} f(q^k z) - f(0) \right| \leq \frac{M q^{\binom{n+1}{2}}}{(q;q)_n} \frac{(|z|/R)^{n+1}}{1 - |z|/R}. \tag{21.13}$$

Proof. For $r \geq 0$,

$$\begin{bmatrix} n+r \\ n \end{bmatrix}_q = \frac{(q^{r+1};q)_n}{(q;q)_n} \leq \frac{1}{(q;q)_n},$$

because every factor in $(q^{r+1};q)_n$ lies in $(0,1)$. Take absolute values in (21.12), use the coefficient bound, and sum the resulting geometric series. □

Remark 21.12. The exact multiplier in (21.12), rather than merely the order estimate $O(z^{n+1})$, is the principal advantage of a geometric grid. The complete tail is controlled by one Gaussian row. This phenomenon is used repeatedly in geometric extrapolation, in formal power-series filters, and in the dyadic identities of the next chapter.

Chapter 22

Half-base arithmetic and Prouhet–Thue–Morse cancellation

The specialization $q = 1/2$ turns factors $1 - q^j$ into normalized Mersenne numbers. It is especially important in exact formulas for the Fabius and Rvachev functions. At this base, the finite q-binomial theorem produces a compact $n + 1$-point Prouhet functional, while the classical Thue–Morse product produces a 2^n-point functional. Both annihilate all polynomials of degree below n, but their weights and node geometries are very different.

22.1 Mersenne normalization

Define

$$\mathfrak{m}_0 := 1, \qquad \mathfrak{m}_n := \prod_{j=1}^{n} (2^j - 1) \quad (n \geq 1). \tag{22.1}$$

Theorem 22.1 (Half-base factorial and Gaussian formulas)**.** *For $n \geq 0$ and $0 \leq k \leq n$,*

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n, \tag{22.2}$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = 2^{-k(n-k)} \frac{\mathfrak{m}_n}{\mathfrak{m}_k \mathfrak{m}_{n-k}}. \tag{22.3}$$

Proof. Each factor satisfies

$$1 - 2^{-j} = 2^{-j}(2^j - 1).$$

Multiplication for $1 \leq j \leq n$ gives (22.2), because $1 + 2 + \cdots + n = \binom{n+1}{2}$. Substitute this expression into

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = \frac{(1/2; 1/2)_n}{(1/2; 1/2)_k (1/2; 1/2)_{n-k}}.$$

The power of 2 in the quotient is

$$-\binom{n+1}{2} + \binom{k+1}{2} + \binom{n-k+1}{2} = -k(n-k),$$

which proves (22.3). $\square$

Theorem 22.2 (Dyadic evaluation of the finite product)**.** *For integers $m, n \geq 0$,*

$$(2^m; 1/2)_n = \begin{cases} 0, & m < n, \\ (-1)^n \dfrac{\mathfrak{m}_m}{\mathfrak{m}_{m-n}}, & m \geq n. \end{cases} \tag{22.4}$$

Equivalently,

$$\sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} 2^{mk} = \begin{cases} 0, & m < n, \\ (-1)^n \dfrac{\mathfrak{m}_m}{\mathfrak{m}_{m-n}}, & m \geq n. \end{cases} \tag{22.5}$$

Proof. By definition,

$$(2^m; 1/2)_n = \prod_{j=0}^{n-1} (1 - 2^{m-j}).$$

If $m < n$, the factor with $j = m$ is zero. If $m \geq n$, every factor is negative and

$$\prod_{j=0}^{n-1} (1 - 2^{m-j}) = (-1)^n \prod_{r=m-n+1}^{m} (2^r - 1) = (-1)^n \frac{\mathfrak{m}_m}{\mathfrak{m}_{m-n}}.$$

The finite q-binomial theorem at $q = 1/2$ and $z = 2^m$ turns the left side into the sum in (22.5). □

Corollary 22.3 (Root locus)**.** *The polynomial*

$$\sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k$$

has the simple roots $1, 2, 4, \ldots, 2^{n-1}$ and no others.

Proof. It is $(z; 1/2)_n = \prod_{j=0}^{n-1} (1 - z/2^j)$, a degree-$n$ polynomial with exactly the displayed roots. □

22.2 A compressed geometric Prouhet functional

Define nodes and weights

$$\xi_k := -2^k, \qquad w_{n,k} := (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} \qquad (0 \leq k \leq n). \tag{22.6}$$

Theorem 22.4 (Geometric Prouhet identity)**.** *For every polynomial p of degree at most n,*

$$\sum_{k=0}^{n} w_{n,k} p(\xi_k) = \mathfrak{m}_n [X^n] p(X). \tag{22.7}$$

In particular, the functional on the left annihilates every polynomial of degree less than n.

Proof. It suffices to take $p(X) = X^d$ for $0 \leq d \leq n$. With $q = 1/2$,

$$\begin{aligned} \sum_{k=0}^{n} w_{n,k} \xi_k^d &= (-1)^d \sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q q^{-kd} \\ &= (-1)^d (q^{-d}; q)_n \end{aligned}$$

by the finite q-binomial theorem. If $d < n$, the factor with index $j = d$ in $(q^{-d}; q)_n$ is zero. At $d = n$, base reversal gives

$$(q^{-n}; q)_n = (-1)^n q^{-\binom{n+1}{2}} (q; q)_n.$$

Therefore

$$(-1)^n (q^{-n}; q)_n = 2^{\binom{n+1}{2}} (1/2; 1/2)_n = \mathfrak{m}_n$$

by (22.2). Linearity now proves (22.7). □

Corollary 22.5 (Affine form)**.** *For constants a, b and every polynomial p of degree at most n,*

$$\sum_{k=0}^{n} w_{n,k} p(a + b\xi_k) = b^n \mathfrak{m}_n [X^n] p(X).$$

Proof. Apply Theorem 22.4 to the polynomial $X \mapsto p(a + bX)$. Its leading coefficient is $b^n [X^n] p(X)$. □

Remark 22.6. The identity uses only $n + 1$ geometrically spaced nodes. The price for this compression is that the weights are nonuniform rational numbers involving the half-base Gaussian row. The Thue–Morse construction below uses 2^n equally weighted signs on an arithmetic grid instead.

22.3 The Thue–Morse product

For a nonnegative integer r, let $s_2(r)$ be the sum of the binary digits of r and put

$$\tau_r := (-1)^{s_2(r)}.$$

This is the signed Thue–Morse sequence.

Theorem 22.7 (Finite Thue–Morse product)**.** *For every $n \geq 0$,*

$$\sum_{r=0}^{2^n - 1} \tau_r z^r = \prod_{j=0}^{n-1} (1 - z^{2^j}). \tag{22.8}$$

Proof. Every r with $0 \leq r < 2^n$ has a unique binary expansion $r = \sum_{j=0}^{n-1} \varepsilon_j 2^j$ with $\varepsilon_j \in \{0, 1\}$. Expanding the product on the right, choose either 1 or $-z^{2^j}$ from the jth factor. The choice vector $(\varepsilon_0, \ldots, \varepsilon_{n-1})$ contributes

$$(-1)^{\varepsilon_0 + \cdots + \varepsilon_{n-1}} z^{\varepsilon_0 + 2\varepsilon_1 + \cdots + 2^{n-1}\varepsilon_{n-1}} = \tau_r z^r.$$

Summing over all binary choices gives the identity. □

Theorem 22.8 (Prouhet cancellation and first surviving moment)**.** *For $n \geq 1$,*

$$\sum_{r=0}^{2^n - 1} \tau_r r^d = \begin{cases} 0, & 0 \leq d < n, \\ (-1)^n 2^{\binom{n}{2}} n!, & d = n. \end{cases} \tag{22.9}$$

More generally, for every polynomial p of degree at most n,

$$\sum_{r=0}^{2^n - 1} \tau_r p(a + br) = (-1)^n 2^{\binom{n}{2}} n!\, b^n [X^n] p(X). \tag{22.10}$$

Proof. Let $P_n(z) = \sum_{r<2^n} \tau_r z^r$. By Theorem 22.7, P_n has a zero of order exactly n at $z = 1$, because every factor $1 - z^{2^j}$ has a simple zero there. The Euler operator $D = z\frac{d}{dz}$ satisfies

$$D^d P_n(1) = \sum_{r<2^n} \tau_r r^d.$$

For $d < n$, this is zero because D^d is a linear combination of ordinary derivatives of orders at most d, all of which vanish at 1.

For $d = n$, the coefficient of the ordinary nth derivative in D^n is z^n, while all lower derivatives still vanish at 1. Hence

$$D^n P_n(1) = P_n^{(n)}(1).$$

The leading coefficient of the local expansion at 1 is the product of the first derivatives of the n factors:

$$P_n^{(n)}(1) = n! \prod_{j=0}^{n-1} \frac{d}{dz}(1 - z^{2^j})\Big|_{z=1} = n! \prod_{j=0}^{n-1}(-2^j) = (-1)^n 2^{\binom{n}{2}} n!.$$

This proves (22.9). Apply it termwise to the polynomial $X \mapsto p(a + bX)$; all terms below degree n vanish, and its leading coefficient is $b^n [X^n] p(X)$. This gives (22.10). □

Corollary 22.9 (Equal power sums). *Partition $\{0, 1, \ldots, 2^n - 1\}$ according to the parity of $s_2(r)$. The two classes have equal sums of dth powers for every $0 \le d < n$.*

Proof. Move the negative-sign terms in (22.9) to the other side. □

22.4 Two cancellation mechanisms

Principle 22.10 (Geometric versus binary Prouhet filters). *There are two complementary exact functionals of order n:*

$$\sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} p(-2^k) = \mathfrak{m}_n [X^n] p,$$

$$\sum_{r=0}^{2^n-1} (-1)^{s_2(r)} p(r) = (-1)^n 2^{\binom{n}{2}} n! [X^n] p,$$

valid for $\deg p \le n$. *The first uses a Gaussian row on a geometric grid; the second uses uniform signs generated by binary digits on an arithmetic grid.*

Proof. These are Theorems 22.4 and 22.8. □

The common product mechanism is now visible. In the geometric case the annihilating polynomial is $(q^{-d}; q)_n$ evaluated at integer d; in the binary case it is the product $\prod_{j<n}(1 - z^{2^j})$ and its order of vanishing at $z = 1$. Both encode a hierarchy of moment cancellations by forcing a product to acquire prescribed zeros.

Chapter 23

Geometric uniform sums, q-cumulants, and the Fabius–Rvachev functions

A second bridge from q-products to the Fabius theory is probabilistic. Replace discrete binary digits by independent uniform coordinates and weight them geometrically. The resulting random series has a smooth compactly supported density, a Fourier transform given by an infinite product of sinc functions, and cumulants containing the factors $1 - q^{2m}$. At $q = 1/2$ its distribution function is the Fabius function and its centered density is the Rvachev up function [37, 39, 34, 38].

23.1 The geometric uniform random series

Let $0 < q < 1$, let $U_0, U_1, \ldots$ be independent random variables uniformly distributed on $[0, 1]$, and define

$$X_q := (1-q)\sum_{j=0}^{\infty} q^j U_j. \tag{23.1}$$

Theorem 23.1 (Convergence, support, symmetry, and self-similarity)**.** *The series* (23.1) *converges absolutely for every outcome and satisfies* $0 \le X_q \le 1$*. Its topological support is* $[0, 1]$*, and*

$$\mathbb{E}X_q = \frac{1}{2}, \qquad X_q \overset{d}{=} 1 - X_q. \tag{23.2}$$

If X_q' is an independent copy of X_q, then

$$X_q \overset{d}{=} (1-q)U_0 + qX_q'. \tag{23.3}$$

Consequently its distribution function $F_q(x) = \mathbb{P}(X_q \le x)$ satisfies

$$F_q(x) = \int_0^1 F_q\left(\frac{x-(1-q)u}{q}\right) du, \tag{23.4}$$

where $F_q(y) = 0$ for $y \le 0$ and $F_q(y) = 1$ for $y \ge 1$.

Proof. Because $0 \le U_j \le 1$,

$$0 \le (1-q)\sum_{j=0}^{N-1} q^j U_j \le (1-q)\sum_{j=0}^{N-1} q^j \le 1.$$

The tail is bounded by

$$(1-q)\sum_{j=N}^{\infty} q^j = q^N,$$

so the series converges absolutely and takes values in $[0,1]$.

Every $x \in [0,1]$ belongs to the support: if all of the first N variables lie sufficiently close to x, then their weighted partial sum lies close to $x(1-q^N)$, and the remaining tail is at most q^N; the corresponding finite cylinder has positive probability. No point outside $[0,1]$ can belong to the support.

Linearity of expectation and the geometric sum give

$$\mathbb{E}X_q = (1-q)\sum_{j\ge 0} q^j \mathbb{E}U_j = \frac{1-q}{2}\frac{1}{1-q} = \frac{1}{2}.$$

Since $1-U_j$ are again independent uniform variables,

$$1-X_q = (1-q)\sum_{j\ge 0} q^j(1-U_j)$$

has the same distribution as X_q. Splitting off the first term and reindexing the tail proves (23.3). Conditioning on $U_0 = u$ then gives

$$\mathbb{P}((1-q)u + qX_q' \le x) = F_q\left(\frac{x-(1-q)u}{q}\right),$$

and integration over $u \in [0,1]$ proves (23.4). □

23.2 Fourier product and cumulants

Use the entire function

$$\operatorname{sinc} z := \begin{cases} \dfrac{\sin z}{z}, & z \ne 0,\\ 1, & z = 0.\end{cases}$$

Theorem 23.2 (Characteristic function)**.** *For real t,*

$$\varphi_q(t) := \mathbb{E}\, e^{itX_q} = e^{it/2}\prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{(1-q)q^j t}{2}\right). \tag{23.5}$$

The product converges uniformly on compact subsets of the complex t-plane and therefore defines an entire function.

Proof. For a uniform variable U on $[0,1]$ and any complex s,

$$\mathbb{E}e^{sU} = \int_0^1 e^{su}\,du = e^{s/2}\frac{\sinh(s/2)}{s/2}.$$

At $s = ita$ this becomes $e^{ita/2}\operatorname{sinc}(at/2)$. Independence gives the finite product for the partial sum. The phases combine because $\sum_{j\ge 0}(1-q)q^j = 1$.

For compact t-sets, the arguments of the sinc factors are $O(q^j)$. Since $\operatorname{sinc} z = 1 + O(z^2)$ near 0, the series of deviations from 1 is bounded by a constant multiple of $\sum q^{2j}$, uniformly on the compact set. The standard uniform convergence criterion for infinite products proves locally uniform convergence to an entire function. Passing to the limit in the finite characteristic functions gives (23.5). □

Let $Y_q = X_q - 1/2$. We use Bernoulli numbers in the convention

$$\frac{z}{e^z - 1} = \sum_{m=0}^{\infty} B_m \frac{z^m}{m!}. \tag{23.6}$$

The cumulants $\kappa_m(Y_q)$ are defined by

$$\log \mathbb{E} e^{tY_q} = \sum_{m \geq 1} \kappa_m(Y_q) \frac{t^m}{m!}$$

near $t = 0$.

Theorem 23.3 (q-cumulants of the geometric uniform sum)**.** *All odd cumulants of Y_q vanish, and for $m \geq 1$,*

$$\kappa_{2m}(Y_q) = \frac{B_{2m}}{2m} \frac{(1-q)^{2m}}{1-q^{2m}}. \tag{23.7}$$

Equivalently,

$$\log \mathbb{E} e^{tY_q} = \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!} \frac{(1-q)^{2m}}{1-q^{2m}} t^{2m} \tag{23.8}$$

for t in a neighborhood of 0.

Proof. For U uniform on $[0, 1]$,

$$\mathbb{E} e^{t(U-1/2)} = \frac{\sinh(t/2)}{t/2}.$$

Differentiating the logarithm gives

$$\frac{d}{dt} \log \frac{\sinh(t/2)}{t/2} = \frac{1}{2} \coth(t/2) - \frac{1}{t}.$$

The Bernoulli generating function implies the classical Laurent expansion

$$\frac{1}{2} \coth(t/2) - \frac{1}{t} = \sum_{m=1}^{\infty} B_{2m} \frac{t^{2m-1}}{(2m)!}.$$

Integrating from 0 to t yields

$$\log \frac{\sinh(t/2)}{t/2} = \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!} t^{2m}. \tag{23.9}$$

Now

$$Y_q = (1-q) \sum_{j \geq 0} q^j (U_j - 1/2).$$

Cumulant generating functions add under independence. Substitute $t(1-q)q^j$ into (23.9), sum over j, and use $\sum_{j \geq 0} q^{2mj} = 1/(1-q^{2m})$. This proves (23.8); comparison of coefficients gives (23.7). Only even powers occur, so all odd cumulants vanish. □

Definition 23.4 (Complete exponential Bell polynomials)**.** The polynomials $\mathcal{B}_n(x_1, \ldots, x_n)$ are defined by

$$\exp\left(\sum_{r \geq 1} x_r \frac{t^r}{r!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \ldots, x_n) \frac{t^n}{n!}. \tag{23.10}$$

Corollary 23.5 (All moments)**.** *Let $\kappa_r = \kappa_r(Y_q)$, with the values in Theorem 23.3. Then*

$$\mathbb{E}(Y_q^n) = \mathcal{B}_n(\kappa_1, \ldots, \kappa_n). \tag{23.11}$$

The moments can also be generated recursively from $\mu_0 = 1$ by

$$\mu_n = \sum_{r=1}^{n} \binom{n-1}{r-1} \kappa_r \mu_{n-r}, \qquad \mu_n := \mathbb{E}(Y_q^n). \tag{23.12}$$

Finally,

$$\mathbb{E}(X_q^n) = \sum_{j=0}^{n} \binom{n}{j} 2^{j-n} \mu_j. \tag{23.13}$$

Proof. The moment generating function is the exponential of the cumulant generating function. Comparing it with the defining identity (23.10) proves (23.11). If $M(t) = \sum \mu_n t^n/n!$ and $K(t) = \sum \kappa_r t^r/r!$, then $M = e^K$ and $M' = K'M$. Comparing the coefficient of t^{n-1} gives (23.12). The last formula is the binomial expansion of $(Y_q + 1/2)^n$. □

Corollary 23.6 (The half-base cumulants)**.** *At $q = 1/2$,*

$$\kappa_{2m}(Y_{1/2}) = \frac{B_{2m}}{2m(2^{2m}-1)}, \qquad \kappa_{2m+1}(Y_{1/2}) = 0. \tag{23.14}$$

Hence all moments of $X_{1/2}$ are rational and are computable to arbitrary order by (23.12)*.*

Proof. In (23.7),

$$\frac{(1-1/2)^{2m}}{1-2^{-2m}} = \frac{1}{2^{2m}-1}.$$

The Bernoulli numbers are rational, so the recurrence uses only rational operations. □

23.3 Superpolynomial Fourier decay and smoothness

Theorem 23.7 (Log-Gaussian decay in frequency)**.** *Put*

$$A = \frac{1-q}{2}, \qquad b = \log(1/q) > 0.$$

For sufficiently large $|t|$, with $L = \log(A|t|)$,

$$|\varphi_q(t)| \le \exp\left(-\frac{L^2}{2b} + \frac{L}{2}\right). \tag{23.15}$$

In particular, for every $M \ge 0$,

$$|t|^M \varphi_q(t) \in L^1(\mathbb{R}).$$

Proof. The phase $e^{it/2}$ has absolute value 1, and $|\operatorname{sinc} u| \le \min(1, 1/|u|)$. Choose

$$N = \lfloor L/b \rfloor + 1.$$

For $0 \le j < N$ we have $A|t|q^j \ge 1$. Hence

$$\begin{aligned}|\varphi_q(t)| &\le \prod_{j=0}^{N-1} \frac{1}{A|t|q^j} \\ &= \exp\left(-NL + \frac{bN(N-1)}{2}\right).\end{aligned}$$

Because $L \ge b(N-1)$, the exponent is at most $-bN(N-1)/2$. For L sufficiently large, $N \ge L/b$ and $N - 1 \ge L/b - 1$, so

$$-\frac{bN(N-1)}{2} \le -\frac{L^2}{2b} + \frac{L}{2}.$$

This proves (23.15).

After the change of variables $L = \log(At)$, the right side is a Gaussian in L multiplied by $e^{L/2}$. It therefore dominates every negative power of t. More explicitly, for each M the exponent $-L^2/(2b) + (M + 3/2)L$ tends to $-\infty$ quadratically, which proves integrability of $|t|^M \varphi_q(t)$. □

Theorem 23.8 (Smooth compactly supported density)**.** *For every* $0 < q < 1$*, the distribution of* X_q *has a* C^∞ *density* f_q *supported on* $[0, 1]$*. For every* $m \ge 0$*,*

$$f_q^{(m)}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} (-it)^m e^{-itx} \varphi_q(t)\, dt. \tag{23.16}$$

All derivatives of f_q *vanish at the endpoints* 0 *and* 1*.*

Proof. By Theorem 23.7, $t^m \varphi_q(t)$ is integrable for every m. Fourier inversion for a finite measure with integrable characteristic function gives the density

$$f_q(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \varphi_q(t)\, dt.$$

For completeness, one may obtain this directly by convolving the distribution with a centered Gaussian of variance ε: its density is the inverse transform of $\varphi_q(t)e^{-\varepsilon t^2/2}$. Dominated convergence as $\varepsilon \downarrow 0$ yields the displayed f_q uniformly. Against every compactly supported continuous test function, the Gaussian convolutions converge both to integration against f_q and weakly to the original measure; hence that measure has density f_q.

Because $t^m \varphi_q(t)$ is integrable, differentiation under the integral is justified by dominated convergence and gives (23.16). Thus $f_q \in C^\infty(\mathbb{R})$. The random variable is supported on $[0, 1]$, so the density is zero on the two open complementary half-lines. Smoothness then forces every derivative to have value 0 at both boundary points. □

23.4 Finite spline approximants

Put

$$X_{q,N} := (1-q) \sum_{j=0}^{N-1} q^j U_j, \qquad a_j := (1-q)q^j. \tag{23.17}$$

For $x \in \mathbb{R}$ and an integer $r \ge 0$, write $x_+^r = (\max(x, 0))^r$; when $r = 0$, interpret x_+^0 as the indicator of $x > 0$ away from the knots.

The following nonidentically scaled uniform-sum formula is a box-spline form of the generalized Irwin–Hall distribution; compare Bradley and Gupta [35].

Theorem 23.9 (Generalized uniform-sum spline)**.** *For $N \geq 1$, the density of $X_{q,N}$ is*

$$f_{q,N}(x) = \frac{1}{(N-1)!\prod_{j=0}^{N-1} a_j} \sum_{A\subseteq\{0,\ldots,N-1\}} (-1)^{|A|} \left(x - \sum_{j\in A} a_j\right)_+^{N-1}. \tag{23.18}$$

It is a compactly supported piecewise polynomial of degree at most $N-1$, and its Fourier transform is the corresponding finite product in (23.5).

Proof. The density of $a_j U_j$ is $a_j^{-1}\mathbf{1}_{[0,a_j]}$. Thus $f_{q,N}$ is the convolution of these N box densities. For $s > 0$, its Laplace transform is

$$\prod_{j=0}^{N-1} \frac{1-e^{-a_j s}}{a_j s}.$$

On the other hand,

$$\int_0^\infty e^{-sx} \frac{(x-c)_+^{N-1}}{(N-1)!}\, dx = \frac{e^{-cs}}{s^N}.$$

Taking the Laplace transform of the right side of (23.18) therefore gives

$$\frac{1}{s^N \prod_j a_j} \sum_A (-1)^{|A|} e^{-s\sum_{j\in A} a_j} = \prod_{j=0}^{N-1} \frac{1-e^{-a_j s}}{a_j s},$$

because the subset sum factors as $\prod_j (1-e^{-a_j s})$. Uniqueness of the Laplace transform proves the density formula. Piecewise polynomiality is immediate from the truncated-power representation. The Fourier product follows either by taking Fourier transforms of the box convolutions or by the finite version of the proof of Theorem 23.2. □

Theorem 23.10 (Dyadic Thue–Morse spline)**.** *At $q = 1/2$,*

$$f_{1/2,N}(x) = \frac{2^{\binom{N+1}{2}}}{(N-1)!} \sum_{r=0}^{2^N-1} (-1)^{s_2(r)} \left(x - \frac{r}{2^N}\right)_+^{N-1}. \tag{23.19}$$

Thus, after translating by the mean $(1-2^{-N})/2$, the inverse Fourier transform of the Nth partial sinc product is a Thue–Morse signed spline on the dyadic grid.

Proof. Here $a_j = 2^{-j-1}$, so

$$\prod_{j=0}^{N-1} a_j = 2^{-\binom{N+1}{2}}.$$

Every subset A corresponds to a binary word $(\varepsilon_0, \ldots, \varepsilon_{N-1})$ and

$$\sum_{j\in A} 2^{-j-1} = \sum_{j=0}^{N-1} \varepsilon_j 2^{-j-1}.$$

Reversing the order of the bits identifies this number with $r/2^N$ for a unique $0 \leq r < 2^N$, while $|A|$ and $s_2(r)$ have the same parity. Substitute these facts into (23.18). □

Corollary 23.11 (A coupling convergence bound)**.** *Almost surely,*

$$0 \le X_q - X_{q,N} \le q^N. \tag{23.20}$$

Consequently,

$$F_q(x - q^N) \le F_{q,N}(x) \le F_q(x + q^N). \tag{23.21}$$

In particular,

$$\sup_x |F_{q,N}(x) - F_q(x)| \le \|f_q\|_\infty q^N. \tag{23.22}$$

Proof. The omitted tail is nonnegative and bounded by $(1-q)\sum_{j \ge N} q^j = q^N$, proving (23.20). The inclusions

$$\{X_q \le x - q^N\} \subseteq \{X_{q,N} \le x\} \subseteq \{X_q \le x + q^N\}$$

give (23.21). Since F_q has bounded density, $|F_q(x+h) - F_q(x)| \le \|f_q\|_\infty |h|$; apply this on both sides. □

23.5 The Fabius and Rvachev specializations

Set $q = 1/2$, write $X = X_{1/2}$, and let

$$F(x) := \mathbb{P}(X \le x),$$

extended by $F(x) = 0$ for $x \le 0$ and $F(x) = 1$ for $x \ge 1$.

Theorem 23.12 (The probabilistic Fabius equation)**.** *The function F is C^∞, satisfies*

$$F(1-x) = 1 - F(x), \tag{23.23}$$

$$F(x) = \int_{2x-1}^{2x} F(u)\,du, \tag{23.24}$$

$$F'(x) = 2F(2x) - 2F(2x-1). \tag{23.25}$$

In particular, for $0 \le x \le 1/2$,

$$F'(x) = 2F(2x). \tag{23.26}$$

Together with $F(0) = 0$, $F(1) = 1$, and symmetry, this is the standard Fabius function.

Proof. Smoothness follows from Theorem 23.8. Since X has no atoms and $X \overset{d}{=} 1 - X$,

$$F(1-x) = \mathbb{P}(1 - X \le 1 - x) = \mathbb{P}(X \ge x) = 1 - F(x),$$

which proves (23.23).

At $q = 1/2$, the fixed-point equation is $X \overset{d}{=} (U + X')/2$. Conditioning on $U = u$ gives

$$F(x) = \int_0^1 F(2x - u)\,du = \int_{2x-1}^{2x} F(v)\,dv,$$

proving (23.24). Differentiate using the fundamental theorem of calculus to obtain (23.25). On $0 \le x \le 1/2$ the term $F(2x-1)$ is zero, giving (23.26). □

Let $f = F'$ and define the centered density

$$\mathrm{up}(y) := \begin{cases} \dfrac{1}{2} f\left(\dfrac{y+1}{2}\right), & -1 \le y \le 1, \\ 0, & |y| > 1. \end{cases} \tag{23.27}$$

This is the Rvachev up function in the probability normalization $\int_{\mathbb{R}} \mathrm{up}(y)\,dy = 1$.

Theorem 23.13 (Infinite sinc product and the up equation)**.** *With the Fourier convention* $\widehat{g}(t) = \int_{\mathbb{R}} e^{ity} g(y)\, dy$*,*

$$\widehat{\mathrm{up}}(t) = \prod_{k=1}^{\infty} \operatorname{sinc}\left(\frac{t}{2^k}\right). \tag{23.28}$$

Moreover,

$$\mathrm{up}'(y) = 2\,\mathrm{up}(2y+1) - 2\,\mathrm{up}(2y-1) \qquad (y \in \mathbb{R}). \tag{23.29}$$

Proof. The random variable with density up is $Y = 2X - 1$. Writing $V_j = 2U_j - 1$, we have independent uniform variables on $[-1, 1]$ and

$$Y = \sum_{j=0}^{\infty} 2^{-j-1} V_j.$$

The characteristic function of V_j is $\operatorname{sinc} t$. Independence therefore gives

$$\mathbb{E} e^{itY} = \prod_{j=0}^{\infty} \operatorname{sinc}(t/2^{j+1}),$$

which is (23.28).

From (23.27), $f(x) = 2\,\mathrm{up}(2x-1)$. Differentiate (23.25) to obtain

$$f'(x) = 4f(2x) - 4f(2x-1).$$

Set $x = (y+1)/2$. Because $\mathrm{up}'(y) = f'((y+1)/2)/4$, substitution of $f(t) = 2\,\mathrm{up}(2t-1)$ gives

$$\mathrm{up}'(y) = 2\,\mathrm{up}(2y+1) - 2\,\mathrm{up}(2y-1).$$

The identity remains valid outside the support because all boundary derivatives vanish. □

Remark 23.14 (Where the q-objects enter). The parameter q controls four structures at once:

1. the geometric coordinate weights $(1-q)q^j$;
2. the factors $1 - q^{2m}$ in the cumulants (23.7);
3. the geometric interpolation nodes and Gaussian residual moments of Chapter 21;
4. at $q = 1/2$, the Mersenne products, dyadic Thue–Morse splines, and the Fabius–Rvachev functional equations.

Thus the appearance of q-Pochhammer and Gaussian coefficients in exact Fabius formulas is not accidental: both arise from the same geometric scaling law.

Appendix A

Formula atlas

This appendix is a navigation table, not a substitute for hypotheses and proofs. Each formula points back to the theorem where its exact domain and derivation are given.

Name	Formula	Location
Concatenation	$(a;q)_{m+n} = (a;q)_m (aq^m;q)_n$	Proposition 1.2
Dissection	$(a;q)_{rn} = \prod_{j=0}^{r-1}(aq^j;q^r)_n$	Proposition 1.3
Dissection with remainder	$(a;q)_{rn+u} = \prod_{s<u}(aq^s;q^r)_{n+1}\prod_{s\ge u}(aq^s;q^r)_n$	Corollary 1.4
Base reversal	$(a;q)_n = (-a)^n q^{\binom{n}{2}}(a^{-1}q^{1-n};q)_n$	Proposition 1.6
Negative index	$(a;q)_{-n} = 1/(aq^{-n};q)_n$	Proposition 1.8
Infinite logarithm	$\log(a;q)_\infty = -\sum_{r\ge 1} a^r/(r(1-q^r))$	Theorem 1.16
Gaussian product	$\begin{bmatrix} n \\ k \end{bmatrix}_q = (q;q)_n/((q;q)_k(q;q)_{n-k})$	Proposition 2.2
Pascal I	$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$	Theorem 2.3
Pascal II	$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k}\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$	Theorem 2.3
Reciprocity	$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)}\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}$	Theorem 2.5
Finite theorem	$(z;q)_n = \sum_k (-1)^k q^{\binom{k}{2}}\begin{bmatrix} n \\ k \end{bmatrix}_q z^k$	Theorem 3.1
Reciprocal theorem	$(z;q)_{n+1}^{-1} = \sum_{k\ge 0}\begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$	Theorem 3.4
q-Vandermonde	$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_k q^{(m-k)(r-k)}\begin{bmatrix} m \\ k \end{bmatrix}_q \begin{bmatrix} n \\ r-k \end{bmatrix}_q$	Theorem 3.5
q-binomial inversion	$b_n = \sum_k \begin{bmatrix} n \\ k \end{bmatrix}_q a_k \iff a_n = \sum_k (-1)^{n-k} q^{\binom{n-k}{2}}\begin{bmatrix} n \\ k \end{bmatrix}_q b_k$	Theorem 3.8
Infinite theorem	$\sum_{n\ge 0}(a;q)_n z^n/(q;q)_n = (az;q)_\infty/(z;q)_\infty$	Theorem 8.1
Euler I	$(z;q)_\infty = \sum_{n\ge 0}(-1)^n q^{\binom{n}{2}} z^n/(q;q)_n$	Corollary 8.2
Euler II	$1/(z;q)_\infty = \sum_{n\ge 0} z^n/(q;q)_n$, $\lvert z\rvert < 1$	Corollary 8.2
q-Gauss	${}_2\phi_1(a,b;c;q,c/(ab)) = (c/a,c/b;q)_\infty/(c,c/(ab);q)_\infty$	Theorem 9.4
Heine	${}_2\phi_1(a,b;c;q,z) = \dfrac{(b,az;q)_\infty}{(c,z;q)_\infty}\,{}_2\phi_1(c/b,z;az;q,b)$	Theorem 9.3
q-Pfaff–Saalschütz	Terminating balanced ${}_3\phi_2$ product evaluation	Theorem 9.6
q-Chu–Vandermonde	Two terminating ${}_2\phi_1$ evaluations	Corollary 9.7
Ramanujan ${}_1\psi_1$	${}_1\psi_1(a;b;q,z) = (q,b/a,az,q/(az);q)_\infty/(b,q/a,z,b/(az);q)_\infty$	Theorem 10.2
Finite triple product	$(z;q)_N(q/z;q)_N = \sum_{k=-N}^{N}(-1)^k q^{\binom{k}{2}}\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k$	Theorem 13.3
Jacobi triple product	$\sum_{n\in\mathbb{Z}}(-1)^n q^{\binom{n}{2}} z^n = (q,z,q/z;q)_\infty$	Theorem 13.1
Pentagonal theorem	$(q;q)_\infty = \sum_{n\in\mathbb{Z}}(-1)^n q^{n(3n-1)/2}$	Theorem 13.7
q-derivative	$D_q f(x) = (f(x) - f(qx))/((1-q)x)$	Proposition 11.1
Jackson integral	$\int_0^a f(x)\,\mathrm{d}_q x = (1-q)a\sum_{n\ge 0} q^n f(aq^n)$	Theorem 11.11
q-gamma	$\Gamma_q(x) = (1-q)^{1-x}(q;q)_\infty/(q^x;q)_\infty$	Theorem 12.2
Cyclotomic factorization	$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_d \Phi_d(q)^{\lfloor n/d\rfloor - \lfloor k/d\rfloor - \lfloor (n-k)/d\rfloor}$	Theorem 16.1
q-Lucas	$\begin{bmatrix} ad+b \\ rd+s \end{bmatrix}_q \equiv \binom{a}{r}\begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}$	Theorem 16.7
q-Babbage	$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}} \pmod{\Phi_n(q)^2}$	Theorem 16.10

Name	Formula	Location
Negative upper index	$\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk-\binom{k}{2}} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q$	Proposition 17.2
Geometric Newton basis	$P(x) = \sum_k c_k (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k$	Theorem 17.10
Central reduction	$\begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} / (-q; q)_{2k} = (q; q^2)_k / (q^2; q^2)_k$	Lemma 18.1
Bernoulli expansion	All-orders expansion of $\log(\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} / \binom{n}{k})$	Theorem 19.2

Appendix B

Limit and specialization dictionary

B.1 Elementary limits

For fixed nonnegative integers, the following transitions are direct consequences of the product formulas:

$$\lim_{q\to 1}[n]_q = n, \qquad \lim_{q\to 1}(1-q)^{-n}(q^a;q)_n = (a)_n,$$

$$\lim_{q\to 1}\begin{bmatrix} n \\ k \end{bmatrix}_q = \binom{n}{k}, \qquad \lim_{n\to\infty}\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q;q)_k} \quad (|q|<1),$$

$$\lim_{q\to 1^-}\Gamma_q(x) = \Gamma(x), \qquad \lim_{q\to 0}\begin{bmatrix} n \\ k \end{bmatrix}_q = 1 \quad (0\le k\le n).$$

Each has been proved in the main text; the relevant references are Propositions 1.19 and 19.11, corollary 2.6, and Theorems 12.5 and 19.6.

Transitions in the size parameter, rather than in q, are already represented above by $\lim_{n\to\infty}\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_q = 1/(q;q)_k$. Their central counterpart is

$$\lim_{N\to\infty}\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \frac{1}{(q;q)_\infty} \qquad (k\in\mathbb{Z} \text{ fixed},\ 0<|q|<1),$$

which is Theorem 19.7 when $k=0$ and follows for every fixed k by the same argument; it is what carries the finite triple product (13.3) to Jacobi's (13.2). See Theorem 13.3 and corollary 13.5, and compare Proposition 14.5, which records the same stabilization coefficientwise.

B.2 Classical shadows of the main summations

Under the standard scaling $a=q^A$, $b=q^B$, and $q\to 1^-$, terminating basic hypergeometric sums tend to ordinary hypergeometric sums. The mechanism is

$$\frac{(q^A;q)_n}{(1-q)^n} \longrightarrow (A)_n, \qquad \frac{(q;q)_n}{(1-q)^n} \longrightarrow n!.$$

Thus:

1. the finite q-binomial theorem tends to Newton's binomial theorem;
2. q-Chu–Vandermonde tends to Chu–Vandermonde;

3. q-Pfaff–Saalschütz tends to Pfaff–Saalschütz;
4. the Jackson q-beta integral tends to Euler's beta integral;
5. Theorem 18.2 tends to Corollary 18.3;
6. Theorem 18.4 tends, at the level of its Lambert tail and prefactor, to Theorem 18.7.

B.3 Roots of unity

At a root of unity, factorial quotients should be replaced by polynomial formulas before evaluation. The basic dictionary is:

$$\begin{aligned}\Phi_d(q) \mid \begin{bmatrix} n \\ k \end{bmatrix}_q &\iff k \bmod d > n \bmod d, \\ \begin{bmatrix} ad+b \\ rd+s \end{bmatrix}_q &\equiv \binom{a}{r}\begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}, \\ \begin{bmatrix} n \\ k \end{bmatrix}_q \Bigg|_{q=-1} &\text{ is given by Corollary 16.8.}\end{aligned}$$

These statements turn apparently singular substitutions into finite enumerations and are the algebraic basis of the subset cyclic-sieving theorem.

Appendix C

Proof-dependency guide

The monograph can be read linearly, but several shorter routes are useful.

Route to Gaussian combinatorics

$$\text{finite products} \longrightarrow q\text{-Pascal} \longrightarrow \text{finite } q\text{-binomial theorem} \longrightarrow \text{partitions, paths, subspaces.}$$

Read Chapters 1 to 3, 5 and 6.

Route to product–sum identities

$$\begin{gathered}\text{infinite } q\text{-binomial theorem} \longrightarrow \text{Heine transformation} \longrightarrow q\text{-Gauss,}\\ \text{negative indices and bilateral convergence} \longrightarrow {}_1\psi_1,\\ \text{Euler expansions} \longrightarrow \text{Jacobi triple product} \longrightarrow \text{Bailey pairs.}\end{gathered}$$

Read Chapters 8 to 10, 13 and 15.

Route to arithmetic

$$1 - q^n = \prod_{d|n} \Phi_d(q) \longrightarrow \text{carry criterion} \longrightarrow q\text{-Lucas} \longrightarrow \text{cyclic sieving and } q\text{-Babbage.}$$

Read Chapter 16.

Route to analytic deformation

$$\begin{gathered}\log(a;q)_\infty \longrightarrow \text{Lambert series and dilogarithms} \longrightarrow q\text{-gamma}\\ \longrightarrow \text{central sums and Bernoulli asymptotics.}\end{gathered}$$

Read Chapters 1, 12, 18 and 19.

Route to geometric filters and the Fabius bridge

$$\begin{gathered}\text{reversed Gaussian rows} \longrightarrow \text{geometric Lagrange moments} \longrightarrow \text{exact coefficient filters}\\ \longrightarrow \text{dyadic Prouhet cancellation} \longrightarrow \text{Fabius–Rvachev sinc products.}\end{gathered}$$

Read Chapters 21 to 23.

Appendix D

Formalization status

Every mathematical result in this book is intended, eventually, to have a machine-checked proof in the Lean development that accompanies the Fabius project [38]. This appendix records how far that has got, result by result, so that what remains can be planned rather than rediscovered.

The classification is deliberately severe, and it is the same rule the project uses to decide whether a claim may leave its quarantine boundary. A result counts as formalized only when a current Lean declaration proves the *exact* statement: same hypotheses, same domain, same normalization, same conclusion. A related definition, a theorem with stronger hypotheses or a weaker conclusion, an executable computation, a numerical match, or an apparently immediate derivation is *not* an exact counterpart. In particular a Lean lemma that appears as a step inside the proof below does not formalize the result the proof concludes.

exact a named Lean declaration proves the same statement; the declaration and its module are given.

partial Lean has the result only in a special case, at a fixed parameter, on a restricted domain, or with a weaker conclusion. The named declaration is the nearest one, and the shortfall is real.

none nothing in the development proves it, even partially.

n/a the environment states a convention or an interface rather than a mathematical claim.

Of the 182 labelled results, 17 are exact, 26 are partial, 136 are none, and 3 are n/a. The formalized part is concentrated exactly where the Fabius development needed it: the finite Gaussian-coefficient core, the geometric Lagrange and filter calculus, half-base and Thue–Morse cancellation, and the geometric uniform law. The classical analytic layer — infinite products, nonterminating basic hypergeometric summation, Jacobi triple product, Bailey pairs — is essentially untouched, and every infinite q-Pochhammer symbol in the development is still a finite product over a `Finset.range`.

D.1 By chapter

Chapter	exact	partial	none	n/a
The algebra of q-shifted factorials	1	2	12	0
q-integers and Gaussian coefficients	1	2	5	0
Finite q-binomial theorems and inversion	1	1	7	0
Weighted binomial coefficients and symmetric functions	0	2	6	0
Partitions, words, paths, and inversion statistics	0	1	8	0
Finite fields, Grassmannians, and rank enumeration	0	1	5	0
Symmetry and unimodality	0	0	3	0
The infinite q-binomial theorem	0	0	5	0
Basic hypergeometric series	0	0	6	0
Bilateral series and Ramanujan's ${}_1\psi_1$ sum	0	0	2	0
q-difference calculus and the quantum binomial	0	0	10	0
The q-gamma and q-beta functions	0	0	8	0
Jacobi's triple product	0	0	6	0
Partitions and product–sum identities	0	0	4	0
Bailey pairs and the Rogers–Ramanujan identities	0	1	6	0
Cyclotomic factorization and congruences	0	0	9	0
Negative upper indices and geometric Newton interpolation	0	2	7	0
Central q-binomial sums and Lambert series	0	0	7	0
Asymptotic regimes and Bernoulli expansions	0	0	8	0
Computation, certification, and formal proof architecture	0	2	8	3
Geometric interpolation and exact q-filters	6	2	1	0
Half-base arithmetic and Prouhet–Thue–Morse cancellation	6	3	0	0
Geometric uniform sums, q-cumulants, and the Fabius–Rvachev functions	2	7	3	0
Total	17	26	136	3

D.2 By result

Result	Status	Lean counterpart
The algebra of q-shifted factorials		
Proposition 1.2	exact	`finiteQPochhammerIn_add` `FiniteQBinomialCore`
Proposition 1.3	none	—
Corollary 1.4	none	—
Proposition 1.6	none	—
Proposition 1.8	none	—
Corollary 1.9	partial	`qPochhammer_two_pow_eq_mersenne_div` `HalfQBinomial`
Theorem 1.10	none	—
Proposition 1.11	none	—

Result	Status	Lean counterpart
Theorem 1.12	none	—
Proposition 1.14	none	—
Proposition 1.15	none	—
Theorem 1.16	partial	`tsum_log_one_sub_geom` `EulerLogTransform`
Corollary 1.17	none	—
Theorem 1.18	none	—
Proposition 1.19	none	—
q-integers and Gaussian coefficients		
Proposition 2.2	partial	`qPochhammer_tail_div_self_eq_qBinomial` `GeometricLagrangeQMoments`
Theorem 2.3	exact	`gaussianBinomial_succ_succ_alt` `FiniteQBinomialCore`
Corollary 2.4	none	—
Theorem 2.5	partial	`gaussianBinomial_symm` `FiniteQBinomialCore (HalfQBinomial)`
Corollary 2.6	none	—
Proposition 2.8	none	—
Proposition 2.9	none	—
Theorem 2.10	none	—
Finite q-binomial theorems and inversion		
Theorem 3.1	exact	`finite_qBinomial_theorem` `FiniteQBinomialCore`
Corollary 3.2	partial	`halfQBinomial_two_pow_sum_eq_zero` `HalfQBinomial`
Corollary 3.3	none	—
Theorem 3.4	none	—
Theorem 3.5	none	—
Corollary 3.6	none	—
Theorem 3.8	none	—
Corollary 3.9	none	—
Theorem 3.10	none	—
Weighted binomial coefficients and symmetric functions		
Theorem 4.2	partial	`completeHomogeneousEval_option_succ` `CompleteHomogeneous`
Theorem 4.3	none	—
Corollary 4.4	none	—
Theorem 4.5	partial	`completeHomogeneousEval_geometric` `GeometricCompleteHomogeneous`
Lemma 4.7	none	—

Result	Status	Lean counterpart
Theorem 4.8	none	—
Corollary 4.9	none	—
Theorem 4.11	none	—
Partitions, words, paths, and inversion statistics		
Theorem 5.1	none	—
Theorem 5.3	none	—
Corollary 5.4	partial	`qBinomial_symm` `HalfQBinomial`
Corollary 5.5	none	—
Theorem 5.6	none	—
Theorem 5.7	none	—
Proposition 5.9	none	—
Proposition 5.10	none	—
Corollary 5.11	none	—
Finite fields, Grassmannians, and rank enumeration		
Theorem 6.1	none	—
Theorem 6.2	partial	`gaussianBinomial_succ_succ` `FiniteQBinomialCore`
Theorem 6.3	none	—
Theorem 6.4	none	—
Theorem 6.5	none	—
Proposition 6.7	none	—
Symmetry and unimodality		
Lemma 7.1	none	—
Proposition 7.2	none	—
Theorem 7.3	none	—
The infinite q-binomial theorem		
Theorem 8.1	none	—
Corollary 8.2	none	—
Corollary 8.3	none	—
Proposition 8.4	none	—
Proposition 8.5	none	—
Basic hypergeometric series		
Proposition 9.2	none	—
Theorem 9.3	none	—
Theorem 9.4	none	—
Lemma 9.5	none	—
Theorem 9.6	none	—

Result	Status	Lean counterpart
Corollary 9.7	none	—
Bilateral series and Ramanujan's ${}_1\psi_1$ sum		
Proposition 10.1	none	—
Theorem 10.2	none	—
q-difference calculus and the quantum binomial		
Proposition 11.1	none	—
Theorem 11.2	none	—
Lemma 11.3	none	—
Theorem 11.4	none	—
Theorem 11.6	none	—
Corollary 11.7	none	—
Corollary 11.9	none	—
Proposition 11.10	none	—
Theorem 11.11	none	—
Corollary 11.12	none	—
The q-gamma and q-beta functions		
Theorem 12.2	none	—
Proposition 12.3	none	—
Theorem 12.4	none	—
Theorem 12.5	none	—
Theorem 12.7	none	—
Corollary 12.8	none	—
Theorem 12.9	none	—
Proposition 12.10	none	—
Jacobi's triple product		
Theorem 13.1	none	—
Corollary 13.2	none	—
Theorem 13.7	none	—
Corollary 13.8	none	—
Theorem 13.9	none	—
Corollary 13.10	none	—
Partitions and product–sum identities		
Theorem 14.1	none	—
Proposition 14.3	none	—
Corollary 14.4	none	—
Proposition 14.5	none	—

Result	Status	Lean counterpart
Bailey pairs and the Rogers–Ramanujan identities		
Lemma 15.2	partial	`halfQBinomial_negativeDyadic_polynomial_sum_eq_zero_of_degre` `HalfQBinomial`
Theorem 15.3	none	—
Corollary 15.4	none	—
Lemma 15.5	none	—
Theorem 15.6	none	—
Corollary 15.7	none	—
Theorem 15.8	none	—
Cyclotomic factorization and congruences		
Theorem 16.1	none	—
Corollary 16.2	none	—
Corollary 16.4	none	—
Lemma 16.6	none	—
Theorem 16.7	none	—
Corollary 16.8	none	—
Theorem 16.9	none	—
Theorem 16.10	none	—
Corollary 16.12	none	—
Negative upper indices and geometric Newton interpolation		
Proposition 17.2	none	—
Theorem 17.3	partial	`gaussianBinomial_succ_succ` `FiniteQBinomialCore`
Corollary 17.4	none	—
Theorem 17.5	none	—
Theorem 17.7	none	—
Corollary 17.8	none	—
Theorem 17.9	partial	`finite_qBinomial_theorem` `FiniteQBinomialCore`
Theorem 17.10	none	—
Corollary 17.11	none	—
Central q-binomial sums and Lambert series		
Lemma 18.1	none	—
Theorem 18.2	none	—
Corollary 18.3	none	—
Theorem 18.4	none	—
Theorem 18.6	none	—
Theorem 18.7	none	—

Result	Status	Lean counterpart
Proposition 18.8	none	—
Asymptotic regimes and Bernoulli expansions		
Lemma 19.1	none	—
Theorem 19.2	none	—
Corollary 19.4	none	—
Theorem 19.6	none	—
Theorem 19.7	none	—
Proposition 19.9	none	—
Corollary 19.10	none	—
Proposition 19.11	none	—
Computation, certification, and formal proof architecture		
Algorithm 20.1	n/a	
Theorem 20.2	partial	`gaussianBinomial` `FiniteQBinomialCore`
Algorithm 20.4	n/a	
Theorem 20.5	none	—
Theorem 20.7	none	—
Corollary 20.8	none	—
Lemma 20.10	none	—
Corollary 20.11	none	—

Lemma 20.12	partial	`finiteQPochhammerIn_self_mul_`
		`gaussianBinomial`
		`FiniteQBinomialCore`

Theorem 20.13	none	—
Corollary 20.15	none	—
Theorem 20.17	none	—
Principle 20.18	n/a	
Geometric interpolation and exact q-filters		
Theorem 21.1	exact	`reversed_finite_qBinomial_theorem` `GeometricQBinomialLagrange`

Theorem 21.3	exact	`geometricLagrangeWeight_eq_`
		`gaussianBinomial_div`
		`GeometricQBinomialLagrange`

Corollary 21.4	exact	`geometricLagrangeWeight_eq_product` `GeometricLagrangeWeights`
Theorem 21.6	exact	`sum_lagrangeEvalWeight_mul_pow_card_add` `LagrangeResidualMoments`

Theorem 21.7	exact	`sum_geometricLagrangeWeight_mul_pow_eq_`
		`gaussianBinomial`
		`GeometricQBinomialLagrange`

Result	Status	Lean counterpart
Corollary 21.8	partial	`sum_geometricLagrangeWeight_mul_scaled_geometric_pow_of_pos` `GeometricResidualMoments`
Theorem 21.9	exact	`coeff_geometricSeriesFilter` `GeometricPowerSeriesFilter`
Corollary 21.10	partial	`geometricLagrangeAnalyticSeriesFilter_eq_constant_add_gaussi` `AnalyticSeriesFilter`
Theorem 21.11	none	—
Half-base arithmetic and Prouhet–Thue–Morse cancellation		
Theorem 22.1	exact	`halfQPochhammer_eq_mersenne_div` `HalfQBinomial`
Theorem 22.2	exact	`qPochhammer_two_pow_eq_ite` `HalfQBinomial`
Corollary 22.3	partial	`halfQBinomial_sum_eq_zero_iff` `HalfQBinomial`
Theorem 22.4	exact	`halfQBinomial_negativeDyadic_polynomial_sum_eq_mersenne` `HalfQBinomial`
Corollary 22.5	partial	`halfQBinomial_negativeDyadic_polynomial_sum_eq_mersenne` `HalfQBinomial`
Theorem 22.7	exact	`prod_one_sub_pow_two_pow` `ThueMorseProductIdentity`
Theorem 22.8	exact	`sum_thueMorseSign_mul_affine_pow_eq_zero` `ThueMorseSparseProuhet`
Corollary 22.9	partial	`sum_thueMorseSign_mul_affine_pow_eq_zero` `ThueMorseSparseProuhet`
Principle 22.10	exact	`halfQBinomial_negativeDyadic_polynomial_sum_eq_mersenne` `HalfQBinomial`
Geometric uniform sums, q-cumulants, and the Fabius–Rvachev functions		
Theorem 23.1	partial	`geometricUniformSeries_mem_Icc` `GeometricUniformLaw`
Theorem 23.2	partial	`charFun_geometricUniformDistribution_prefix_sinc` `GeometricSincFactorization`
Theorem 23.3	none	—
Corollary 23.5	partial	`completeBellPolynomial` `MomentCumulantAlgebra`
Corollary 23.6	none	—
Theorem 23.7	partial	`norm_rvachevFourierProduct_le_decayGauge_abs` `GlobalDecayEnvelope`

Result	Status	Lean counterpart
Theorem 23.8	partial	`geometricUniformDistribution_eq_withDensity` `GeometricUniformCDF`
Theorem 23.9	none	—
Theorem 23.10	partial	`fabiusUniformSpline_eq_tsum_positivePart` `FabiusUniformSpline`
Corollary 23.11	partial	`abs_uniformPartialSum_sub_weightedCoordinateSum_le` `FabiusUniformSpline`
Theorem 23.12	exact	`weightedSumCDF_eq_fabiusReal` `ProbabilityRepresentation`
Theorem 23.13	exact	`rvachevFourier_eq_product` `FourierProduct`

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